SWINGERsim

User Guide

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Contents

1 Introduction **4**
- 1.1 What SWINGERsim is for . . . 4
- 1.2 Questions it answers . . . 6
- 1.3 What it does not do . . . 6
- 1.4 The documentation set . . . 6
- 1.5 How to read this guide . . . 7

2 How SWINGERsim works **9**
- 2.1 The processing chain . . . 9
- 2.2 Two backends . . . 11
- 2.3 How the parts are connected . . . 13
- 2.4 Module map . . . 13
- 2.5 What SWINGERsim implements and what it uses . . . 16
- 2.6 Design principles . . . 18

3 Installation **19**
- 3.1 Requirements . . . 19
- 3.2 Steps . . . 19
- 3.3 Check the installation . . . 21
- 3.4 Optional: stability classifiers of Vynatheya et al. (2023) . . . 21
- 3.5 Optional: MSE for independent comparisons . . . 21
- 3.6 Optional: MESA for own stellar-model grids . . . 21
- 3.7 The worker for long computations . . . 22
- 3.8 Building this documentation . . . 23

4 Quick start **24**
- 4.1 The example system . . . 24
- 4.2 The web application way . . . 24
- 4.3 The command-line way . . . 25
- 4.4 Change one thing . . . 25
- 4.5 Eclipse times of the same system . . . 25

5 The web application **27**
- 5.1 Starting the application . . . 27
- 5.2 The frame of every page . . . 27
- 5.3 Configuration . . . 28
- 5.4 Jobs . . . 32
- 5.5 Outputs . . . 33
- 5.6 Tools . . . 50
- 5.7 Folders used by the application . . . 51

6 The command line **52**
- 6.1 Common conventions . . . 52
- 6.2 Simulations . . . 53
- 6.3 Observables . . . 54
- 6.4 Fits of measured data . . . 57
- 6.5 Ensembles and inference . . . 58

6.6 Stability, chaos and scattering . . . 60
6.7 Validation and own stellar models . . . 61
6.8 Running and controlling . . . 61

7 Workflows **63**
7.1 Forward model of a published system . . . 63
7.2 Fitting measured data . . . 64
7.3 Ensembles: what we do not know . . . 66
7.4 Evolution over millions of years . . . 68
7.5 Stability, chaos maps and scattering . . . 69
7.6 Long computations on the worker . . . 69
7.7 Reproducibility . . . 70

8 Worked examples **71**
8.1 Example 1: a precessing 2+2 quadruple . . . 71
8.2 Example 2: ZLK cycles . . . 73
8.3 Example 3: stars with structure . . . 74
8.4 Example 4: into the past, and stability . . . 75
8.5 Example 5: an ensemble – V994 Her . . . 75
8.6 Example 6: a fit of times of minima, stage 1 . . . 76
8.7 Example 7: two binaries meet . . . 76
8.8 Example 8: where is a system stable? A chaos map . . . 77
8.9 Example 9: what would observers measure? Eclipse timings . . . 78
8.10 Example 10: six stars – V994 Her . . . 79
8.11 Example 11: an evolving quadruple and its light curve . . . 79
8.12 Example 12: more kinds of systems . . . 80

9 The configuration file **82**
9.1 Units and conventions . . . 82
9.2 Top level . . . 82
9.3 `orbits.<name>` . . . 83
9.4 `stars.<name>` . . . 84
9.5 `discs.<name>` . . . 85
9.6 `stellar_model` . . . 86
9.7 `physics` . . . 86
9.8 `run` . . . 89
9.9 `ensemble` . . . 89
9.10 `fit` . . . 90
9.11 The `scatter` configuration . . . 93
9.12 A complete example . . . 94

10 Outputs **96**
10.1 The summary of a run . . . 96
10.2 Files of a run . . . 98
10.3 Ensembles . . . 100
10.4 Eclipse times, minima, light curves and radial velocities . . . 102
10.5 Fits . . . 102
10.6 Scattering experiments and chaos maps . . . 104
10.7 Machine learning . . . 104
10.8 How to interpret the values . . . 104
10.9 Reading the files in Python . . . 104

11 Troubleshooting and FAQ **106**
11.1 The configuration is refused . . . 106
11.2 The run is slow or stops early . . . 107
11.3 The results look wrong . . . 107
11.4 Ensembles . . . 108
11.5 Fits of measured data . . . 108
11.6 The worker and the web application . . . 109
11.7 Frequently asked questions . . . 110

12 Glossary **111**
12.1 Multiple stars and their hierarchy . . . 111
12.2 Orbits and geometry . . . 111
12.3 Dynamics . . . 112
12.4 Stars, tides and interaction . . . 113
12.5 Stability and fates . . . 114
12.6 Observations . . . 114
12.7 Fitting and statistics . . . 115
12.8 Machine learning . . . 115
12.9 Software . . . 116
12.10 References . . . 116

Chapter 1

Introduction

SWINGERsim is a program for modelling **specific, observed multiple-star systems**: a close binary with a third star, two binaries that orbit each other (a 2+2 quadruple), a triple with a distant fourth star (3+1), and larger hierarchies of up to about ten stars, such as the triply eclipsing sextuple V994 Her. You describe one system – the masses of the stars, how they are paired, the orbits on which they move and the physics that matters – and SWINGERsim computes how the system evolves and what an observer would measure: eclipses that appear and vanish, times of minima and their O−C diagram, light curves, radial velocities, and whether the system survives or two of its stars merge.

This User Guide explains what the software is for, how its parts fit together, how to install and control it, and what goes in and comes out of every workflow. It does not derive the physics; that is the subject of the companion document *Physics and Methods*.

1.1 What SWINGERsim is for

A **hierarchical multiple star** is a system whose orbits have very different sizes, so that each orbit is close to a Kepler ellipse that is slowly perturbed by the others. Every orbit of such a system joins two components, each of which is a star or a smaller subsystem; the whole system is a tree of pairs (Figure 1). The slow perturbations are what makes these systems interesting for an observer: the planes of the orbits precess, so the inclination of an eclipsing binary to the line of sight changes and its eclipses become deeper, shallower or disappear; eccentricities oscillate; the times of minima shift because the binary moves on its outer orbit (the light-travel-time effect, LTTE) and because the other stars pull on it.

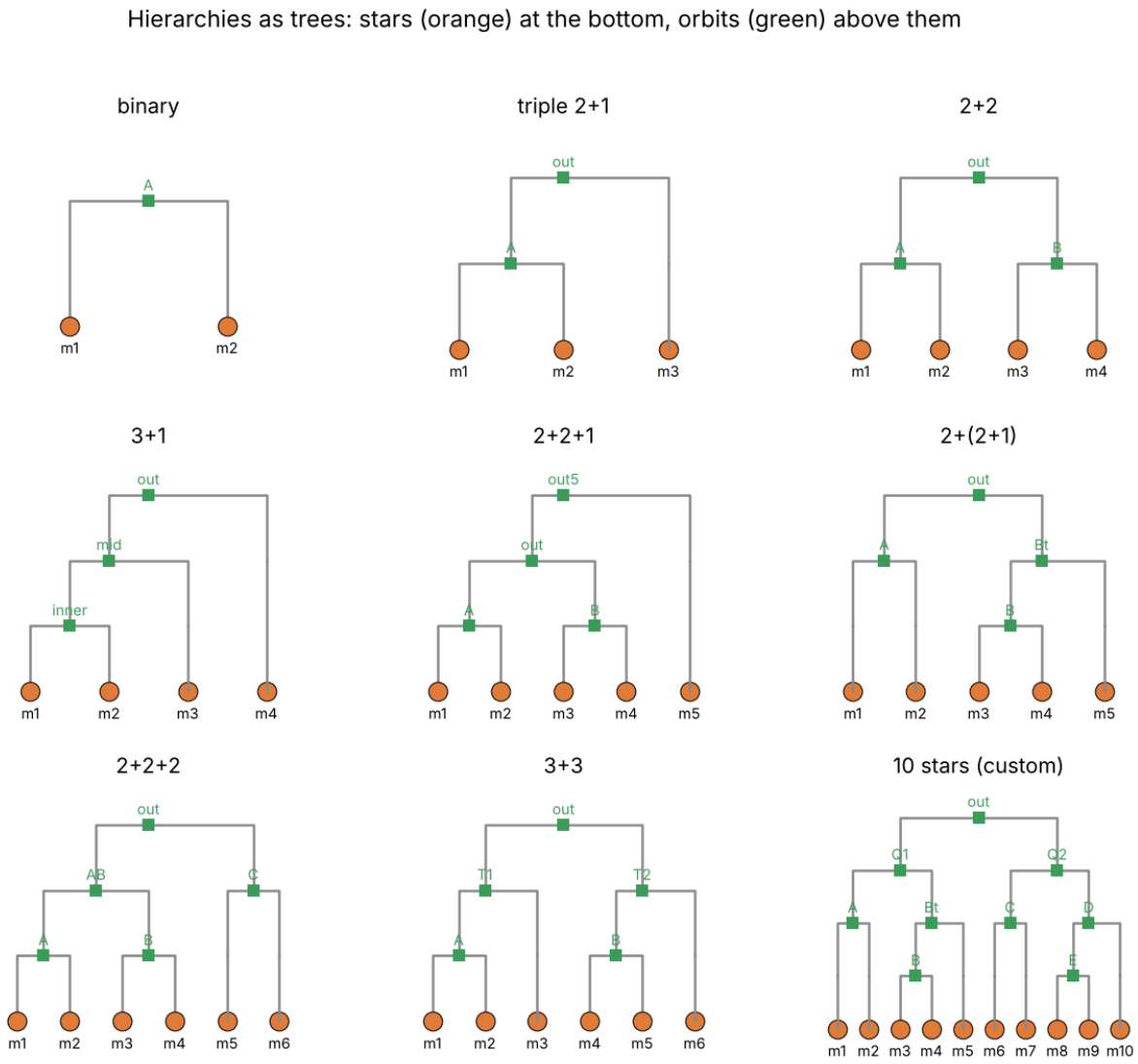


Figure 1. The hierarchies SWINGERsim describes, drawn as trees. Orange circles are stars (named `m1`, `m2`, …), green squares are orbits (named `A`, `B`, `out`, …); every orbit joins the two components drawn below it. The panels show a binary, a triple (2+1), the two kinds of quadruples (2+2 and 3+1), three quintuples and sextuples that have ready-made names in the configuration (2+2+1, 2+(2+1), 2+2+2, 3+3) and a ten-star tree written by hand (`custom`). Read a tree from the bottom: in 3+1, `m1` and `m2` form the orbit `inner`, `inner` and `m3` form `mid`, and `mid` and `m4` form the outermost orbit `out`.

Figure 1 is the key to every configuration file and every output table: the names of the orbits in the tree (`A`, `B`, `out`, `inner`, `mid`, …) are the names under which the elements of the orbits are given in the input and reported in the output. The presets in the top two rows and the two sextuples have fixed names; a custom tree uses the names you write.

SWINGERsim works with one system at a time and offers three ways of asking questions about it.

Way of working	Question it answers	Typical result
forward model	“What does a system with these parameters do, and what would we observe?”	the evolution of the orbits over years to millions of years, eclipse windows, an O−C diagram, a light curve

Way of working	Question it answers	Typical result
fit of measured data	“Which parameters of this system reproduce the measured minima, radial velocities, light curves and inclinations?”	fitted elements with errors, residuals, predictions of future minima, a fitted configuration that runs like any other
ensemble	“Some parameters are unknown – what is possible, and what do the measurements allow?”	distributions of outcomes over thousands of drawn systems, weighted by the measurements

Table 1. The three ways of working with SWINGERsim. Each row names the question, and the right-hand column what you get. The three are connected: a fit produces a configuration that is then run forward, and an ensemble is a set of forward models whose members are weighted by the same kinds of measurements that a fit uses.

1.2 Questions it answers

The following questions are typical for the systems SWINGERsim is made for. Each is answered by one of the workflows of chapter Workflows.

- When did the eclipses of this binary begin, and when will they end?
- Is the observed change of the eclipse depth explained by the precession driven by the known third body, or does it need another, unseen companion?
- Which outer orbits and masses of an unseen companion reproduce a measured inclination and its rate of change?
- Does an N-body model of the whole system reproduce the measured times of minima of all eclipsing pairs at once, and with which elements?
- Is the system dynamically stable, and for how long? Is it near a resonance or in a chaotic region?
- What does the system look like in a TESS light curve, and how precisely would the minima be measured?
- How do the stars and orbits change over millions of years: when does a star fill its Roche lobe, and what follows (mass transfer, a contact phase, a common envelope, a merger)?
- Can the wave in an O−C diagram be magnetic activity instead of a third body?

1.3 What it does not do

SWINGERsim is deliberately narrow. It is not a population-synthesis code: it does not draw whole galaxies of multiple stars and does not model the selection effects of a survey (an ensemble can draw broad distributions, but the interpretation as a population is the user’s). It does not reduce photometry or spectra and does not measure times of minima on real data; it starts from tables of measured quantities. Its light curves are computed from the positions of the stars in the dynamical model (a photodynamical model) with spherical, rotating or Roche-shaped stars; it is not a replacement for a dedicated light-curve code when the detailed surface physics of one binary is the aim (spots, detailed atmospheres). Supernovae and the evolution of neutron stars and black holes are outside its scope, and non-hierarchical (democratic) systems appear only as the outcome of scattering experiments. The limits of each piece of physics are listed in *Physics and Methods*, chapter Limitations.

1.4 The documentation set

The documentation of SWINGERsim consists of seven documents.

Document	Content	Read it when
User Guide (this document)	what the software does, installation, the web application, the command line, workflows, configuration, outputs	you want to use the software
Physics and Methods	the equations and algorithms: hierarchies and orbits, the stars, what the observer sees, the N-body and secular dynamics, stellar evolution, tides, mass transfer, stability, observables, fitting, ensembles, limitations	you need to know what is computed and how accurately
Testing, Part I: Validation against the literature	benchmarks: published results reproduced by SWINGERsim	you want to know how far the physics is verified
Testing, Part II: Three published multiple stars	BU CMi, TIC 120362137 and V994 Her modelled from their data	you want worked applications to real systems
Testing, Part III: Eclipsing binaries with a changing inclination	V907 Sco, QX Cas and AH Cep	you study binaries whose eclipses come and go
Testing, Part IV: Special cases	questions without a published answer: the future of CzeV343 (mass transfer, instability, ZLK merger), the hierarchy of CzeV4315 (2+2 against 2+(2+1) by Bayesian evidence), and the convergence of the MCMC fits	you combine evolution, dynamics and ensembles, or compare hypotheses
Open questions and future work	what is not solved and what could come next	you plan further work

Table 2. The seven documents of the SWINGERsim documentation. The middle column says what each contains, the right column when to read it. References between documents are given in the text by the title of the document and of its chapter.

1.5 How to read this guide

The guide is written for astronomers and students of astrophysics; no programming beyond editing a text file and typing a command is needed. Every technical term is explained at its first use or in the Glossary.

If you want to …	read
know how the program is built and which part does what	How SWINGERsim works
install it	Installation
see results in a few minutes	Quick start
work with forms in a web browser	The web application
work from a terminal or a script	The command line
do a concrete task (fit minima, run an ensemble, follow evolution)	Workflows and Worked examples
look up a key of the configuration file	The configuration file
understand an output file or column	Outputs

If you want to …	read
solve a problem	Troubleshooting and FAQ
look up a term	Glossary

Table 3. Where to find what in this guide. The left column is a task, the right column the chapter that covers it; the chapters can be read in any order after the first two.

Commands are written as they are typed, e.g. `swingersim run config.yaml`; configuration keys are written with dots for nesting, e.g. `run.t_max_yr` is the key `t_max_yr` inside the section `run`, and `orbits.<name>.P` is the period of any orbit. Units follow the conventions of the field: masses in $M_\odot$, radii in $R_\odot$, periods in days (d), semi-major axes in au, angles in degrees and times in years (yr) unless stated otherwise.

Chapter 2

How SWINGERsim works

This chapter describes what happens between a configuration file and the finished results: the processing chain, the parts of the program that carry out each step, how they are connected, and the two ways of computing the motion (the N-body and the secular backend). It ends with a map of all modules and a table that separates what SWINGERsim computes with its own code from what it takes from external libraries and published data.

2.1 The processing chain

Every computation follows the same chain (Figure 2). A **configuration** – a text file in the YAML format – describes one system. SWINGERsim checks it, builds the initial state of the system, integrates the motion with one of two **backends**, analyses the result with a set of **diagnostics**, and writes **products** (tables, figures, an HTML report) and **observables** (eclipse times, light curves, radial velocities). Ensembles, fits and maps are layers on top of this chain: they call it many times with different parameters. The command line and the web application start the computations and show their results.

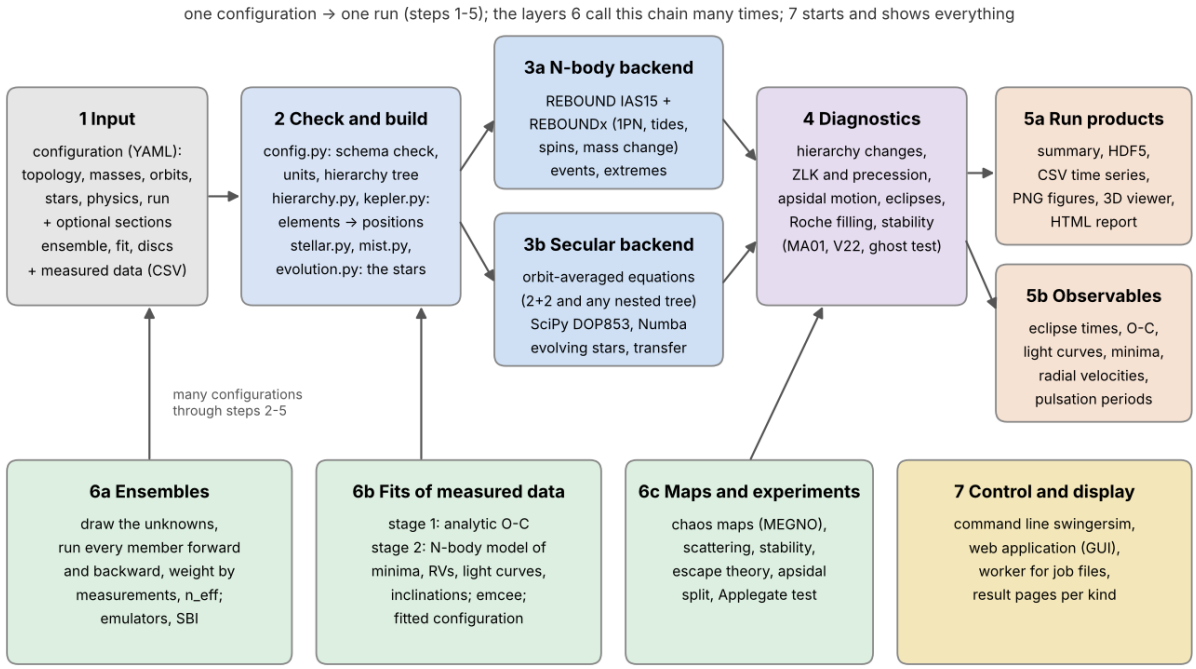


Figure 2. The processing chain of SWINGERsim. Boxes 1–5 (top) are the path of one configuration: the input file (grey) is checked and turned into an initial state (blue, box 2), integrated by the N-body backend (3a) or the secular backend (3b), analysed (violet, box 4) and turned into run products and observables (orange, 5a and 5b). The green boxes at the bottom (6a–6c) are the methods that run this chain many times – ensembles, fits of measured data, maps and experiments; the arrows from them point into the chain they call. The yellow box (7) is the control layer: the command line, the web application and the worker that runs job files. Each box lists the main tasks of that step; the modules that carry them out are in Table 5.

Figure 2 is read from left to right for one run. The arrows from box 2 split because each run uses exactly one backend, chosen in the configuration; both backends hand the same kind of result to the diagnostics, which is why everything to the right of them works with either. The bottom row shows that ensembles, fits and maps are not separate programs: each of them produces many configurations (or many variants of one) and sends them through boxes 2–5.

The steps in detail:

1. **Input.** The configuration names the hierarchy (`topology`), the masses, the Keplerian elements of every orbit at the start ($t = 0$), optional properties of the stars, the physics to include and the length of the run. Optional sections describe an ensemble (`ensemble`), a fit of measured data (`fit`) and occulting dust discs (`discs`). Measured data (times of minima, radial velocities, light curves, inclinations) are CSV tables that the configuration refers to. The complete reference is the chapter The configuration file.
2. **Check and build.** The file is validated against a schema before anything is computed: every key must be known, every value must have the right type and range, the orbits must match the hierarchy, and combinations that do not make sense are refused with a message naming the key. The hierarchy becomes a binary tree; the elements of every orbit are converted to positions and velocities in Jacobi coordinates (each orbit is the relative motion of the centres of mass of its two components). Every star receives a radius, an effective temperature, an apsidal-motion constant k_2, a moment of inertia and, if requested, a tidal time lag, from a stellar model at the age of the system or from values given in the file. With mass transfer switched on, the history of every interacting binary is computed from the stellar tracks at this point (see *Physics and Methods*, chapter Mass transfer, common envelope and mergers).
3. **Backend.** The N-body backend integrates the positions and velocities of all stars; the secular backend integrates orbit-averaged elements. Both are described below. With evolving stars the run is cut into segments within which the stars are held fixed, and

the stars are updated between segments (operator splitting).

4. **Diagnostics.** The time series of the elements are turned into numbers that answer questions: which stars are bound to which at every output (and whether the hierarchy changed), the ranges of eccentricity and mutual inclination compared with the analytic theory of von Zeipel–Lidov–Kozai (ZLK) cycles, the precession and apsidal periods, the fraction of time each binary eclipses, the largest Roche-lobe filling, and the stability ratios of every nested triple.
5. **Products and observables.** A run writes its summary, an HDF5 file with all time series, a CSV table, PNG figures, an interactive 3D viewer and an HTML report. Observables are computed from the same dynamics on request: eclipse times and the O−C diagram, synthetic light curves, times of minima measured on them, radial velocities and pulsation periods.
6. **Layers.** Ensembles draw unknown parameters and run every member; fits adjust free parameters until the model reproduces measured data; chaos maps and scattering experiments run grids or random sets of configurations.
7. **Control.** The command `swingersim` with one sub-command per task, the web application (`swingersim gui`) and the worker, which runs job files on a computer that stays on for long computations.

2.2 Two backends

The motion of a hierarchical system has two very different time scales: the revolutions of the orbits (days to years) and their slow secular changes (precession, ZLK cycles; decades to millions of years). SWINGERsim offers two backends for this reason.

	N-body (`run.backend: nbody`)	secular (`run.backend: secular`)
what is integrated	positions and velocities of all stars	orbit-averaged elements (angular-momentum and eccentricity vectors of every orbit)
integrator	IAS15 of REBOUND, adaptive, error at machine precision	DOP853 of SciPy; the 2+2 equations compiled with Numba when it is installed
relativity	1PN for all pairs of bodies (REBOUNDx `gr_full`)	periastron advance of every orbit
stars	tides and rotation as forces on extended bodies (REBOUNDx `tides_spin`), tidal dissipation, mass change	distortion as apsidal and nodal precession, tidal dissipation, evolving stars, mass loss and mass transfer
events	collisions, changes of the hierarchy, escapes, Roche-lobe overflow, energy-error limit, wall-clock limit	Roche-lobe overflow and mass-transfer events; the hierarchy cannot change
outputs	osculating elements	mean elements
observables (eclipse times, light curves, radial velocities) and fits	yes	no
cost	seconds to hours, set by the number of revolutions of the shortest orbit	about a second per run, independent of the inner periods

	N-body (`run.backend: nbody`)	secular (`run.backend: secular`)
use it for	decades to thousands of years; mild hierarchies; close encounters; instability; everything an observer measures	millions of years; ensembles of thousands of members; strongly hierarchical systems

Table 4. The two backends. Each row compares one property; the last row says when to use which. "Osculating" elements are the instantaneous Kepler ellipse computed from positions and velocities; "mean" elements are averaged over the orbital motion and lack the short-period wiggles.

Both backends return the same structure – time series of the elements of every orbit, mutual inclinations, energy and angular-momentum errors, events – so every diagnostic, output file and result page works with either, and a result can always be checked by running it with the other backend. Figure 3 shows the difference on one system.

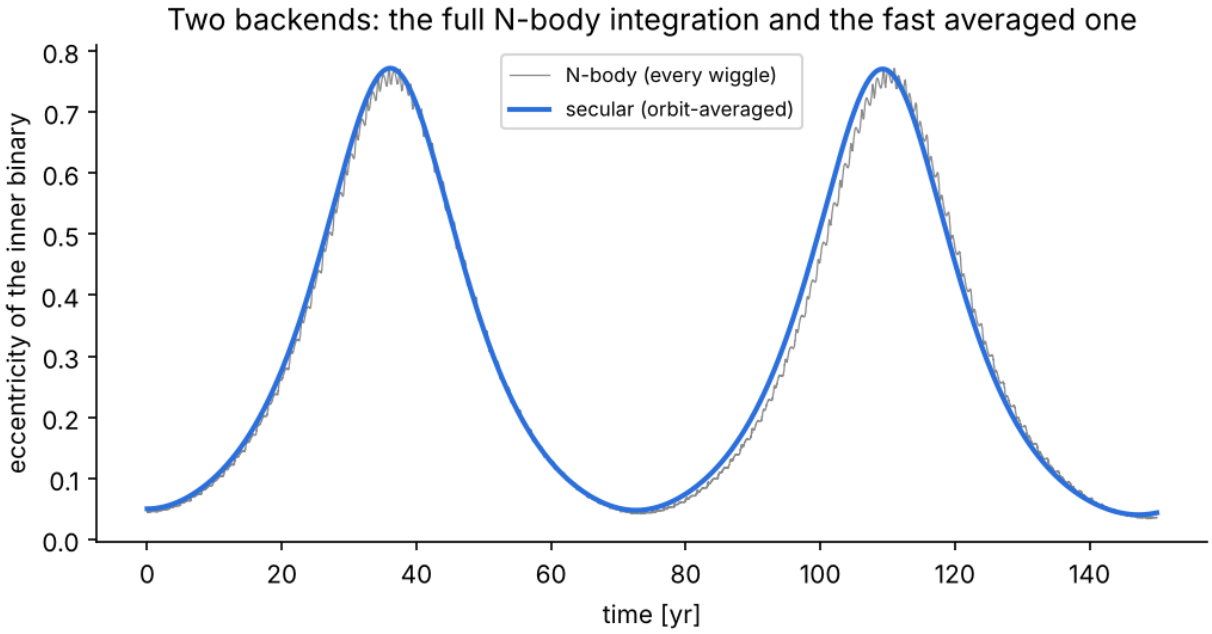


Figure 3. The same hierarchical system computed with both backends. Horizontal axis: time in years; vertical axis: eccentricity of the inner binary. The thin grey line is the N-body backend, the thick blue line the secular backend. Both show two ZLK cycles in which the eccentricity grows from 0.05 to 0.77 and returns. The N-body curve carries small fast wiggles – the osculating elements respond to every revolution of the outer orbit – which the averaged equations leave out; the slow cycle, which matters for the long-term evolution, is the same in both.

Figure 3 illustrates the rule of thumb: when the period ratios are large and the system stays hierarchical, the averaged equations follow the slow evolution closely at a tiny fraction of the cost. They lose accuracy when the hierarchy is mild (the outer period only a few tens of inner periods), near mean-motion resonances and in chaotic systems; the summary of a run then warns that the "secular estimates" are approximate (the parameter $\varepsilon_{\mathrm{SA}}$, see Outputs). The secular equations of a 2+2 quadruple follow Vokrouhlický (2016); every other hierarchy uses the general formulation for nested binaries of Hamers et al. (2015), of which the 2+2 equations are a special case. The equations, their validity and the tests against the N-body backend are in *Physics and Methods*, chapter Dynamics: N-body, relativity and the secular backend.

Which backend to choose. Start with N-body for anything an observer measures or for runs of up to a few thousand years. Use the secular backend for evolution over millions of

years, for long ensembles, and when the inner binaries are so short that N-body would need billions of steps. When in doubt, run both over a short time and compare the summaries.

2.3 How the parts are connected

The parts communicate only through files and one internal result structure, which keeps them independent:

- **Configuration → everything.** Every command reads the same YAML file and validates it with the same schema. A configuration written in the web application, by hand or by a fit (`stage2_fitted.yaml`) is the same kind of file.
- **Backend → diagnostics → files.** Both backends return the same result structure; the diagnostics read only this structure; the HDF5 file stores the result together with the complete configuration, so a run file always says how it was made.
- **Observables from the dynamics.** Eclipse times, light curves, radial velocities and the fitted model observables all integrate the same N-body equations as a run; a light curve is therefore consistent with the O−C diagram of the same configuration.
- **Layers call the chain.** An ensemble writes one summary row per member and direction; a fit calls the N-body model once per evaluation of its free parameters; a chaos map runs one integration per grid point.
- **Jobs.** The web application does not compute anything itself: it writes job files (small YAML files that name a command, a configuration and options) into a folder, and the worker executes them with the same command line that a user would type. Results are read back from the output files.

2.4 Module map

The program is a Python package (`swingersim`, source in `src/swingersim/`). Table 5 lists every module, grouped by its role, with one line on what it does and whether it is SWINGERsim's own code or a wrapper around an external library. "Own" means that the method is implemented in SWINGERsim (it may use NumPy and SciPy for arrays, root finding and integrators); "wrapper" means that the essential computation is done by the named external package and the module prepares its input and reads its output; "own + external" means that SWINGERsim's own algorithm uses an external package for one well-defined part.

Module	What it does	Own code or wrapper
Configuration and set-up		
`config.py`	schema of the configuration, units, presets of hierarchies, all consistency checks	own (schema written with pydantic)
`units.py`	internal units ($M_\odot$, au, yr) and conversions	own
`kepler.py`	Kepler's equation, elements ↔ position and velocity	own
`hierarchy.py`	hierarchy tree, Jacobi initialisation, elements of every node	own
`tables.py`, `rootpath.py`	one rule for reading the CSV tables and for paths of jobs	own
Stars		
`stellar.py`	radius, $T_{\rm eff}$, k_2, moment of inertia, tidal lag of every star; ZAMS relations, polytropes, Claret tables, user tables, Roche-lobe radius	own; data of Tout et al. (1996), Claret (2004, 2023)

Module	What it does	Own code or wrapper
`evolution.py`	SSE tracks (radius, luminosity, envelope) at any mass, age and metallicity by interpolation	own interpolation; grid computed from the published SSE routines of Hurley et al. (2000)
`mist.py`	MIST tracks from the pre-main sequence to the end of the AGB, interpolated at equivalent evolutionary points	own interpolation; data of Choi et al. (2016)
`gridclip.py`	one policy for values outside the grids of tabulated models	own
`massloss.py`	stellar winds in the dynamics: masses from tracks or rates, the mass-loss index Ψ	own
`masstransfer.py`	Roche-lobe overflow, stability, accretion, orbit response, rejuvenation, contact binaries, transfer from a tertiary	own (prescriptions of Hurley et al. 2002 and others)
`commonenv.py`	outcome of a common envelope (energy formalism) and the response of the outer orbits	own
`merger.py`	continuation of a run after two stars merge	own
`tides.py`, `apsidal.py`	analytic apsidal motion (tidal, rotational, relativistic) and tidal time scales	own
`inertial_waves.py`	dissipation of inertial waves in convective envelopes	own
`magnetic.py`	magnetic braking and the Applegate energy test	own
`pulsation.py`	pulsation periods from the stellar models and the flux modulation	own
`mesa.py`	own stellar-model grids: writes MESA input, runs MESA, reduces profiles to k_2, envelopes and dissipation	wrapper of MESA (external, installed separately) + own reduction
`gw.py`, `environment.py`, `tidalcapture.py`, `pdot.py`	gravitational-wave decay, Galactic tide and passing stars, tidal capture in encounters, an imposed period change	own
Dynamics		
`backends/nbody.py`	the N-body backend: sets up the simulation, events, extremes between outputs, spins	own + external (REBOUND, REBOUNDx)
`backends/secular.py`, `backends/secular_nested.py`, `backends/secular_common.py`	the secular backends for 2+2 and for any nested hierarchy	own (SciPy integrator)
`backends/evolving.py`, `backends/chunked.py`	evolving stars and mass loss by operator splitting; integration in chunks that fit into memory	own

Module	What it does	Own code or wrapper
`secular/twotwo.py`, `secular/kernel.py`, `secular/nested.py`, `secular/quadrupole.py`	the secular equations (2+2, any nested tree), their compiled kernel, quadrupole predictions	own (kernel compiled by Numba if installed)
Diagnostics and stability		
`diagnostics/hierarchy.py`, `zlk.py`, `binary.py`, `extremes.py`, `orbits.py`	hierarchy detection, ZLK and precession, apsidal motion and eclipse fractions, extremes, mutual inclinations	own
`stability.py`	Mardling–Aarseth and Vynatheya et al. (2022) criteria, the N-body ghost test, the optional neural classifiers of Vynatheya et al. (2023)	own; classifiers are the authors' published networks (optional)
`chaosmap.py`	MEGNO maps over one or two parameters	own driver; MEGNO computed by REBOUND
`scatter.py`, `escape_theory.py`	few-body scattering experiments; statistical theory of three-body break-up (Stone & Leigh 2019)	own (scattering integrated with REBOUND)
Observables		
`timing.py`	eclipse times from the dynamics and their O−C	own
`photometry.py`	passbands, limb darkening, occultation of limb-darkened discs, synthetic light curves	own; limb-darkening tables of Claret & Bloemen (2011) and Reeve & Howarth (2016)
`roche.py`, `mesh.py`, `shapes.py`	Roche geometry, light of distorted stars on surface meshes, choice of the star shapes	own
`discs.py`	dust discs and rings as occulters, their precession, thermal and scattered light	own
`phenom.py`	phenomenological eclipse profiles (Mikulášek 2015) and times of minima measured on synthetic light curves	own
`radvel.py`, `rvlight.py`	radial velocities of the centres of mass and of the centres of light	own
`phoebe_lc.py`	the light of the inner binaries computed by PHOEBE 2 in a hybrid with the N-body positions	wrapper of PHOEBE 2 (optional)
`systemview.py`	the model system as data for pictures	own
Fitting of measured data		
`ocfit.py`	stage 1: analytic O−C model of every pair, least squares, MCMC, predictions	own + external (SciPy least squares, emcee, astropy time conversion)

Module	What it does	Own code or wrapper
`dynfit.py`	stage 2: joint N-body model of all pairs, error scaling, outliers, priors, anchoring, MCMC, predictions	own + external (SciPy least squares, emcee)
`obsmodel.py`, `lcfit.py`, `incfit.py`	model observables of one integration; light curves and inclinations in the joint fit	own
Ensembles and machine learning		
`ensemble.py`, `parallel.py`	drawing members, running them in parallel, summary rows, weights by measurements, reports	own
`emulator.py`	neural emulators and classifiers trained on an ensemble	own features and evaluation; networks of scikit-learn
`sbi_fit.py`	neural posterior estimation on an ensemble	wrapper of `sbi` and PyTorch
Products, control and display		
`io.py`, `runproducts.py`	HDF5 file of a run; summary, CSV, PNG figures next to it	own (h5py, matplotlib)
`viewer.py`, `report.py`	interactive 3D viewer of a run; self-contained HTML report of every command	own (plotly for the plots)
`cli.py`, `__main__.py`	the command line	own
`worker.py`	the job worker	own
`gui/server.py`, `gui/digest.py`, `gui/assist.py`, `gui/modelviews.py`, `gui/app.html`, `gui/help.yaml`	the web application: server, result pages, help texts and quick checks, model curves, the page itself	own (Python standard-library web server; plotly.js in the browser)
`cnoise.py`	filter of messages that compiled libraries print	own
Validation		
`validation.py` and its companions, `_benchmark.py`	the benchmarks run by `swingersim validate`	own

Table 5. Module map of SWINGERsim. Bold rows are the groups. The middle column states the task of each module, the right column whether the computation is SWINGERsim's own code, a wrapper, or own code with one external part (named in brackets). Paths are relative to `src/swingersim/`.

The package contains about 45 000 lines of Python plus the web page. Its own code covers the physics that couples dynamics, stars and observables; the external libraries are used for well-defined numerical tasks.

2.5 What SWINGERsim implements and what it uses

Table 6 separates the two explicitly. It answers the question a referee asks first: which results depend on code written for SWINGERsim, and which on established external software and published data.

Task	Implemented in SWINGERsim	External software or data used
hierarchies	builder of any tree of pairs, Jacobi initialisation, detection of the hierarchy at every output and of its changes	–
N-body integration	set-up, events (collisions, Roche overflow, escapes, wall clock), sampling of extremes between outputs, coupling to evolving stars and mass change, spins	REBOUND with IAS15 (Rein & Liu 2012; Rein & Spiegel 2015); REBOUNDx forces `gr_full`, `tides_spin` and operator `modify_mass` (Tamayo et al. 2020; Lu et al. 2023)
secular integration	the equations for 2+2 (Vokrouhlický 2016) and for any nested tree (Hamers et al. 2015), tides, gas damping, Galactic tide and passing stars, chunked integration	SciPy integrator DOP853; Numba compiler (optional)
stars	interpolation of tracks, structure constants, envelopes, tidal lags, rejuvenation, pulsation periods	tracks: MIST (Choi et al. 2016), SSE grid from the routines of Hurley et al. (2000); structure: Claret (2004, 2023); ZAMS: Tout et al. (1996); optional own grids computed with MESA (Paxton et al. 2011; Jermyn et al. 2023)
interacting binaries	coupling of mass transfer, contact phases, common envelopes, tertiary transfer and mergers to the dynamics	prescriptions from the literature (Hurley et al. 2002 and others; *Physics and Methods*)
eclipses and light curves	eclipse geometry, occultation of limb-darkened discs, ellipsoidal, beaming and reflection terms, Roche and rotating surfaces on meshes, dust discs, light-travel time, pulsations	limb-darkening tables of Claret & Bloemen (2011) and the TLUSTY coefficients of Reeve & Howarth (2016); passband curves; PHOEBE 2 (Prša et al. 2016) as an optional engine for the inner binaries
eclipse timing and O−C	conjunctions found in the integration, light-time, O−C, phenomenological minima (Mikulášek 2015)	–
radial velocities	velocities at the emission time, centre of light with the Rossiter–McLaughlin effect	–
fit chain	stage 1 and stage 2 models, error scaling and jitter, outlier rejection, priors, red-noise factors, anchoring, predictions with uncertainties	SciPy least squares; emcee (Foreman-Mackey et al. 2013) for MCMC; astropy for time conversions
ensembles	drawing, parallel runs, resumable rows, weights, effective number of members, Wilson intervals, re-weighting	pandas and pyarrow only to write Parquet (optional)
machine learning	features, training sets, evaluation, calibration plots	scikit-learn networks; `sbi` and PyTorch (optional)
stability	Mardling–Aarseth and Vynatheya et al. (2022) criteria for every nested triple, N-body ghost test, chaos maps, scattering, escape theory	MEGNO of REBOUND; the neural classifiers published by Vynatheya et al. (2023) (optional, downloaded separately)
configuration	schema, defaults, checks, error messages	pydantic (validation framework), PyYAML

Task	Implemented in SWINGERsim	External software or data used
outputs	HDF5 layout, products, 3D viewer, HTML reports	h5py, matplotlib, plotly
web application, worker	server, result pages, help, job queue, pause and stop	Python standard library; plotly.js in the browser

Table 6. What SWINGERsim implements itself (middle column) and which external libraries and published data it relies on (right column), task by task. A dash means that no external component is involved beyond NumPy and SciPy.

2.6 Design principles

- **Forward model first.** The core only predicts. Fits, ensemble weights and machine learning are layers that call the core and never change it.
- **Two backends, one result.** The N-body backend is the reference; the secular backend is the fast approximation; both produce the same structure, so one can check the other.
- **Any hierarchy as a tree.** Nothing in the chain assumes four stars: presets are trees, diagnostics walk the tree, stability is tested on every nested triple.
- **Validated physics.** Every physical ingredient is compared with a published result (*Testing, Part I*).
- **Reproducibility.** Every run file stores its complete configuration and the versions of the software that computed it; an ensemble is fixed by its seed; resuming an ensemble checks a fingerprint of the configuration, so rows of different set-ups are never mixed.
- **Safe remote computing.** The worker runs only SWINGERsim commands with the arguments their parser allows, and only with paths inside the SWINGERsim folder; a job file cannot run shell commands.

Chapter 3

Installation

SWINGERsim is a Python package that runs on Linux and macOS (on Windows through the Windows Subsystem for Linux, WSL). This chapter lists what it needs, how to install it and check the installation, the optional parts, the worker for long computations and how to build this documentation.

3.1 Requirements

Item	Requirement	Why
Python	3.11 or newer	the language of the package
C compiler	`gcc` or `clang`	REBOUND and REBOUNDx, the N-body libraries, are compiled during installation where pip finds no ready-made package
operating system	Linux or macOS; Windows through WSL	the worker uses process groups and signals of POSIX systems
disk	about 100 MB for the stellar tracks and tables of the package, plus the optional libraries (the `sbi` group alone about 1 GB)	the data are part of the package

Table 7. Requirements. The left column names the item, the middle the requirement, the right column why it is needed. Memory depends on the task: a parallel ensemble needs memory for every process, which is why the worker watches the free memory (below).

3.2 Steps

```
git clone https://github.com/uhlarmilan-create/SWINGERsim.git
cd SWINGERsim
./install.sh
source .venv/bin/activate
```

The installation script `install.sh` finds Python 3.11 or newer, checks that the stellar tracks and tables of the package are present, creates an isolated Python environment in the folder `.venv` (so that the packages of SWINGERsim do not mix with the rest of the system), installs SWINGERsim with the groups `all` and `dev` of Table 8, makes a test run and runs the fast benchmarks. It never uses `sudo`: when a system package is missing (Python, its `venv` module, a C compiler), it stops and prints the command that installs it. Its options:

Option	Effect
`--minimal`	only the core and the test suite (groups `dev`)
`--with-sbi`	also the group `sbi` (PyTorch, about 1 GB)
`--with-phoebe`	also the group `phoebe`
`--venv DIR`	the environment in `DIR` instead of `.venv`
`--python PY`	this Python interpreter
`--no-check`	no test run and no benchmarks at the end

Table 7a. Options of `install.sh` (`./install.sh --help` lists them). They can be combined, e.g. `./install.sh --with-sbi --no-check`.

Activate the environment with `source .venv/bin/activate` in every new terminal; the command `swingersim` is then available. **Windows:** install WSL once (PowerShell as administrator: `wsl --install`, restart, open *Ubuntu* once to create a user) and run the steps above in the Ubuntu terminal; alternatively `powershell -ExecutionPolicy Bypass -File install.ps1` checks WSL and runs `install.sh` inside it. The web application started in WSL opens in the Windows browser at the printed address.

The same by hand:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e ".[all,dev]"
swingersim --help
```

`pip install -e` installs the package in editable mode: the source in `src/swingersim/` is used directly, so the configurations, data and outputs in the folder of the repository are found. The names in square brackets are optional groups of dependencies (Table 8). The last command lists the sub-commands; if it prints them, the installation works.

Group	Packages	What it enables
(always)	numpy, scipy, pydantic, pyyaml, h5py, rebound, reboundx	runs, all physics, HDF5 output
`dev`	pytest, ruff	the test suite
`all`	the groups `viz`, `ensemble`, `fast`, `emulator`, `fit`	everything except `sbi` and `phoebe`; installed by `install.sh`
`viz`	plotly, kaleido, matplotlib	the interactive 3D viewer, the plots inside the HTML reports, PNG figures (matplotlib; kaleido with a Chrome browser for PNG files of plotly figures); the web application uses plotly if present, otherwise it loads plotly.js from the internet
`fit`	emcee, corner, astropy	MCMC errors of fits, corner plots, conversion of HJD/JD (UTC) to BJD_TDB
`fast`	numba	the compiled secular equations, about a hundred times faster secular runs and ensembles
`ensemble`	pandas, pyarrow	ensemble summaries also as Parquet files
`emulator`	scikit-learn	neural emulators (`emulator-train`, `emulator-report`) and the stability classifiers below
`sbi`	sbi, torch	simulation-based inference (`swingersim sbi`); about 1 GB

Group	Packages	What it enables
`phoebe`	phoebe	PHOEBE 2 as the light-curve engine of the inner binaries (`--engine phoebe`)

Table 8. Optional groups of dependencies. Install a group with `pip install -e ".[<group>]"`; several groups are separated by commas. The right column says which commands or options need the group; without it, they stop with a message naming the missing package.

3.3 Check the installation

```
pytest -m "not slow and not validation"     # unit tests, a few minutes
swingersim validate                          # fast benchmarks against published results
swingersim run configs/examples/example_2p2.yaml -o output/check/example.h5
```

The unit tests check the conversions between elements and positions, the hierarchical initialisation, energy conservation, the relativistic periastron advance, the configuration schema, the web application and the documentation. `swingersim validate` reproduces a set of published results and prints PASS, EXPLAINED (a difference whose reason is documented) or FAIL for each; the benchmarks and their verdicts are discussed in *Testing, Part I*. The last command is the first run of the Quick start.

3.4 Optional: stability classifiers of Vynatheya et al. (2023)

The machine-learning stability classifiers for 2+2 and 3+1 quadruples (Vynatheya et al. 2023) are published by their authors and are not part of SWINGERsim. To use them, download their repository into `data/`:

```
git clone https://github.com/pavanvyn/quadruple-stability data/quadruple-stability
pip install -e ".[emulator]"
```

`swingersim stability config.yaml` then reports their verdict for every sub-quadruple. The networks were saved with scikit-learn 1.2.2; newer versions read them, and the version warning is suppressed. Another location can be given with the environment variable `SWINGERSIM_VYNATHEYA_DIR`.

3.5 Optional: MSE for independent comparisons

The Multiple Stellar Evolution code MSE (Hamers et al. 2021) is an independent code for hierarchical multiples that was used to cross-check SWINGERsim (*Testing, Part I*). It is not needed to run SWINGERsim. To make own comparisons, install it from its repository; it needs C, C++ and Fortran compilers:

```
sudo apt install gfortran
git clone https://github.com/hamers/mse ~/mse
cd ~/mse && make
```

3.6 Optional: MESA for own stellar-model grids

SWINGERsim can compute its own grids of stellar models with MESA (Modules for Experiments in Stellar Astrophysics; Paxton et al. 2011; Jermyn et al. 2023): pre-main-sequence and envelope structure, k_2 and the dissipation of inertial waves from full stellar profiles. This is needed only when the built-in MIST tracks and the Claret tables do not cover a star (an unusual metallicity, a particular physics set-up). MESA is installed once by hand; SWINGERsim then runs it (also through the worker).

Requirements: Linux x86_64, at least 8 GB memory, about 30 GB of free disk (MESA about 20 GB, its software development kit about 2 GB, a grid a few GB) and one to two hours.

1. System packages (the MESA software development kit, SDK, brings its own compilers):

   ```
   sudo apt install binutils make perl libx11-6 libx11-dev zlib1g zlib1g-dev tcsh unzip
   ↪ python3
   ```

2. Download the MESA SDK for Linux x86_64 from http://user.astro.wisc.edu/~townsend/static.php?ref=mesasdk and unpack it into `~/mesa`.
3. Download a MESA release (the Zenodo record https://doi.org/10.5281/zenodo.2602941 or https://github.com/MESAHub/mesa/releases) and unpack it into `~/mesa`. The SDK must be the version the MESA release asks for.
4. Add to `~/.bashrc` and open a new terminal:

   ```
   export MESASDK_ROOT=$HOME/mesa/mesasdk
   source $MESASDK_ROOT/bin/mesasdk_init.sh
   export MESA_DIR=$HOME/mesa/mesa-<release>
   export OMP_NUM_THREADS=2
   ```

5. Compile MESA (30–60 minutes): `cd $MESA_DIR && ./install`.
6. Test with SWINGERsim – one 1 $M_\odot$ track from the pre-main sequence to the end of the main sequence:

   ```
   swingersim mesa-grid configs/mesa/lowmass_pms_ms.yaml --test
   ```

7. Compute the whole grid and reduce it (hours; both steps can run as worker jobs):

   ```
   swingersim mesa-grid configs/mesa/lowmass_pms_ms.yaml -j 6
   swingersim mesa-reduce configs/mesa/lowmass_pms_ms.yaml
   ```

The reduced grid is `mesa/lowmass_pms_ms/lowmass_pms_ms.npz`; a configuration uses it with `stellar_model.structure: mesa:lowmass_pms_ms`.

3.7 The worker for long computations

Large ensembles (tens of thousands of members, N-body ensembles), fits of many free parameters and MESA grids take hours to days. They are best computed on a computer with many cores that stays switched on. `swingersim worker` turns such a computer into a job runner: it watches a folder, runs every job file that appears there and writes the results into `output/`.

```
cd swingersim
source .venv/bin/activate
swingersim worker                 # watches jobs/, Ctrl+C stops it
swingersim worker --cores 8       # use at most 8 cores for all jobs together
```

The worker creates the folders of Table 9 and writes a heartbeat file `jobs/worker.json` (name of the computer, number of cores, the versions it was started with, the current job, the time of its last check) every few seconds. Jobs that need one core (a run, a light curve, eclipse times) run side by side up to the number of cores; a job that needs all cores (an ensemble, a fit or a chaos map with `processes: 0`, `validate --all`) waits for the others and runs alone.

Folder or file	Content
`jobs/drafts/`	configurations written in the web application (not run by the worker)
`jobs/submitted/`	frozen copies of the configurations of submitted jobs
`jobs/inbox/`	`<id>.yaml`: queued jobs, taken in the order of their names; `<id>.yaml.hold`: jobs waiting until they are queued

Folder or file	Content
`jobs/running/`	the jobs being computed, each with `<id>.yaml`, its screen output `<id>.log` and its progress `<id>.status.json`
`jobs/done/`, `jobs/failed/`	finished and failed jobs with the same three files
`jobs/cancel/`	an empty file `<id>` stops that job; its output inside `output/` is deleted and its files are kept here
`jobs/pause/`	an empty file `<id>` pauses the running job; removing the file lets it continue exactly where it stopped
`jobs/stop`	an empty file stops the worker; running jobs go back to the inbox
`jobs/worker.json`, `jobs/worker.lock`	heartbeat; lock that prevents a second worker on the same folder

Table 9. The job folder of the worker. The left column is the folder or file inside the SWINGERsim folder, the right column what it holds. The web application uses the same layout.

To keep the worker running after logging out, start it in `tmux` or as a user service, e.g. `systemd-run --user --unit=swingersim-worker swingersim worker --root ~/swingersim`. When the free memory of the computer falls below 8 % of the total (at least 1.5 GB), the worker stops the job that uses the most memory and marks it failed with an explanation, so that the computer stays usable. After an update of SWINGERsim, restart the worker; `worker.json` shows the version it runs. Job files and the rules for them are described in Workflows.

3.8 Building this documentation

The documentation is written in Markdown in `docs/en/` (English) and `docs/cs/` (Czech), one folder per document. The PDF files are built with pandoc and XeLaTeX:

```
sudo apt install pandoc texlive-xetex texlive-fonts-recommended texlive-lang-czechslovak
python tools/build_docs_pdf.py guide --lang en      # this guide:
↪  docs/pdf/SWINGERsim-guide-en.pdf
python tools/build_docs_pdf.py                      # every document in both languages
```

Chapter 4

Quick start

This chapter takes you from an installed SWINGERsim to the first results in a few minutes, once with the web application and once from the command line. Both compute the same thing: the example 2+2 quadruple `configs/examples/example_2p2.yaml` over 200 years.

4.1 The example system

```yaml
topology: "2+2"
masses: {m1: 1.2, m2: 1.0, m3: 0.9, m4: 0.8}        # solar masses
orbits:
  A:   {P: 6.74, e: 0.05, i: 85.0, Omega: 0.0,  omega: 90.0,  M0: 0.0}
  B:   {P: 2.68, e: 0.02, i: 60.0, Omega: 40.0, omega: 10.0,  M0: 120.0}
  out: {P: 500,  e: 0.3,  i: 70.0, Omega: 10.0, omega: 120.0, M0: 45.0}
physics: {gr: true}                                  # general relativity on
run: {t_max_yr: 200, n_output: 2000}                 # 200 years, 2000 snapshots
```

Four Sun-like stars form two binaries, A (periods in days: 6.74) and B (2.68), which orbit each other on the outer orbit `out` (500 d, eccentricity 0.3). Each orbit has a period `P` in days, an eccentricity `e` and three angles in degrees: the inclination `i` to the plane of the sky ($i = 90°$ is edge-on), the longitude of the ascending node `Omega` and the argument of periastron `omega`; `M0` is the mean anomaly at the start, i.e. where on the orbit the stars are at $t = 0$. The stars have no radii given, so they get zero-age main-sequence radii.

4.2 The web application way

```
swingersim gui
```

The command starts a small web server and opens `http://127.0.0.1:8765` in the browser.

1. On the **Configuration** page choose `configs/examples/example_2p2.yaml` in the list *Open…*. The form and the YAML text show the system; the badge at the top right says *valid*.
2. Type a name for the draft (e.g. `my_2p2`) and press **Save**. The configuration is now a draft in `jobs/drafts/`.
3. In the panel **What to compute** tick **Run** (and, if you like, **Eclipse times (O − C)**), then press **Add and run**. The jobs are queued and the worker starts in the background.
4. The **Jobs** page shows the job running with a progress bar. After about a minute it moves to *Finished*.
5. On the **Outputs** page open the card of the run. The result page shows the model system, when each binary eclipses, the orbital elements over time and the stability ratios.

The pages are described panel by panel in The web application; Figures 4, 7, 8 and 9 there show what you will see.

4.3 The command-line way

```
swingersim run configs/examples/example_2p2.yaml -o output/example.h5
```

After about a minute and a half the program prints a summary (the tutorial run of this configuration took 80 s on one core). The most important lines are:

Line of the summary	What it tells you
`out a = 1.94..1.9418 AU e = 0.2995..0.3007`	the outer orbit keeps its size (about twice the Earth–Sun distance) and shape
`i_mut A-B 25.59..46.67 deg`	the angle between the planes of A and B swings between 26° and 47°: the binaries tilt against each other
max |dE/E| = 3.23e-13	the energy is conserved to 13 digits: the integration is accurate
`hierarchy: configured [[1,2],[3,4]], no change detected`	no star was ejected or changed partner
`precession period : 283 yr estimated`	the plane of binary A turns around once in about 283 years
`stability ratio Mardling-Aarseth = 3.67 (stable)`	the orbits are far enough apart for a long life (above 1 is stable)
`eclipses (observer along z): A 2 %, B 43 %`	B eclipses during 43 % of the run, A only briefly at the start
`m1 R = 1.14 Rsun, …, Teff = 6144 K, main sequence`	the stars: radius, temperature, evolutionary phase

Table 10. The key lines of the summary of the example run. The left column quotes the line as printed, the right column says what it means. Every line of the summary is explained in Outputs.

The run has also written its standard products next to the HDF5 file:

```
ls output/example*
# example.h5  example.html  example_summary.txt  example_timeseries.csv
# example_orbits.png  example_inclinations.png  example_conservation.png
 ↪  example_viewer.html
```

Open `output/example.html` (the report) and `output/example_viewer.html` (an animated 3D view of the run) in a web browser. Both are single files that work without SWINGERsim and without an internet connection.

4.4 Change one thing

The typical way of working is to change one parameter or switch one effect on or off and compare. A prepared variant tilts binary B by 65° against the outer orbit:

```
swingersim run configs/examples/example_2p2_zlk.yaml -o output/zlk.h5
```

The eccentricity of B now swings between 0.01 and 0.81 every 14 years: these are ZLK cycles (von Zeipel–Lidov–Kozai), the exchange of eccentricity and inclination of an orbit perturbed by a distant inclined body (Worked examples).

4.5 Eclipse times of the same system

```
swingersim timings configs/examples/example_2p2.yaml --years 20 -o
 ↪  output/example_timings.csv \
  --plot output/example_oc.png
```

The command finds every eclipse of A and B in the next 20 years, including the light-travel time across the outer orbit, and writes their O−C against a linear ephemeris (a table and a figure). Continue with Workflows for complete tasks, or with Worked examples for more systems.

Chapter 5

The web application

`swingersim gui` starts a small web server on your computer and opens a page in the browser from which everything SWINGERsim does can be controlled: writing configurations, choosing what to compute, running and controlling jobs, and reading the results on pages made for each kind of result. The page is in English. It does not compute anything itself: it writes job files that the worker executes (chapter Installation) and reads the files the jobs write. A result computed from the command line therefore looks the same as one started from the page, and the browser can be closed while jobs run.

The screenshots of this chapter show a real session. The system is V907 Sco, a 3.78-d eclipsing binary with a third star on a 142-d orbit whose eclipses come and go (Zasche et al. 2023); its runs, light curve, eclipse times and fit are those of *Testing, Part III*. The ensemble is that of QX Cas from the same document.

5.1 Starting the application

```
swingersim gui                              # http://127.0.0.1:8765, opens the browser
swingersim gui --port 8800 --no-browser     # another port, no browser
swingersim gui --root ~/swingersim          # another SWINGERsim folder
swingersim gui --host 0.0.0.0               # reachable from other computers
```

Option	Default	Meaning
`--root`	the current folder, or the nearest folder above it that contains `configs/`	the SWINGERsim folder whose `configs/`, `jobs/` and `output/` are used
`--host`	`127.0.0.1`	address the server listens on; `0.0.0.0` makes it reachable from the network
`--port`	`8765`	port of the server
`--no-browser`	off	do not open a browser

Table 11. Options of `swingersim gui`. With `--host 0.0.0.0` anyone on the network can read the outputs, write configurations and start jobs; use it only on a trusted network. The server serves only files below `--root`.

5.2 The frame of every page

The bar at the top carries the name of the program and four tabs – **Configuration**, **Jobs**, **Outputs** and **Tools** – which are the four pages of the application. The work goes in one

direction: on *Configuration* you describe a system and tick what to compute, *Jobs* runs the computations, *Outputs* shows what they gave; *Tools* holds the calculators that need no configuration. On the right of the bar, **?** opens the help drawer for the current page and ◐ switches between a light and a dark colour scheme. Below the bar each page starts with a blue box that says in two sentences what the page is for; × closes it. Every section, field, computation, chart and warning has its own **?**, which opens the help drawer with an explanation (Figure 5).

5.3 Configuration

The Configuration page writes and edits configuration files and says what to compute with them (Figure 4).

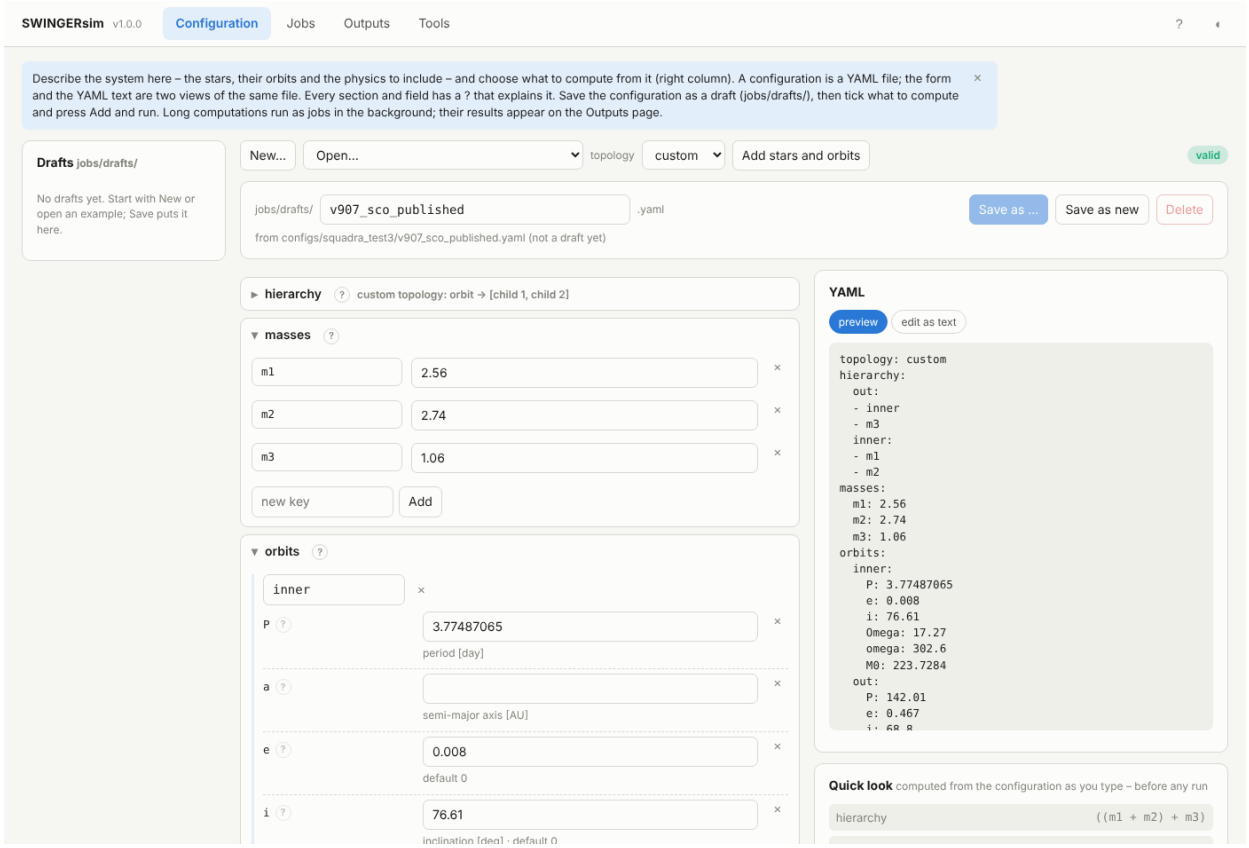


Figure 4. The Configuration page with the configuration of V907 Sco opened. Left: the list of drafts (empty here). Top: the toolbar – *New…*, *Open…*, the topology selector (`custom`), *Add stars and orbits* and the status badge (*valid*, green). Below it the name bar: the draft will be saved as `jobs/drafts/v907_sco_published.yaml`; *Save as …*, *Save as new* and *Delete*; the line under it says where the configuration came from. Middle: the form, here with the sections *hierarchy* (collapsed), *masses* (three stars) and *orbits* (the orbit `inner` with its period, semi-major axis, eccentricity and inclination fields; each field has a **?**, a unit and its default below it, and × removes it). Right: the YAML text of the same configuration (*preview*), and the top of the *Quick look* panel.

Drafts. The list on the left shows the configurations in `jobs/drafts/`. A click opens a draft. A configuration from elsewhere (an example in `configs/`, a fitted configuration, a member of an ensemble) becomes a draft when it is saved.

Toolbar. *New…* starts a configuration, empty or from one of five templates (an eclipsing binary with a third body, a 2+2 quadruple, a 3+1 quadruple, an ensemble for an unseen companion and a fit of times of minima). *Open…* loads any configuration under `configs/`, a draft or a submitted job. The **topology** selector sets the hierarchy (the presets of Figure 1 or `custom`); *Add stars and orbits* fills in the masses and orbits that the topology needs. The

badge on the right shows the state of the configuration: *valid*, *not checked* or the number of errors.

Name bar. The name of the draft (`jobs/drafts/<name>.yaml`). *Save* writes the draft under this name (the button says *Save as …* while the name differs from the opened file), *Save as new* asks for a new name and never overwrites an existing draft, *Delete* moves the draft to `jobs/drafts/.deleted/`. A dot next to the name marks unsaved changes; jobs always use the saved version.

Form. The form is generated from the schema of the configuration, so it always offers every key with its unit, default and a one-line description; it is the same schema that the command line uses (chapter The configuration file). Sections that are not used show *not set* and say which computations use them (e.g. *stars – used by Light curve, Eclipse times, Tides*). A section opens with a click; *new key* and *Add* add an entry to a dictionary (a star to `masses`, an orbit to `orbits`); × removes a field. Every change is validated at once with the same checks as the command line, and errors are shown next to the field and listed under the YAML text.

YAML. The same configuration as text. *preview* shows it; *edit as text* turns it into an editor, and *Apply text to the form* reads the edited text back into the form. Comments in the text are not kept by the form; keep annotated files under `configs/`.

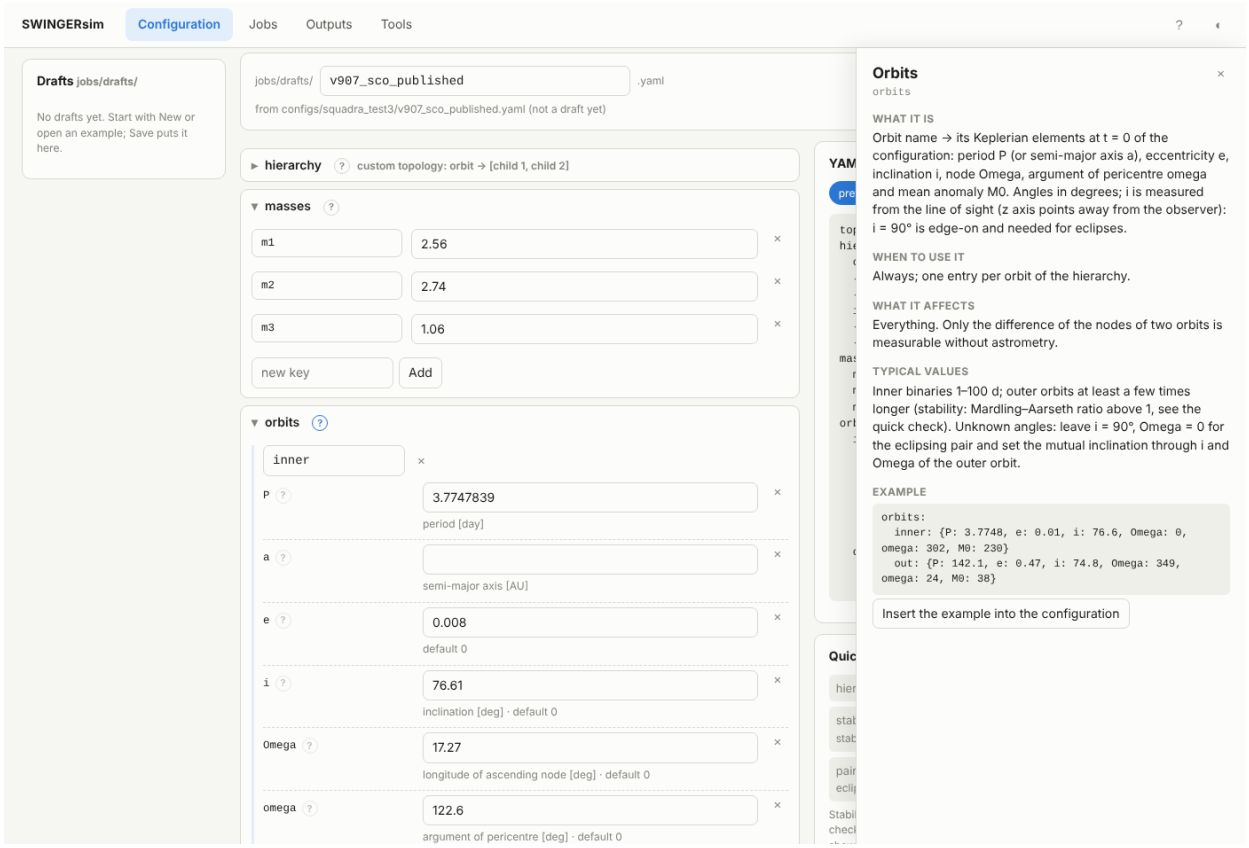


Figure 5. The help drawer, opened with the **?** of the section *orbits*. It covers the right third of the page and has the parts *What it is*, *When to use it*, *What it affects*, *Typical values* and *Example*; *Insert the example into the configuration* copies the example into the form. The form on the left stays usable; × closes the drawer.

The help drawer (Figure 5) is the built-in manual. For a configuration key it explains what the key means, when it is needed, what it affects, typical values, an example and, where relevant, the literature; for a computation it says what is computed, what the configuration needs, how long it takes and what the result page shows; for a chart it explains how to read it; for a warning it says what it means and what to do.

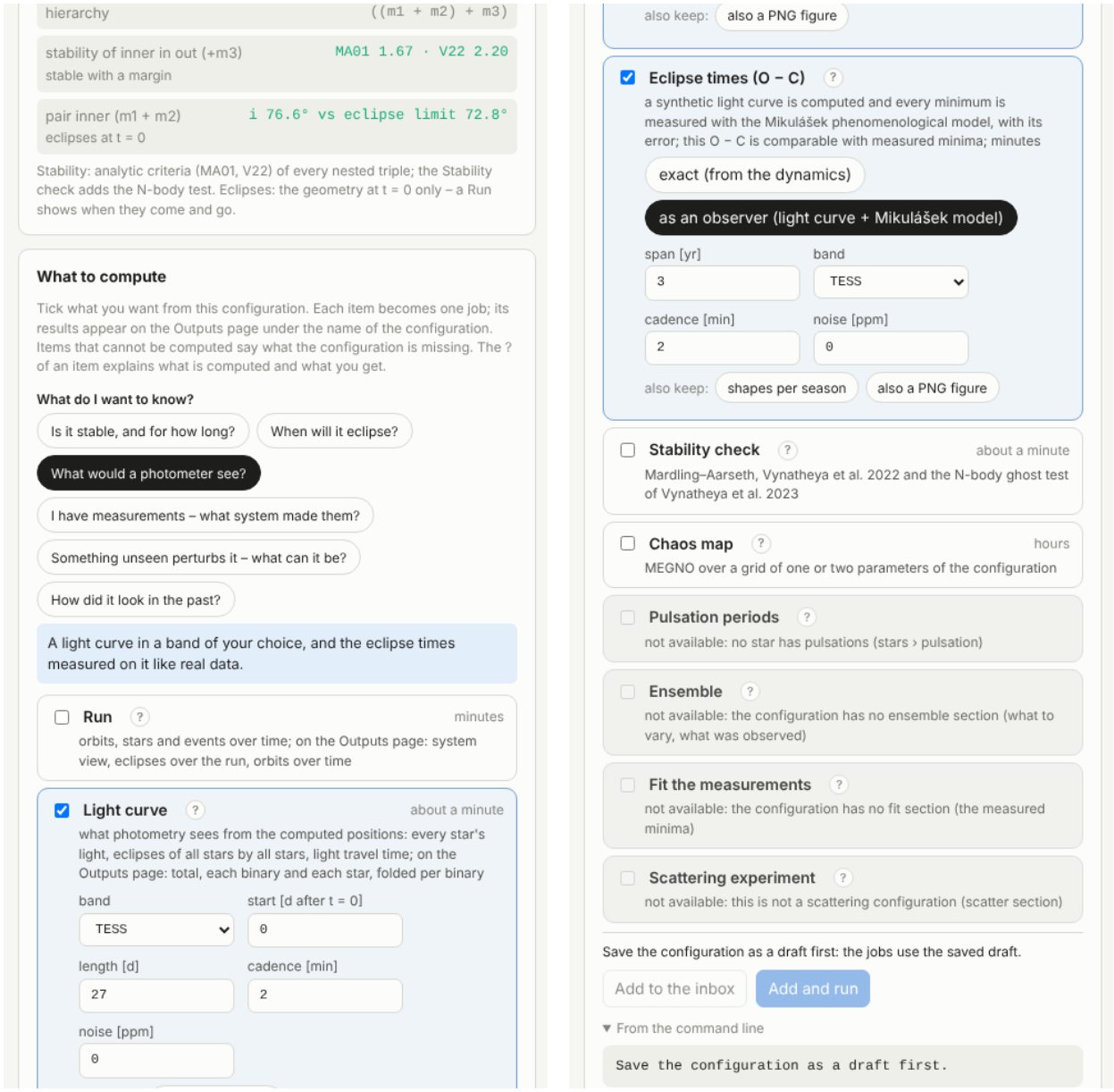


Figure 6. The right column of the Configuration page for V907 Sco, in two parts placed side by side. Left part, from the top: the *Quick look* panel with the hierarchy `((m1 + m2) + m3)`, the stability ratios of the only nested triple (MA01 1.67, V22 2.20, "stable with a margin") and the eclipse geometry of the pair `inner` (inclination 76.6° against the eclipse limit 72.8°: "eclipses at t = 0"); then *What to compute* with the six questions of *What do I want to know?* – the question *What would a photometer see?* is selected and its answer is shown in the blue box – and the items *Run* (not ticked) and *Light curve* (ticked, with its options band, start, length, cadence and noise). Right part: *Eclipse times (O − C)*, ticked, switched to *as an observer* with its options (span, band, cadence, noise; *shapes per season* and *also a PNG figure* as extra files); *Stability check* and *Chaos map* (not ticked); *Pulsation periods*, *Ensemble*, *Fit the measurements* and *Scattering experiment* greyed out with the reason (the configuration has no pulsating star, no `ensemble`, `fit` or `scatter` section); at the bottom *Add to the inbox* and *Add and run* (disabled until the draft is saved) and *From the command line*.

Quick look (Figure 6, top left) is computed from the configuration as you type, before anything is run: the hierarchy in bracket notation, the stability ratios of every nested triple after Mardling & Aarseth (2001, MA01) and Vynatheya et al. (2022, V22) – above 1 is stable, green from 1.5 – and, for every pair of two stars, its inclination against the eclipse limit $|\cos i| < (R_1 + R_2)/a$ and the Roche-lobe filling of its stars. It warns early, e.g. that an observed eclipsing binary has radii too small to eclipse at the given inclination.

What to compute lists every computation a configuration can give. Each ticked item

becomes one job of the saved draft, named `<number>_<draft>_<item>`; its results appear on the Outputs page under the name of the configuration. Items the configuration cannot give are greyed out and say what is missing. The questions under *What do I want to know?* tick the items that answer them and say in one sentence how (Table 12).

Question	Items it ticks	Note
Is it stable, and for how long?	Stability check, Run	the check gives the verdict of three criteria in a minute; a long run shows what actually happens
When will it eclipse?	Run, Eclipse times (exact)	the run shows the eclipsing geometry over time, the eclipse times every eclipse and its O−C
What would a photometer see?	Light curve, Eclipse times (as an observer)	a light curve and minima measured on it like real data
I have measurements – what system made them?	Fit the measurements	opens the `fit` section to fill in
Something unseen perturbs it – what can it be?	Ensemble	opens the `ensemble` section to fill in
How did it look in the past?	Run	with a negative `run.t_max_yr`

Table 12. The questions of *What do I want to know?*. Each row is a question (left), the computations it selects (middle) and the hint shown with it (right).

Item	Command	Options in the panel	Needs
Run	`run`	backend (empty: as configured)	masses and orbits
Light curve	`lightcurve`	band, start [d after $t = 0$], length [d] (default 27), cadence [min] (default 2), noise [ppm]; a PNG figure	radii and temperatures (from `stars` or the stellar model)
Eclipse times (O − C), exact	`timings`	span [yr] (default 10); a PNG figure	radii for the eclipse limit
Eclipse times (O − C), as an observer	`minima`	span [yr] (default 3), band, cadence, noise; shapes per season, a PNG figure	as the light curve
Stability check	`stability`	outer orbits of the N-body test (default 100)	masses and orbits
Chaos map	`chaosmap`	x parameter, x range, x points (30); optional y parameter, range, points	masses and orbits
Pulsation periods	`pulsation`	–	a star with `pulsation`
Ensemble	`ensemble`	members (empty: `ensemble.n`)	an `ensemble` section
Fit the measurements	`fit`	stage (empty: 2 if `fit.stage2` exists)	a `fit` section
Scattering experiment	`scatter`	–	a scattering configuration

Table 13. The items of *What to compute*. Columns: the label in the panel, the command the job runs (chapter The command line), the options shown when the item is ticked, and what the configuration must contain.

The two kinds of eclipse times answer different questions. *Exact* takes the moments of conjunction from the integration, with the light-travel time: the true O−C of the model, without a light curve or noise, in seconds of computing. *As an observer* computes a synthetic light curve and measures every minimum on it with the phenomenological model of Mikulášek (2015), with an error: an O−C of the same kind as one from real minima, in minutes. They differ by what moves a measured minimum away from the conjunction (asymmetric eclipses of eccentric orbits, the light of the other stars, pulsations, noise and sampling).

Add to the inbox writes the jobs as waiting jobs (`.yaml.hold`), *Add and run* queues them and starts the worker. *From the command line* shows the commands the ticked items correspond to, for use in a terminal or a script.

5.4 Jobs

The Jobs page is the control panel of the worker and of the job queue (Figure 7).

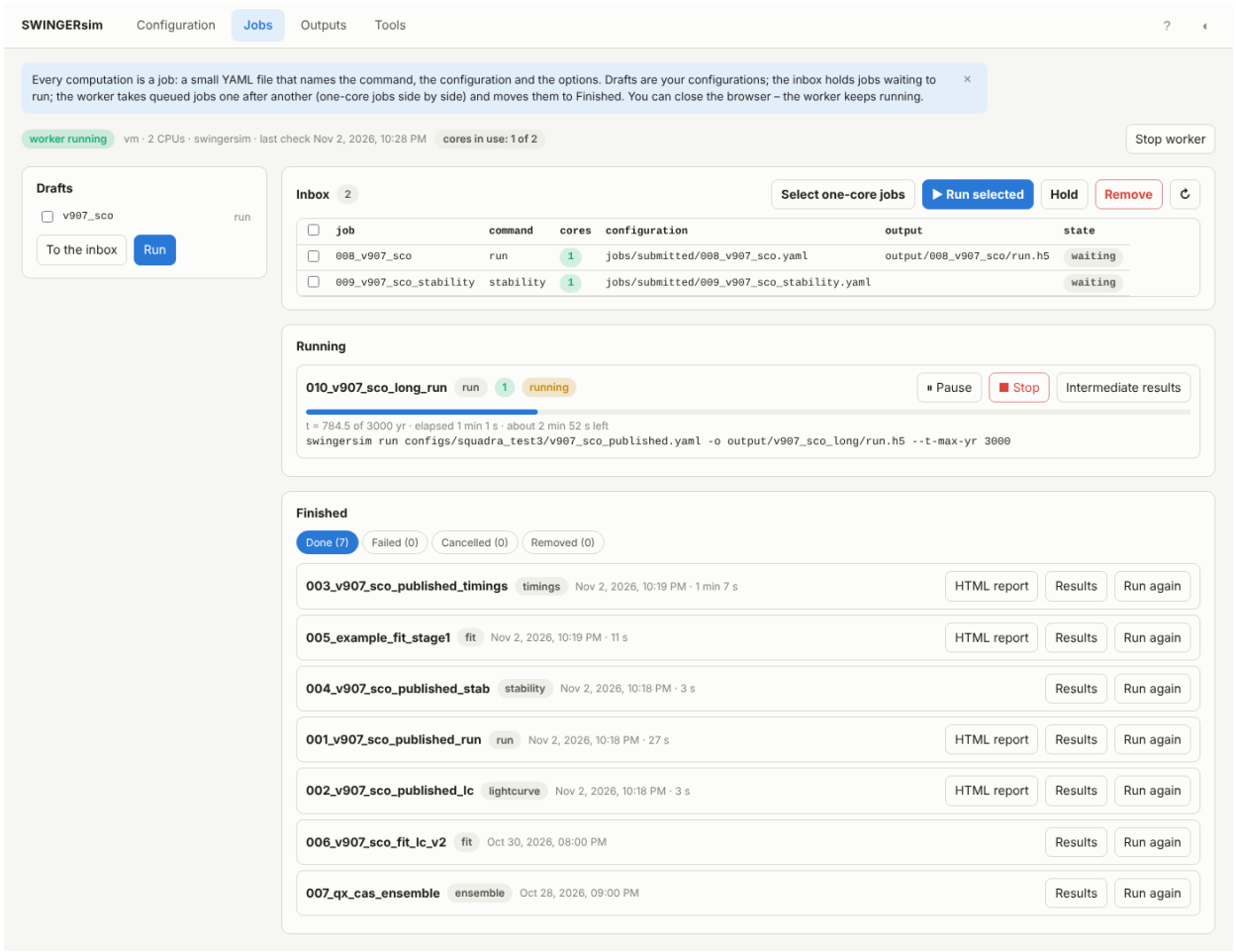


Figure 7. The Jobs page while a job runs. Top: the worker bar – *worker running* (green), the computer (`vm`), its 2 CPUs, the time of the last check, *cores in use: 1 of 2* – and *Stop worker*. Left: *Drafts* with the draft `v907_sco` (tick box; *run* says which item would be run), *To the inbox* and *Run*. Right, from the top: the *Inbox* with two waiting jobs (columns job, command, cores, configuration, output, state) and the buttons *Select one-core jobs*, *Run selected*, *Hold*, *Remove* and ↻; *Running* with the job `010_v907_sco_long_run` – a 3000-yr N-body run – its progress bar (*t = 784.5 of 3000 yr · elapsed 1 min 1 s · about 2 min 52 s left*), the command it runs and the buttons *Pause*, *Stop* and *Intermediate results*; *Finished* with the tabs *Done (7)*, *Failed (0)*, *Cancelled (0)*, *Removed (0)* and one line per finished job with its command, finishing time, duration and the buttons *HTML report*, *Results* and *Run again*.

Worker bar. Whether the worker runs, on which computer, with how many cores, when it last reported and how many cores the running jobs use. *Start worker* starts it in the

background, so that it keeps running when the browser or the web server is closed; *Stop worker* stops it, and a running job goes back to the inbox (an ensemble with `resume` continues where it stopped). A worker started with an older version of SWINGERsim is marked: restart it after an update.

Drafts. Tick drafts and press *To the inbox* (waiting jobs) or *Run* (queued, worker started). This is the same as *Add to the inbox* and *Add and run* on the Configuration page.

Inbox. Waiting (`.yaml.hold`) and queued (`.yaml`) jobs. The *cores* column says whether a job needs one core or all of them. *Run selected* runs only the ticked jobs: other queued jobs are put on hold, and the page says which. *Select one-core jobs* ticks every job that needs one core, so that they run side by side. *Hold* puts ticked jobs on hold, *Remove* moves them to `jobs/cancel/`, ↻ refreshes the list.

Running. One block per running job: its command and cores, the progress bar with the estimated time left (paused time is not counted), the command line, and three buttons. *Pause* stops the processes of the job (SIGSTOP) and *Continue* lets them go on exactly where they were. *Stop* ends the job, deletes its results so far (only inside `output/`) and moves the job to `jobs/cancel/`. *Intermediate results* opens what the job has written so far: a long N-body run writes its state every 20 s, an ensemble writes one row per finished member.

Finished. Done, failed, cancelled and removed jobs on four tabs. *HTML report* opens the report of the job, *Results* its result page on the Outputs page, *Run again* puts a copy of the job back into the queue. A failed job shows the end of its log and, for common errors (a missing file, an invalid configuration, too little memory, the energy-error limit), what the error means and what to do.

Jobs sent from the page that can use several cores (`ensemble`, `scatter`, `chaosmap`, a stage-2 `fit`, the MESA commands) get `processes: 0` (all cores of the job) unless another value is set. A single run, light curve or set of eclipse times is one calculation on one core.

5.5 Outputs

Every finished, failed or running job is a **result**. The Outputs page lists them as cards and opens each on a page made for its kind (Figure 8).

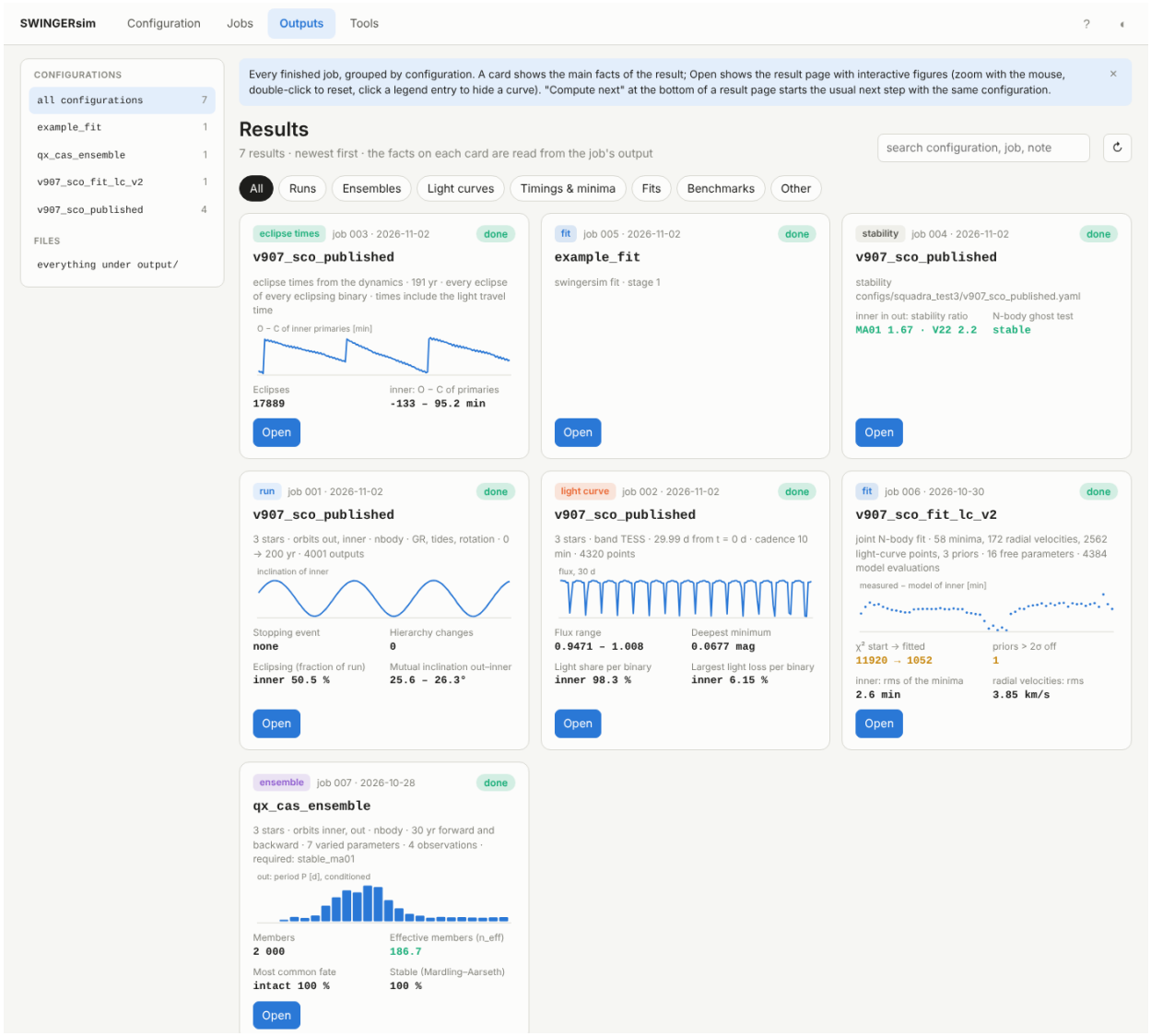


Figure 8. The list of results. Left: *Configurations* – the results grouped by the configuration they were computed from, with their number (*all configurations* 7, `example_fit` 1, `qx_cas_ensemble` 1, `v907_sco_fit_lc_v2` 1, `v907_sco_published` 4), and *everything under output/*. Top: the title *Results*, the search field and ↻. Below it the filter chips *All*, *Runs*, *Ensembles*, *Light curves*, *Timings & minima*, *Fits*, *Benchmarks* and *Other*. Each card names its kind (coloured label), job number and date, its state (*done*), the configuration, one line on what was computed, a small plot (O−C of the eclipses, the inclination of the inner binary over the run, the light curve, the measured minus model residuals of a fit, the conditioned distribution of an ensemble parameter) and two to four facts read from the output, e.g. *Eclipsing (fraction of run) inner 50.5 %*. *Open* opens the result page.

The facts on the cards are read from the output files; they state what the output contains (ranges, fractions, events, stability ratios, effective numbers of members) and never interpret the system. Colours mark the quality of a number consistently on all pages (Table 14).

Quantity	green	amber	red
largest relative energy error $\lvert\Delta E/E\rvert$ of a run (not marked for dissipative runs)	$< 10^{-6}$	$< 10^{-3}$	$\geq 10^{-3}$
stability ratio (Mardling–Aarseth, Vynatheya et al. 2022)	≥ 1.5	1–1.5	< 1
effective number of members n_{eff} of an ensemble	≥ 100	20–100	< 20

reduced χ^2, MCMC length, priors of a fit	as stated on the fit page

Table 14. The colour code of the result pages. Each row is a quantity, the columns the ranges shown in green (good), amber (look at it) and red (do not trust the number without checking). The ranges are printed under every *Data quality* box.

Every result page has the same frame: a header with the name of the configuration, the kind of result, its state, a line on what was computed and the job's note; the main facts in boxes; a *Data quality* box on the right; a blue box *What am I looking at?*; the parts made for this kind of result; *Other results of this configuration*; *Compute next*; *For a paper*; and *Advanced*. *Compute next* offers what usually follows (a light curve or the O−C of a run, more members of an ensemble, the fitted configuration of a fit) and opens the Configuration page with the item ticked; greyed-out actions say why they are not available. *For a paper* explains that every figure can be saved as a vector graphic (the camera icon above it) and its data as CSV (the disk icon), and *Methods paragraph* writes a paragraph that describes how the result was computed, with the references it needs. *Advanced* holds the files of the job, its log, its configuration and job file, free plots of any columns, all time series of a run and the 3D viewer.

5.5.1 The result page of a run

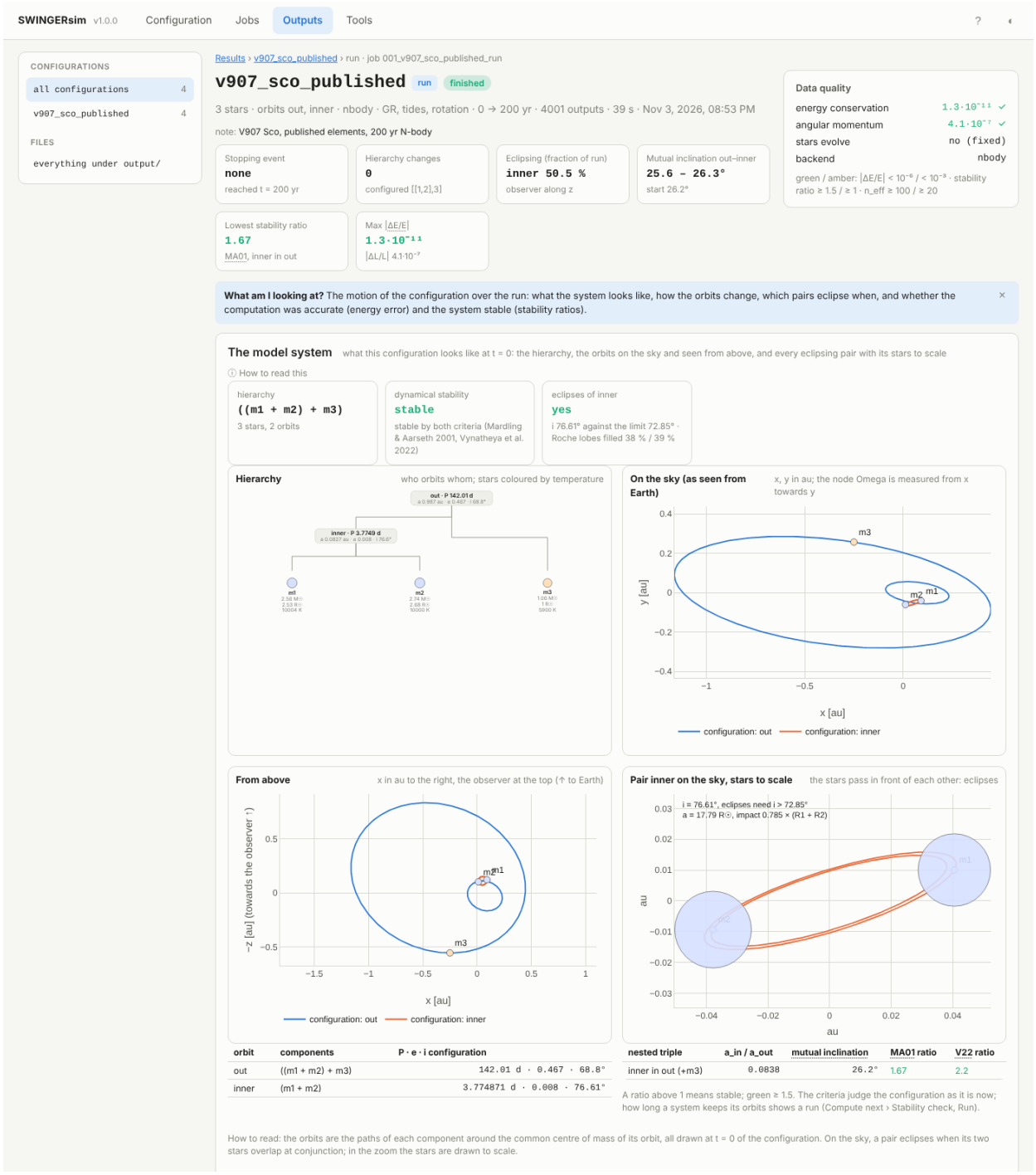


Figure 9. Top of the result page of a run (V907 Sco, 200 yr N-body with relativity, tides and rotation). Header: `v907_sco_published`, *run*, *finished*, "3 stars · orbits out, inner · nbody · GR, tides, rotation · 0 → 200 yr · 4001 outputs · 39 s". Fact boxes: *Stopping event* none, *Hierarchy changes* 0, *Eclipsing (fraction of run)* inner 50.5 %, *Mutual inclination out–inner* 25.6–26.3°, *Lowest stability ratio* 1.67 (MA01), *Max* $|\Delta E/E|$ 1.3×10^{-11}. *Data quality*: energy conservation 1.3×10^{-11}, angular momentum 4.1×10^{-7}, stars evolve no, backend nbody. *The model system*: the summary boxes (hierarchy, dynamical stability, eclipses of inner at $t = 0$); *Hierarchy* – the tree with the orbits as boxes and the stars as circles coloured by temperature; *On the sky* – the orbits of both components of every orbit projected on the sky (x, y in au, the node Ω measured from x towards y; blue: out, orange: inner); *From above* – the same seen from the pole of the sky plane, the observer at the top; *Pair inner on the sky, stars to scale* – the two stars of the eclipsing pair drawn with their radii along their apparent orbit, with the inclination, the eclipse limit, a and the impact parameter. Below: tables of the orbits (P, e, i) and of the nested triple ($a_{\rm in}/a_{\rm out}$, mutual inclination, MA01 and V22 ratios).

The top of a run page (Figure 9) answers “what is this system and did the computation work”. The fact boxes say whether the run ended early, whether the hierarchy changed, which binaries eclipse and for how much of the run, the range of the mutual inclinations, the lowest stability ratio and the energy error. *The model system* draws the configuration at $t = 0$ four ways. In the sky view, a pair eclipses when its two stars overlap on the sky at conjunction; in the zoomed panel the stars are drawn to scale, so the eclipse geometry can be read directly (here the inclination 76.6° is just above the limit 72.85°, and the impact parameter is 0.785 of $R_1 + R_2$: grazing, partial eclipses).

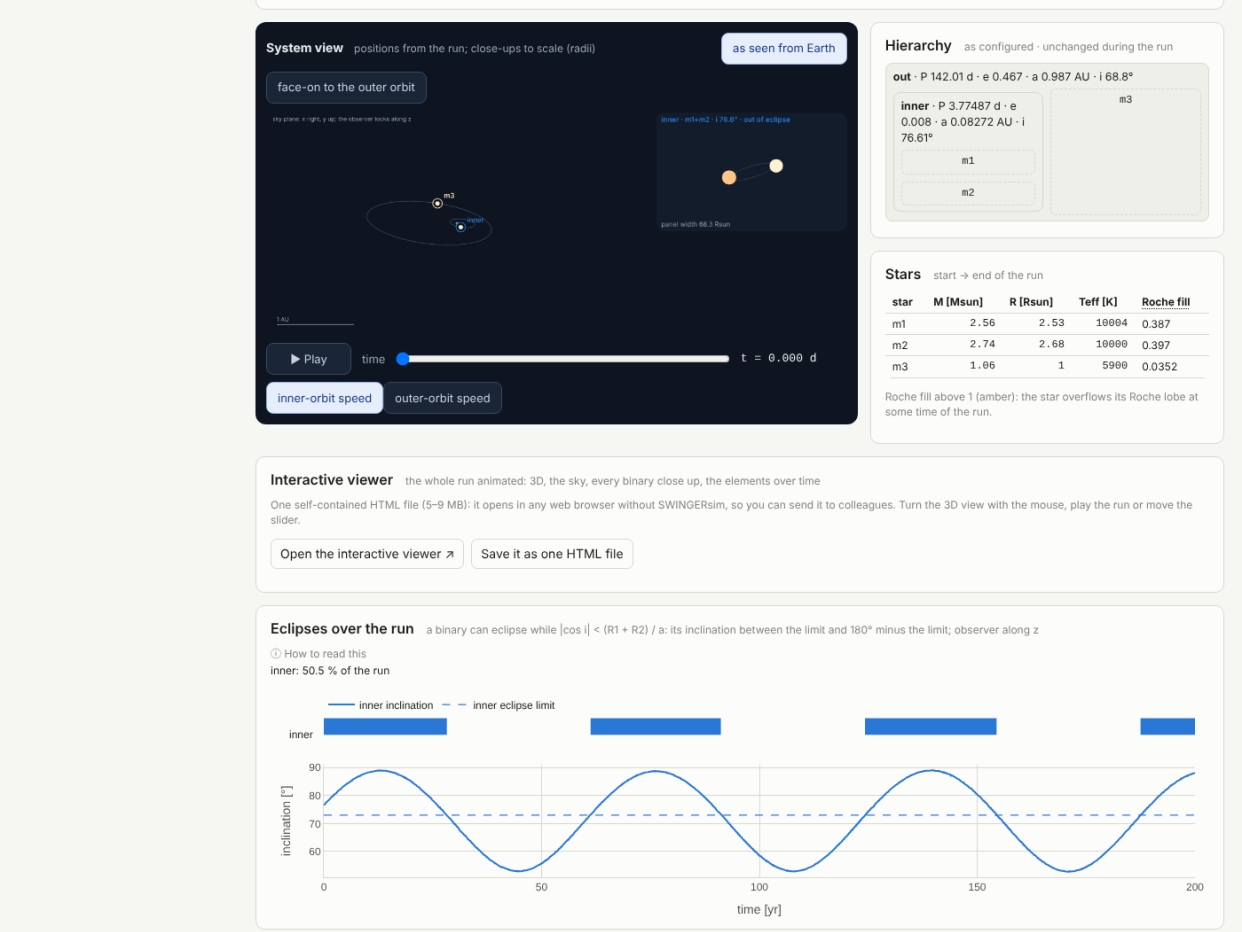


Figure 10. The middle of the run page. *System view* (dark panel): the stars at the time of the slider, as seen from Earth (*as seen from Earth*) or face-on to the outer orbit; on the right a close-up of the pair `inner` with the stars to scale and the label *out of eclipse* or *eclipse now*; *Play*, the time slider and the playback speed (*inner-orbit speed* or *outer-orbit speed*). *Hierarchy*: the configured hierarchy and whether it changed during the run. *Stars*: mass, radius, T_{eff} and the largest Roche-lobe filling of every star at the start and the end. *Interactive viewer*: opens or saves the 3D viewer of the run as one HTML file. *Eclipses over the run*: the inclination of the binary `inner` to the sky (blue curve, degrees) against time (years), its eclipse limit (dashed line, 72.85°) and the time windows in which it can eclipse (blue bars at the top); here 50.5 % of the run in four windows.

The *System view* (Figure 10, top) animates the run: positions come from the stored osculating elements, with Kepler motion between outputs. *Eclipses over the run* is the central observational result: the inclination of the inner binary oscillates between about 53° and 89° with the nodal precession period (63 yr measured in this run), and the binary eclipses only while the inclination is above its eclipse limit $\arccos[(R_1 + R_2)/a]$. The bars at the top are these windows; a gap is a time when the orbit has turned away and the eclipses are absent.

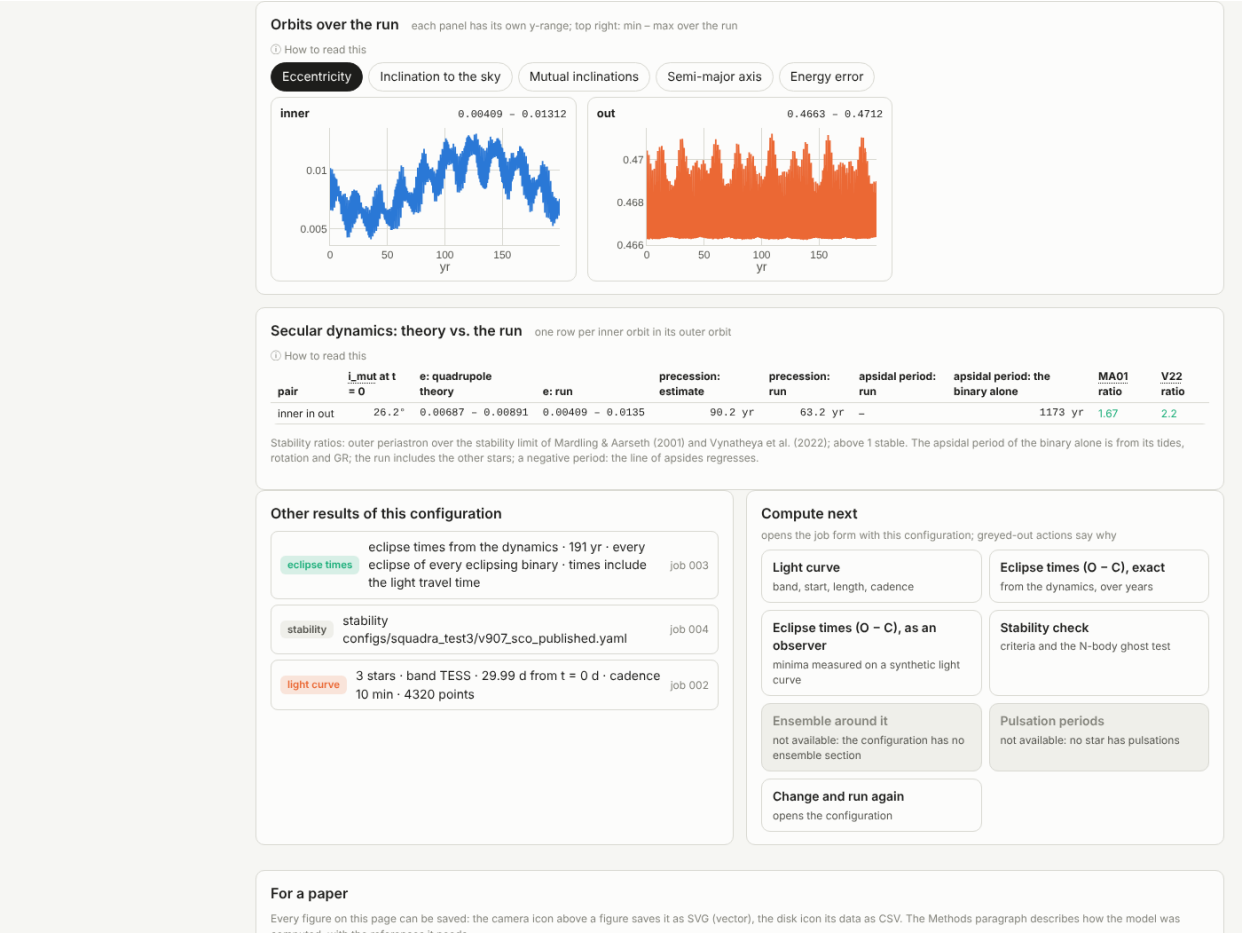


Figure 11. The bottom of the run page. *Orbits over the run*: tabs *Eccentricity*, *Inclination to the sky*, *Mutual inclinations*, *Semi-major axis* and *Energy error*; one small panel per orbit, each with its own vertical range (min–max printed at the top right); here the eccentricity of `inner` (0.0041–0.0131) and of `out` (0.4663–0.4712) against time in years. *Secular dynamics: theory vs. the run*: for every inner orbit the mutual inclination at $t = 0$, the eccentricity range of the quadrupole theory and of the run, the precession period estimated and measured, the apsidal period measured and expected for the binary alone, and the MA01 and V22 ratios. *Other results of this configuration* (eclipse times, stability check, light curve) and *Compute next* (Light curve, Eclipse times exact or as an observer, Stability check; *Ensemble around it* and *Pulsation periods* greyed out with the reason; *Change and run again*).

Orbits over the run (Figure 11) shows the osculating elements, sampled at the outputs; the fast oscillations of the outer eccentricity (orange) are the response to the inner binary on every outer revolution, not a secular change. The table *Secular dynamics* compares the run with the orbit-averaged quadrupole theory: agreement says that the system behaves as the analytic theory predicts; large differences point to octupole, resonant or non-hierarchical effects (*Physics and Methods*, chapter Dynamics).

5.5.2 Light curves



Figure 12. The light-curve page (V907 Sco, TESS band, 30 d at 10-min cadence). *Light curve*: the relative flux (1 = all stars outside eclipse) against time in days after $t = 0$, with *show:* buttons for the total light, the binary *inner (m1 + m2)* and each star, and the choice *relative flux* or *magnitude*; the bar under the plot selects a time range. *Folded on the period of a binary*: the light of `inner` folded on its period (3.77487 d, the period in the configuration; phase 0 = its deepest point), with the choice *light of this binary* or *total light*. *Light of each star and binary*: the share of the light outside eclipse and the deepest loss of its own light (inner 98.3 % and 6.18 %; m1 46.5 % and 12.1 %; m2 52.1 % and 10.0 %; m3 1.69 % and 0.0413 %). *Compare with a measured light curve*: a CSV file (time in BJD or days, flux or magnitude) drawn over the model, with the BJD of model $t = 0$.

The light-curve page (Figure 12) separates the light into its parts, so that each feature can be traced to the star that causes it. The folded curve shows the primary and secondary eclipses of the binary on its own period; scatter around the mean shape comes from the other stars and from changes of the orbit during the window. Measured data are drawn over the model after giving the BJD of model $t = 0$, because the model has no calendar: its time starts at $t = 0$, the epoch of the elements.

5.5.3 Eclipse times and minima

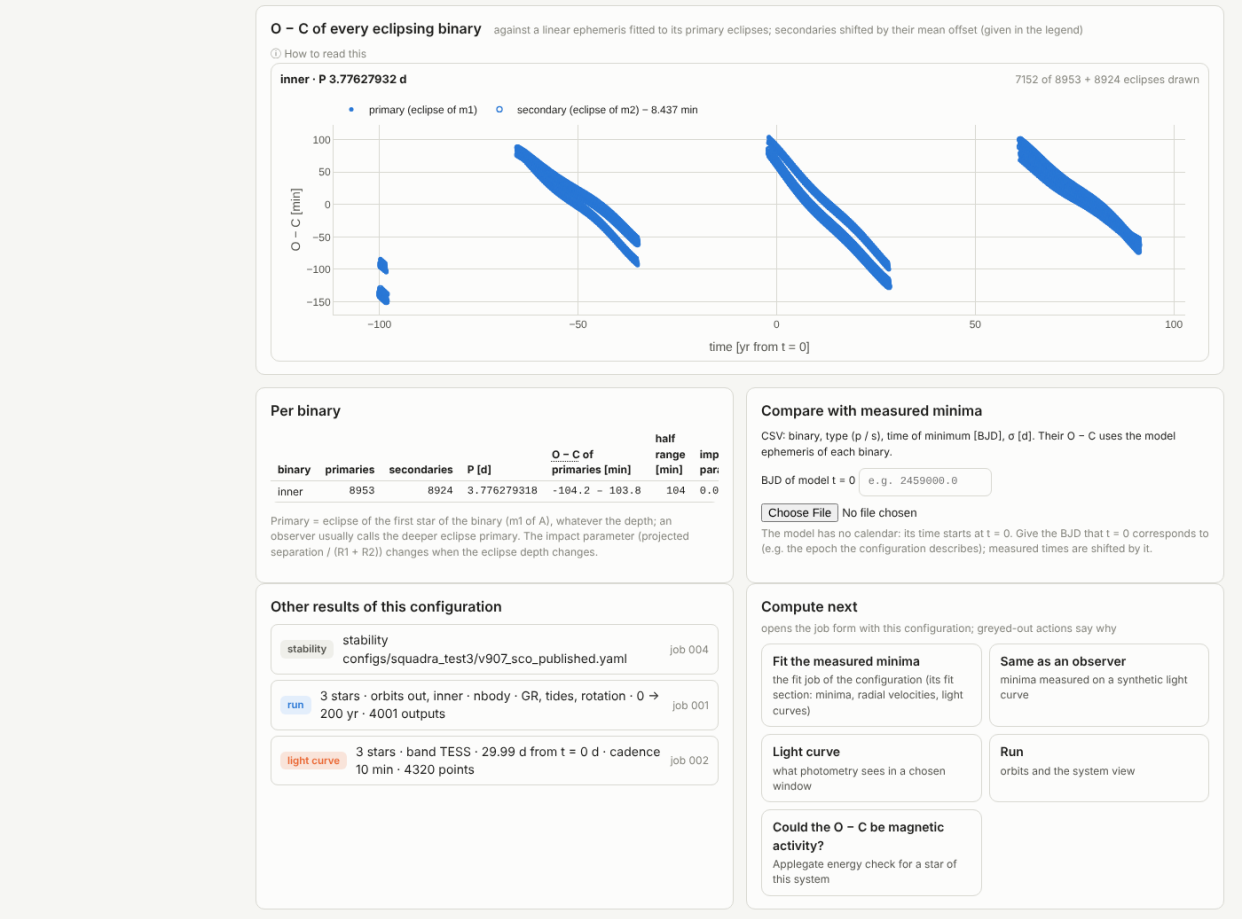


Figure 13. The eclipse-times page (exact times from the dynamics, V907 Sco, 100 yr before and after $t = 0$). *O − C of every eclipsing binary*: the O−C in minutes against a linear ephemeris fitted to the primary eclipses, against time in years from $t = 0$; filled dots: primary eclipses (eclipses of m1), open circles: secondary eclipses, shifted by their mean offset (−8.44 min, given in the legend). The eclipses exist only in four windows; within each window the O−C runs downward over about 200 min. *Per binary*: the numbers of primary and secondary eclipses (8953, 8924), the period of the ephemeris (3.776279318 d), the O−C range (−104.2 to 103.8 min), its half range and the impact parameter. *Compare with measured minima*: a CSV file of measured minima and the BJD of model $t = 0$. *Other results* and *Compute next*.

Figure 13 shows how the dynamics appears in an O−C diagram. A linear ephemeris fitted to all eclipses cannot follow the slow drift of the eclipse period inside each window, which is caused by the precession of the orbit with respect to the line of sight; within a window, primary and secondary eclipses run almost parallel, so the apsidal motion of the nearly circular binary is small. For a binary with a light-time orbit, the O−C instead shows a wave with the outer period. The page of minima measured *as an observer* has the same layout and adds the error of every minimum, the shape of the eclipses (depth, width and kurtosis of the Mikulášek profile, per season) and the shift between the measured minimum and the geometric conjunction.

5.5.4 Stability checks and other results

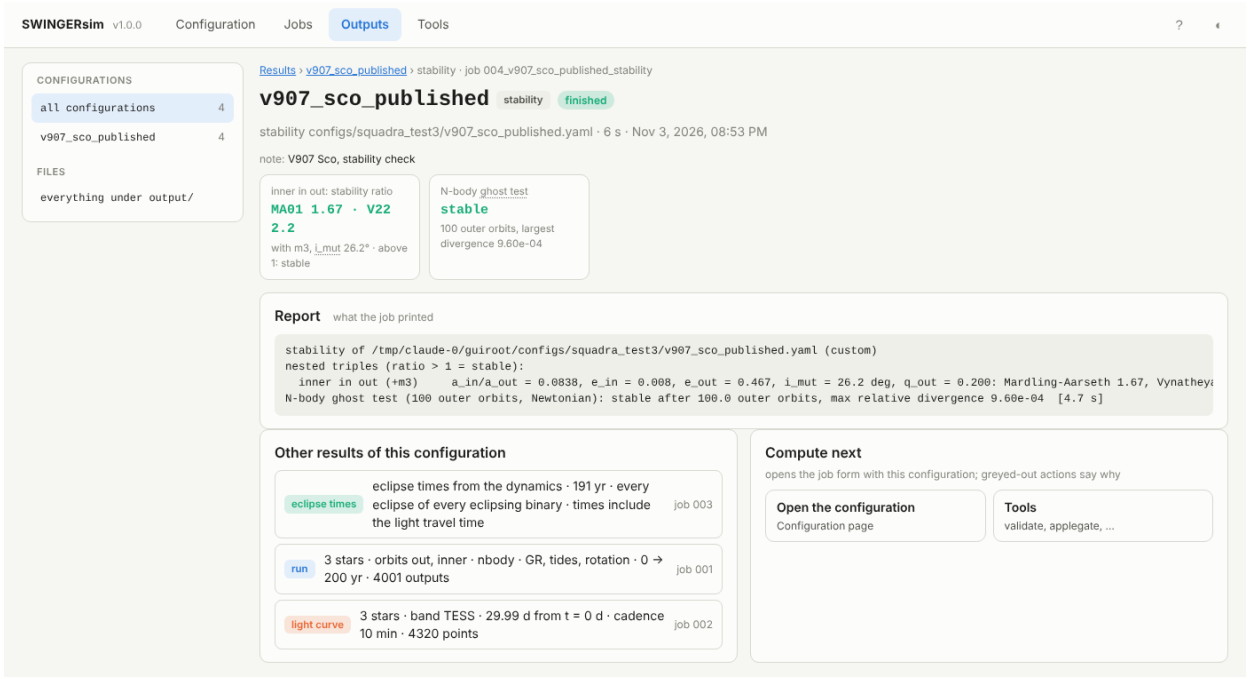


Figure 14. The page of a stability check, an example of the pages of the commands without a dedicated layout. Fact boxes: *inner in out: stability ratio* MA01 1.67 · V22 2.2 ("with m3, i_mut 26.2° · above 1: stable") and *N-body ghost test* stable ("100 outer orbits, largest divergence 9.60×10^{-4}"). *Report*: the text the job printed – the nested triple with $a_{\mathrm{in}}/a_{\mathrm{out}}$, the eccentricities, the mutual inclination, the outer mass ratio and both ratios, and the result of the N-body test. *Other results of this configuration*, *Compute next* (*Open the configuration*, *Tools*), *For a paper* and *Advanced*.

Commands without a page of their own (stability checks, chaos maps, scattering experiments, pulsation periods, emulators) show the text they printed and the facts recognised in it (Figure 14), with their files under *Advanced*.

5.5.5 The result page of an ensemble

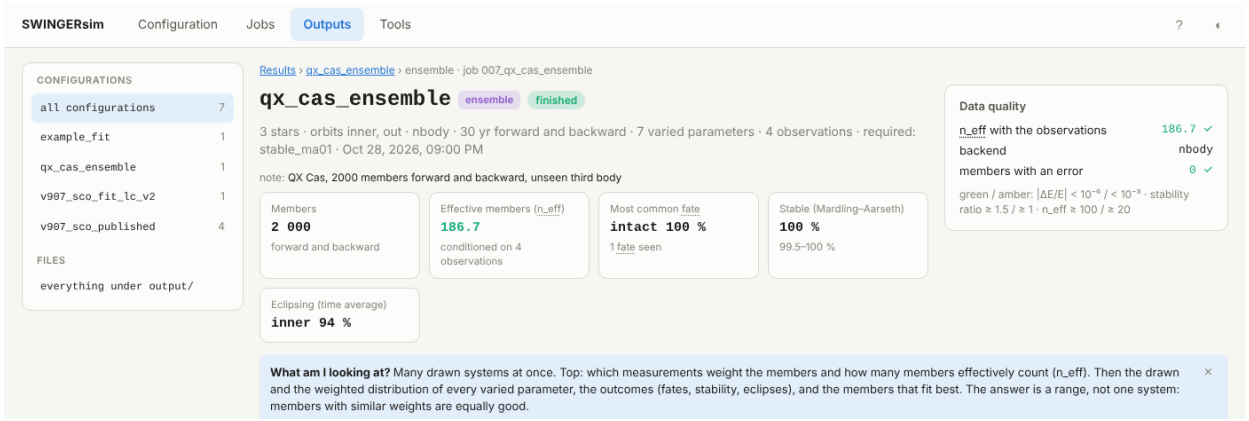


Figure 15. Top of an ensemble page (QX Cas: 2000 members, forward and backward, 30 yr N-body each; the mass and orbit of the third star drawn). Header: 3 stars, the backend, the run length, "7 varied parameters · 4 observations · required: stable_ma01". Fact boxes: *Members* 2 000, *Effective members (n_eff)* 186.7 ("conditioned on 4 observations"), *Most common fate* intact 100 %, *Stable (Mardling–Aarseth)* 100 %, *Eclipsing (time average)* inner 94 %. *Data quality*: n_{eff} with the observations (green), backend, members with an error (0).

An ensemble page (Figures 15–18) answers “which systems are possible, and which do the measurements allow”. The model system at the top (not shown) is the nominal configuration.

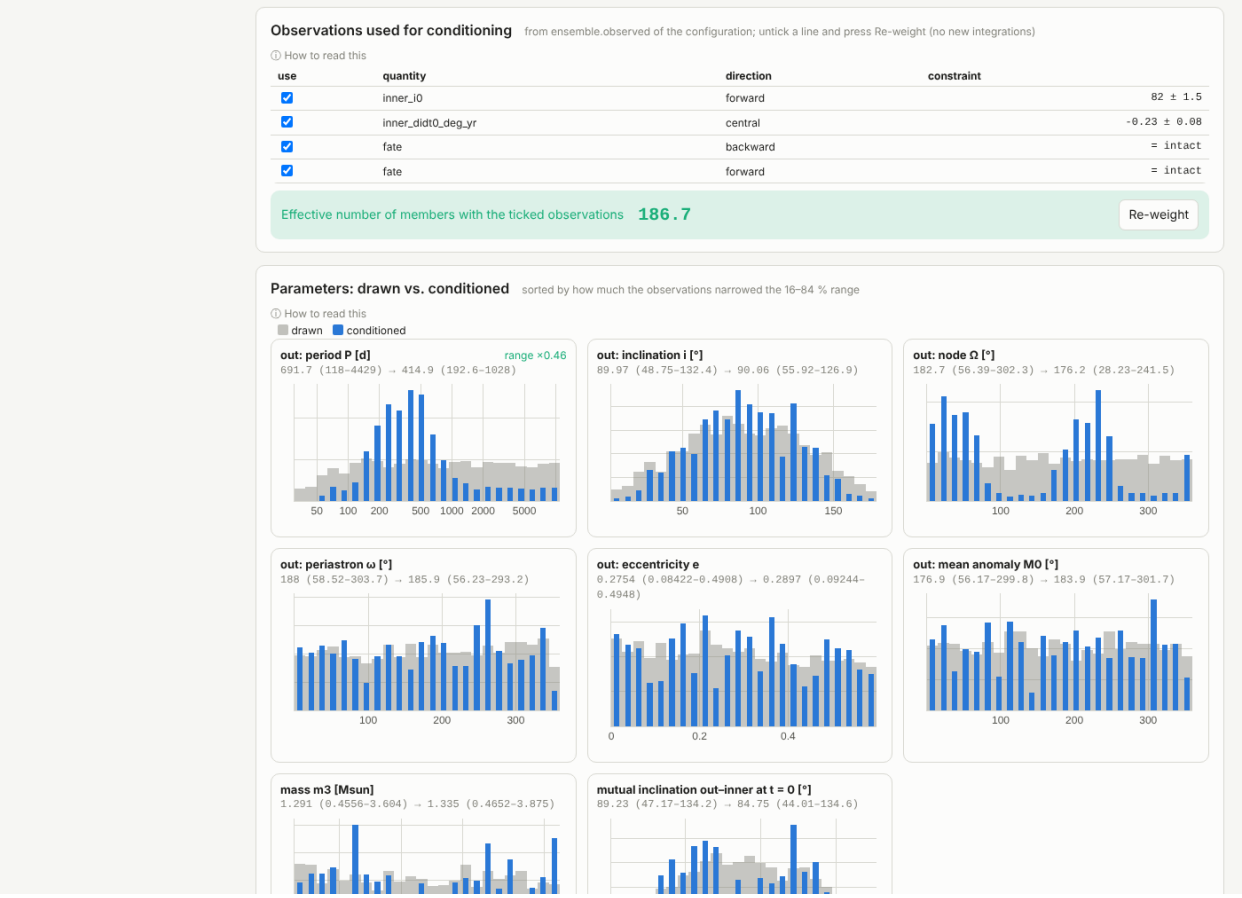


Figure 16. The conditioning and the parameters of the ensemble. *Observations used for conditioning*: one row per entry of `ensemble.observed` with a tick box *use*, the quantity (`inner_i0`, `inner_didt0_deg_yr`, `fate` backward, `fate` forward), the direction (forward, central, backward) and the constraint (82 ± 1.5, −0.23 ± 0.08, = intact); the effective number of members with the ticked observations (186.7) and *Re-weight*. *Parameters: drawn vs. conditioned*: one histogram per drawn parameter and for the initial mutual inclination, grey = drawn (prior), blue = weighted by the observations; above each, the median and 16–84 % range before → after; the panels are sorted by how much the observations narrowed the range (the outer period first, its range ×0.46; logarithmic axis for a log-uniform draw).

The observations table (Figure 16) is interactive: untick a measurement and press *Re-weight* to see what it contributes; nothing is integrated again. The weight of a member is $w = \exp(-\chi^2/2)$ for Gaussian measurements (1 for an exact match, 0.61 at 1σ, 0.14 at 2σ) and 0 when it violates a limit; the effective number of members $n_{\mathrm{eff}} = (\sum w)^2 / \sum w^2$ (Kish 1965) says how many members carry the weight. Below about 20 the conditioned histograms are noise and are hidden. Where the blue histogram is narrower than the grey one, the measurements constrain the parameter: here they prefer outer periods of 190–1030 d over the drawn 120–4400 d, and leave the angles almost unconstrained.

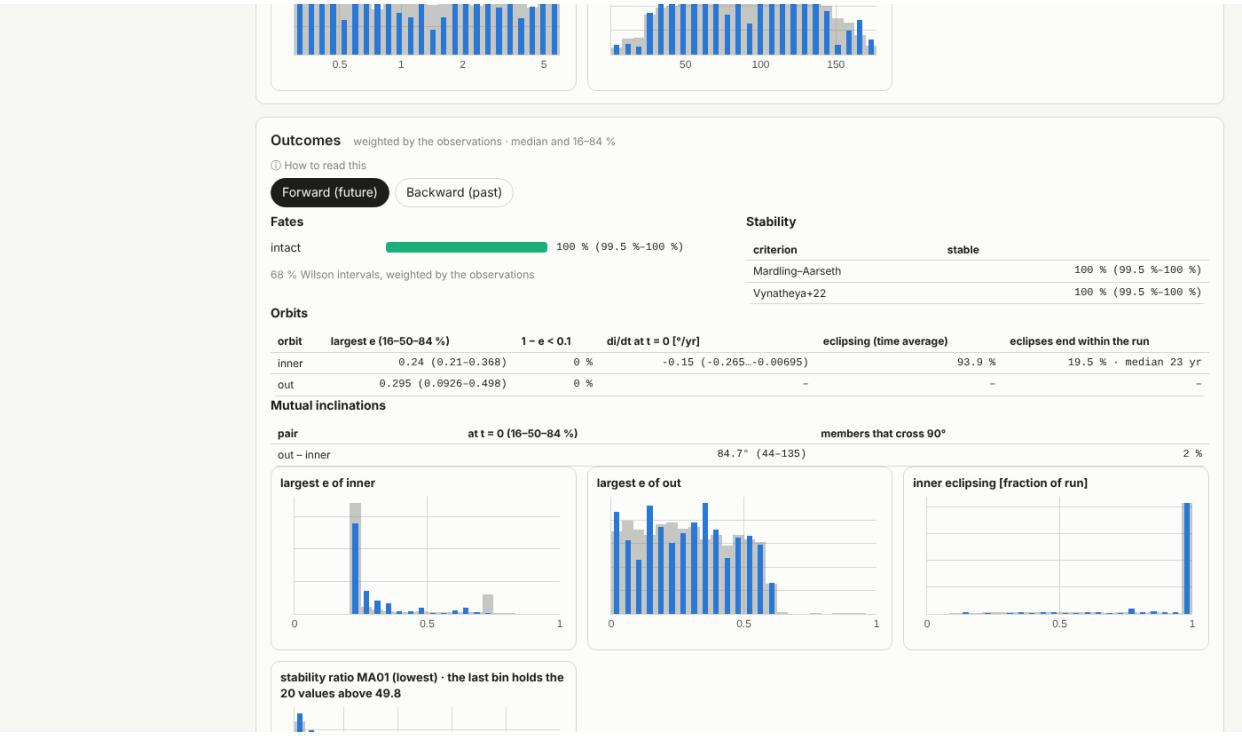


Figure 17. Outcomes of the ensemble, weighted by the observations. Tabs *Forward (future)* and *Backward (past)*. *Fates*: intact 100 % (99.5–100 %, 68 % Wilson interval). *Stability*: the fractions stable by Mardling–Aarseth and Vynatheya et al. (2022). *Orbits*: per orbit the largest eccentricity (median and 16–84 %), the fraction with $1-e<0.1$, the present rate of change of the inclination di/dt at $t=0$, the time-averaged eclipsing fraction and the fraction of members whose eclipses end within the run (19.5 %, median 23 yr). *Mutual inclinations*: at $t=0$ (84.7°, 44–135°) and the fraction of members that cross 90°. Histograms (grey drawn, blue weighted): largest e of inner and of out, eclipsing fraction of inner, lowest MA01 stability ratio.

The outcomes (Figure 17) are the answer of the ensemble: fractions with their 68 % Wilson intervals (Wilson 1927), which behave well also for fractions close to 0 or 1, and medians with 16–84 % ranges, all weighted. The *Backward* tab gives the same for the runs into the past.

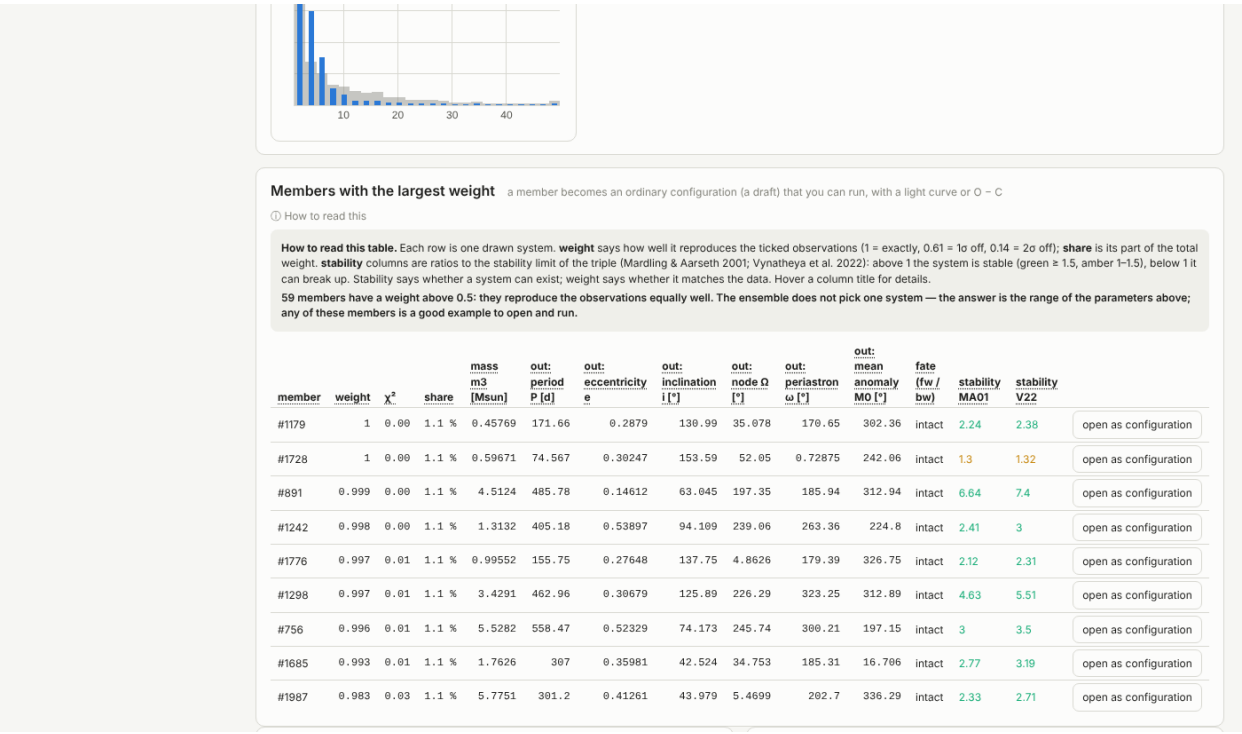


Figure 18. *Members with the largest weight*. The grey box explains the table: each row is one drawn system; *weight* says how well it reproduces the ticked observations; *share* is its part of the total weight; the stability columns are the MA01 and V22 ratios (green ≥ 1.5, amber 1–1.5); "59 members have a weight above 0.5: they reproduce the observations equally well". The columns: member number, weight, χ^2, share, the drawn values (m_3, outer P, e, i, Ω, ω, M_0), the fate forward/backward, the two stability ratios and *open as configuration*.

Open as configuration (Figure 18) turns a member into an ordinary draft with its drawn values, ready to be run, with a light curve or an O−C diagram. The answer of an ensemble is the range of the parameters, not one member: members with similar weights are equally good examples.

5.5.6 The result page of a fit

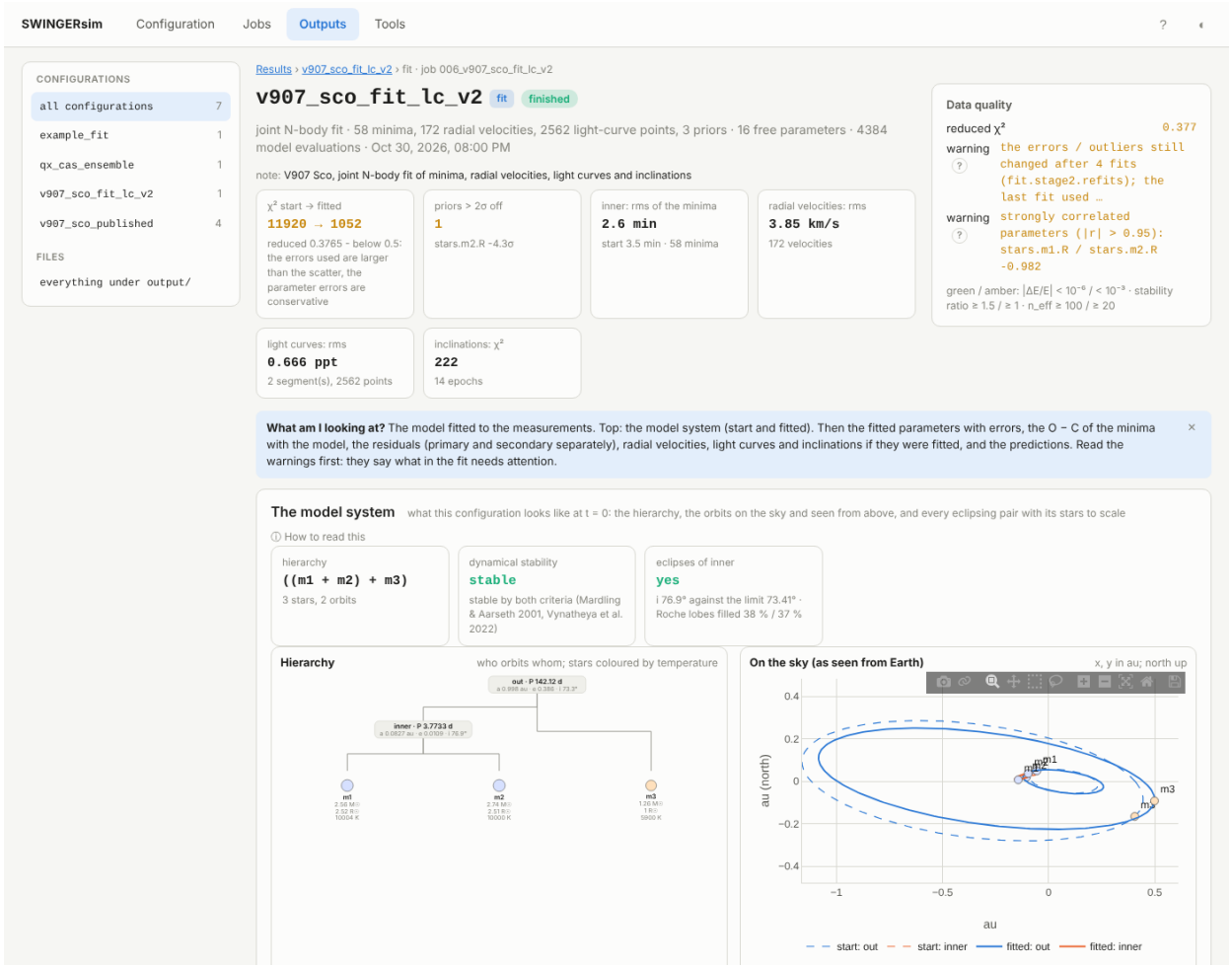


Figure 19. Top of the page of a stage-2 fit (V907 Sco: 58 minima, 172 radial velocities, 2562 light-curve points, 14 inclinations, 3 priors, 16 free parameters, 4384 model evaluations). Fact boxes: *χ^2 start → fitted* 11 920 → 1052 ("reduced 0.3765 – below 0.5: the errors used are larger than the scatter, the parameter errors are conservative"), *priors > 2σ off* 1 (`stars.m2.R` −4.3σ), *inner: rms of the minima* 2.6 min (start 3.5 min), *radial velocities: rms* 3.85 km/s, *light curves: rms* 0.666 ppt (2 segments), *inclinations: χ^2* 222 (14 epochs). *Data quality*: reduced χ^2 and two warnings (the error factors still changed after 4 fits; a strongly correlated pair of parameters). *The model system*: as for a run, with the start configuration dashed and the fitted one solid.

The fit page (Figures 19–23) shows the model fitted to the measurements and everything needed to judge it. Read the warnings first: they say what needs attention.

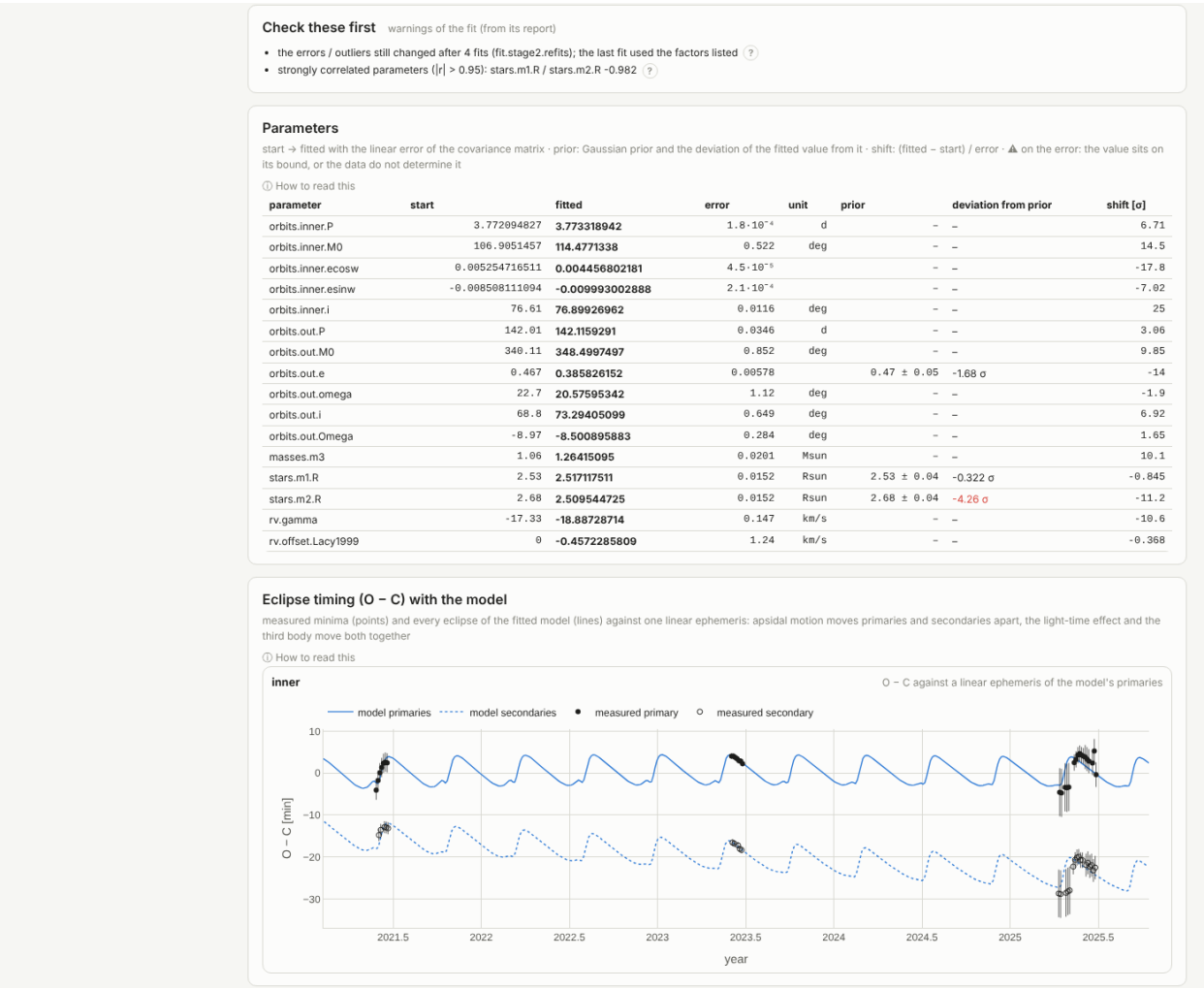


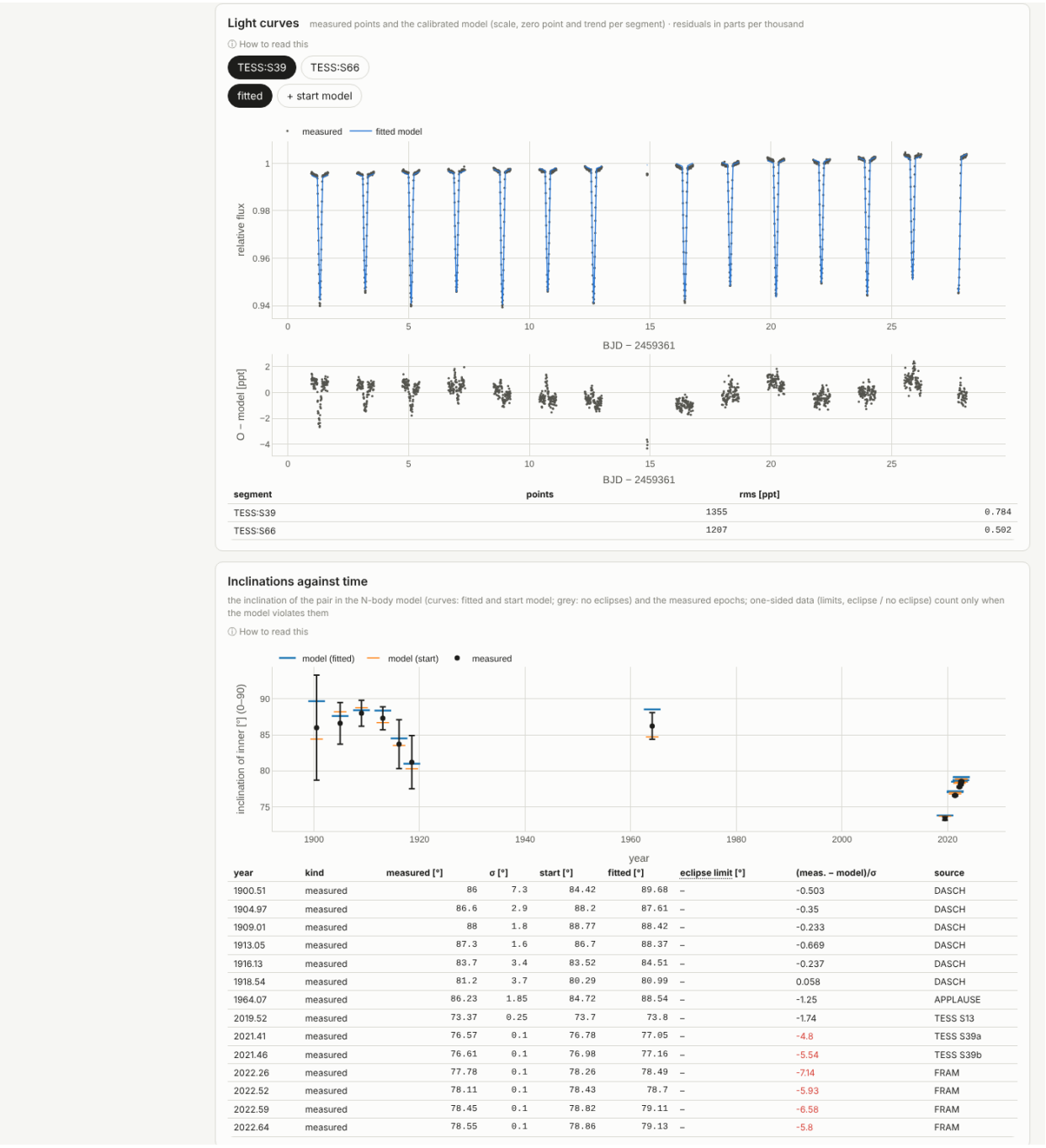
Figure 20. Warnings, parameters and the O−C of the fit. *Check these first*: the warnings of the fit with their **?**. *Parameters*: one row per free parameter with its start value, fitted value, linear error from the covariance matrix, unit, prior (mean ± σ) and the deviation of the fitted value from the prior, and the shift (fitted − start)/error; deviations from a prior above 2σ are red. *Eclipse timing (O − C) with the model*: the measured minima (dots: primary filled, secondary open, with error bars) and every eclipse of the fitted model (solid: primaries, dashed: secondaries), as O−C in minutes against one linear ephemeris over the years 2021–2025.

In the O−C panel of Figure 20, apsidal motion moves the primary and secondary curves apart, while the light-time effect and the perturbation by the third star move both together; here the secondary curve lies about 15–25 min below the primary one and both carry the 142-d wave of the outer orbit.



Figure 21. Residuals of the minima and the radial velocities. *Times of minima*: measured minus fitted model in minutes against time, filled primary, open secondary, crosses for minima left out as outliers; the buttons switch between the fitted and the start model; the error bars are the errors the fit used (with the error floor of the source). Tables: the rms per source (TESS sectors) and the mean residual of primary and secondary minima with their difference (0.435 min). *Radial velocities*: measured points and the model at the same times for m1 and m2, against the phase of the binary (tabs *phase of its own binary*, *against time*, *phase of inner*, *phase of out*); blue CHIRON 2022, orange Lacy (1999); the grey ticks are the model, which includes the motion around the third star and is therefore not one sine curve; below, the residuals in km/s.

The difference between the mean residuals of primary and secondary minima (Figure 21) is a sensitive check: a difference of several σ means that the model puts one kind of eclipse at the wrong phase, usually a wrong $e \cos \omega$ of the pair.



year	kind	measured [°]	σ [°]	start [°]	fitted [°]	eclipse limit [°]	(meas. − model)/σ	source
1900.51	measured	86	7.3	84.42	89.68	–	-0.503	DASCH
1904.97	measured	86.6	2.9	88.2	87.61	–	-0.35	DASCH
1909.01	measured	88	1.8	88.77	88.42	–	-0.233	DASCH
1913.05	measured	87.3	1.6	86.7	88.37	–	-0.669	DASCH
1916.13	measured	83.7	3.4	83.52	84.51	–	-0.237	DASCH
1918.54	measured	81.2	3.7	80.29	80.99	–	0.058	DASCH
1964.07	measured	86.23	1.85	84.72	88.54	–	-1.25	APPLAUSE
2019.52	measured	73.37	0.25	73.7	73.8	–	-1.74	TESS S13
2021.41	measured	76.57	0.1	76.78	77.05	–	-4.8	TESS S39a
2021.46	measured	76.61	0.1	76.98	77.16	–	-5.54	TESS S39b
2022.26	measured	77.78	0.1	78.26	78.49	–	-7.14	FRAM
2022.52	measured	78.11	0.1	78.43	78.7	–	-5.93	FRAM
2022.59	measured	78.45	0.1	78.82	79.11	–	-6.58	FRAM
2022.64	measured	78.55	0.1	78.86	79.13	–	-5.8	FRAM

Figure 22. Light curves and inclinations of the fit. *Light curves*: one tab per segment (TESS S39, S66), *fitted* or *+ start model*; the measured flux (dots) and the calibrated model (line, scale, zero point and trend solved per segment) in windows around the eclipses, against BJD − 2459361; below, the residuals in parts per thousand (ppt); the table gives the points and the rms per segment. *Inclinations against time*: the inclination of the pair in the N-body model (blue: fitted, orange: start) and the measured epochs (black dots with error bars) from 1900 to 2022; the table lists every epoch with the measured value, its error, the start and fitted model values, the eclipse limit (for one-sided data), the normalised residual and the source (DASCH, APPLAUSE, TESS, FRAM).

The inclination panel (Figure 22) shows a property that only a dynamical model can fit: the eclipses of V907 Sco disappeared and reappeared during the 20th century because the inclination of its orbit changes, and the N-body model of the whole system follows the measured epochs.

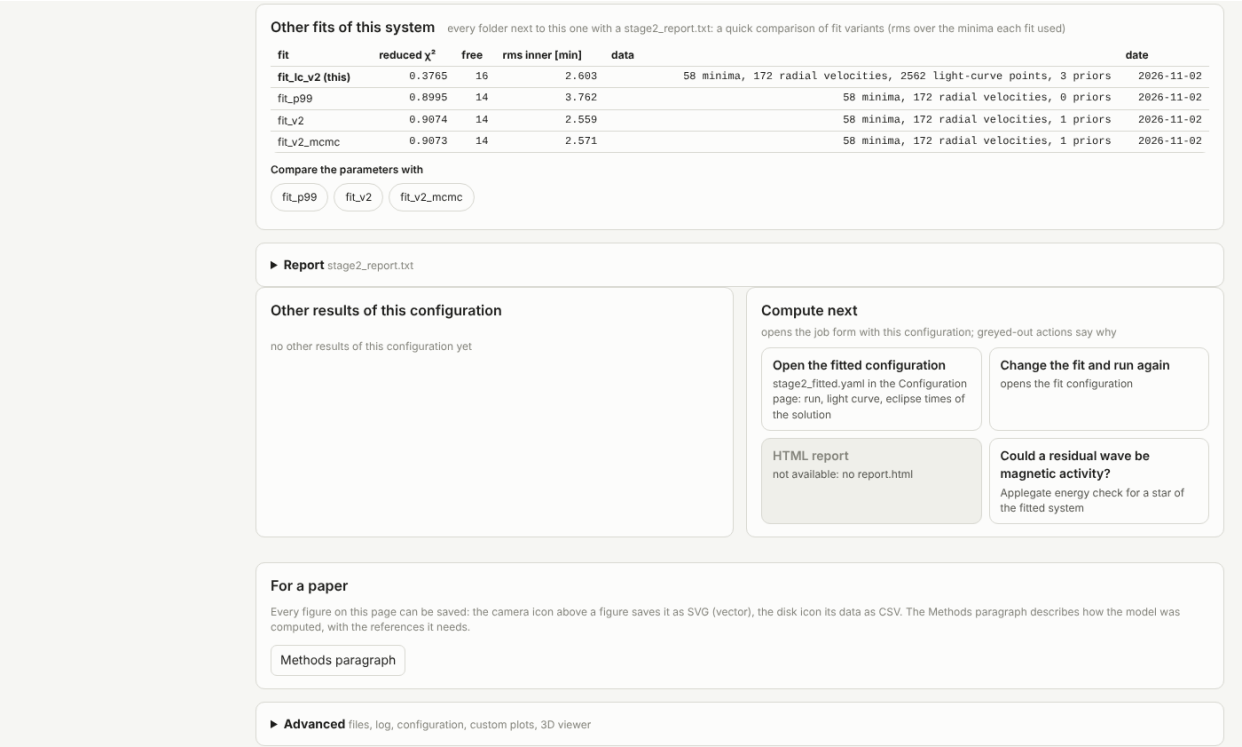


Figure 23. The end of the fit page. *Predicted minima* (above, cut): the next minima in BJD_TDB and as a calendar date, with their uncertainty σ in minutes, the eclipse duration and the impact parameter. *Other fits of this system*: every fit folder next to this one with its reduced χ^2, number of free parameters, rms of the minima and data used; *Compare the parameters with* adds a second fit to the parameter table. *Report*: the text report of the fit. *Compute next*: *Open the fitted configuration* (run, light curve, eclipse times of the solution), *Change the fit and run again*, *HTML report* (greyed out here) and *Could a residual wave be magnetic activity?* (the Applegate test). *For a paper* and *Advanced*.

The predictions (Figure 23) are computed from many parameter sets drawn from the MCMC samples or the covariance matrix, one N-body integration each, so their errors include the correlations of the parameters. *Other fits of this system* makes variants (with and without light curves, priors or radial velocities) comparable at a glance.

5.6 Tools

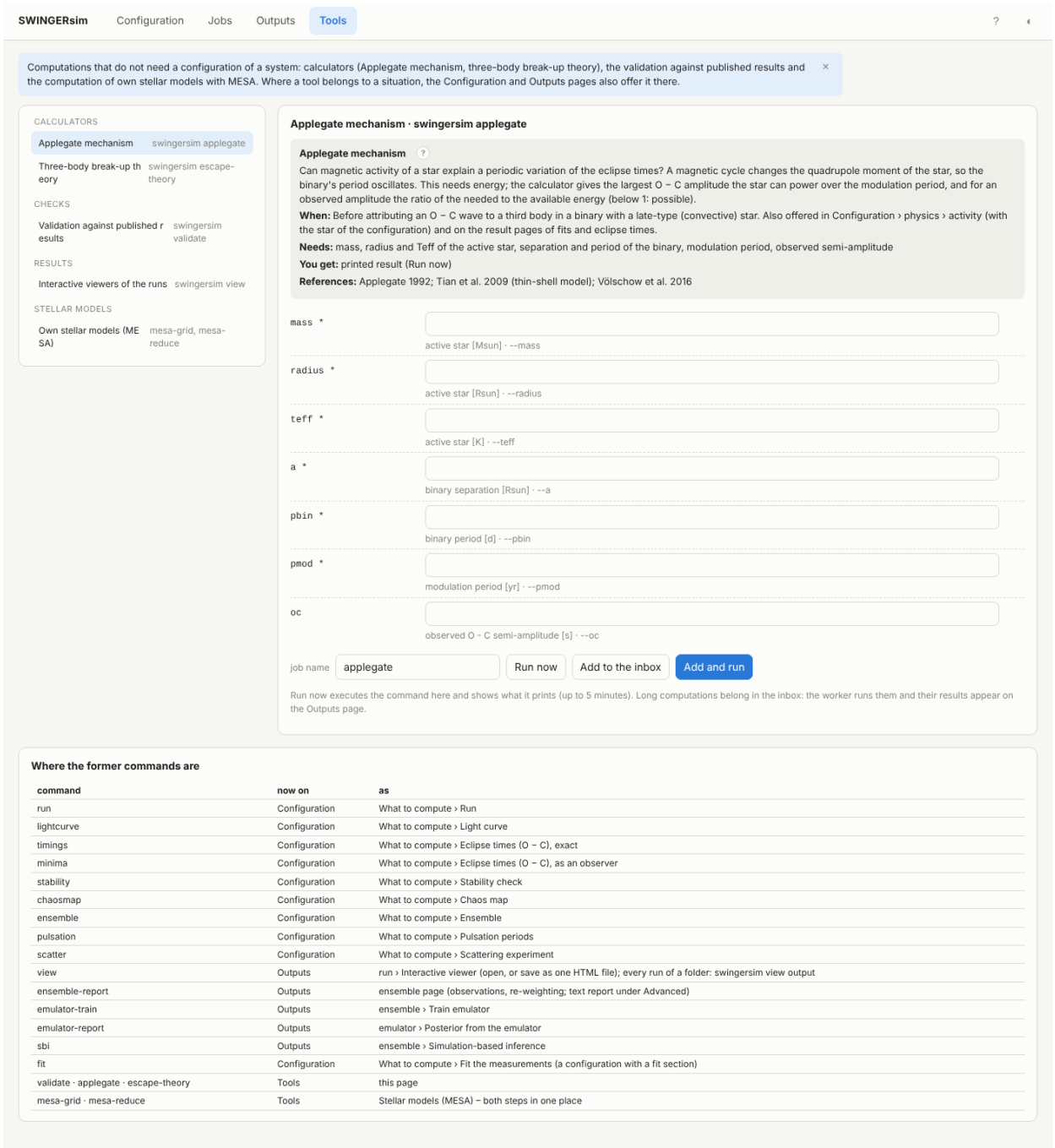


Figure 24. The Tools page with the Applegate calculator selected. Left: the tools grouped as *Calculators* (Applegate mechanism, three-body break-up theory), *Checks* (validation against published results), *Results* (interactive viewers of the runs) and *Stellar models* (own models with MESA); each with the command it runs. Right: the help of the tool (what it does, when to use it, what it needs, what you get, references), its form (mass, radius, $T_{\rm eff}$, a, $P_{\rm bin}$, $P_{\rm mod}$ of the active star and binary; the observed O−C semi-amplitude), the job name and *Run now*, *Add to the inbox*, *Add and run*. Bottom: *Where the former commands are* – for every command the page and the item where it is found.

The Tools page (Figure 24) holds what belongs neither to a configuration nor to a result. Each tool has a form generated from the options of its command (chapter The command line). *Run now* runs the command at once and shows what it prints (at most 5 minutes); longer computations go to the inbox. The table at the bottom says where each command is found in the application (Table 15).

Command	Where in the application
`run`, `lightcurve`, `timings`, `minima`, `stability`, `chaosmap`, `ensemble`, `pulsation`, `scatter`, `fit`	Configuration › What to compute
`view`	Outputs › a run › Interactive viewer; all runs of a folder: Tools › Interactive viewers
`ensemble-report`	Outputs › an ensemble (text report under Advanced)
`emulator-train`, `sbi`	Outputs › an ensemble › Compute next
`emulator-report`	Outputs › an emulator › Compute next
`validate`, `applegate`, `escape-theory`, `mesa-grid`, `mesa-reduce`	Tools
`rv`, `apsidal`, `worker`, `gui`	command line only (the worker is started on the Jobs page)

Table 15. Where every command of SWINGERsim is found in the web application. Left: the sub-command; right: the page and the place on it.

5.7 Folders used by the application

Folder	Content	Written by
`configs/`	configurations that come with SWINGERsim and your own annotated files	you
`jobs/drafts/`	configurations written in the application, members of ensembles opened as configurations	Configuration, Outputs
`jobs/submitted/`	frozen copies of the drafts as they were sent	Configuration, Jobs
`jobs/inbox/`, `running/`, `done/`, `failed/`, `cancel/`	the job queue (Table 9)	the application and the worker
`output/<job>/`	the results of a job sent from the application	the command of the job
`output/.gui_cache/`	cached cards, pages and viewers; rebuilt when a job or its output changes	the application

Table 16. The folders the web application reads and writes, all inside the SWINGERsim folder given by `--root`.

Chapter 6

The command line

Everything SWINGERsim computes is started by one command, `swingersim`, followed by a sub-command and its arguments. `swingersim --help` lists the sub-commands and `swingersim <sub-command> --help` lists the options of one. The web application and the worker build their forms and job command lines from the same definitions, so an option described here is available everywhere.

6.1 Common conventions

- The first argument of most sub-commands is a configuration file (YAML). Relative paths inside a configuration (tables of minima, light curves, output folders of a fit) are relative to the configuration file.
- `-o` / `--output` names the output file or folder; missing folders are created.
- Every sub-command that produces results also writes an **HTML report**: one self-contained file with the command line, the configuration, the printed summary, interactive plots and the figures, which opens offline in any browser. It is written next to the output; `--html PATH` chooses another name and `--no-html` switches it off. `gui`, `worker`, `view`, `pulsation`, `applegate`, `mesa-grid` and `mesa-reduce` write no report.
- `-j` / `--processes` sets the number of parallel processes of the commands that can use several cores; `0` means all cores (of the job, when run by the worker).

Sub-command	Purpose	Main input	Main output
`run`	one simulation	configuration	summary, HDF5, CSV, PNG, 3D viewer, report
`view`	3D viewer of saved runs	HDF5 files or folders	HTML viewer
`stability`	stability criteria and the N-body ghost test	configuration	report
`timings`	eclipse times from the dynamics and their O−C	configuration	CSV, PNG
`minima`	minima measured on a synthetic light curve	configuration	CSV, PNG
`lightcurve`	synthetic light curve	configuration	CSV, PNG
`rv`	radial-velocity curves	configuration	CSV, PNG
`apsidal`	apsidal motion split into its parts	configuration	table
`pulsation`	periods of pulsating stars	configuration	printed table

Sub-command	Purpose	Main input	Main output
`fit`	fit of measured minima, radial velocities, light curves, inclinations	configuration with `fit`	parameters, residuals, fitted configuration, predictions
`ensemble`	Monte Carlo ensemble	configuration with `ensemble`	folder of rows and summary
`ensemble-report`	statistics of a finished ensemble, weighted by measurements	ensemble folder	report, figure
`emulator-train`, `emulator-report`	neural emulators of an ensemble	ensemble folder	network, posterior summary
`sbi`	simulation-based inference on an ensemble	ensemble folder	posterior samples, corner plot
`chaosmap`	MEGNO map over one or two parameters	configuration	map, table
`scatter`	few-body scattering experiments	scattering configuration	outcome table
`escape-theory`	statistical break-up of a three-body system	numbers	outcome distributions
`applegate`	can magnetic activity explain an O−C wave?	numbers	energy ratio
`validate`	benchmarks against published results	benchmark ids	verdicts, report
`mesa-grid`, `mesa-reduce`	own stellar-model grids with MESA	grid file	MESA runs, reduced grid
`worker`	runs job files from a folder	job folder	results of the jobs
`gui`	the web application	SWINGERsim folder	–

Table 17. All sub-commands. Columns: the name, what it is for, what it reads and what it writes. The sections below give the options and an example of each.

6.2 Simulations

6.2.1 `swingersim run`

Integrates one configuration over `run.t_max_yr`, prints a summary and, with `-o`, writes the run file and its standard products (chapter Outputs).

Option	Meaning
`config`	the configuration file
`-o`, `--output`	HDF5 output file; the products are written next to it
`--backend nbody` or `secular`	replace `run.backend` of the configuration
`--t-max-yr T`	replace `run.t_max_yr`; a negative value integrates into the past
`--no-viewer`	do not write the 3D viewer `<output>_viewer.html`
`--no-plots`	do not write the PNG figures `<output>_*.png`

Table 18. Options of `swingersim run`.

```
swingersim run configs/examples/example_2p2.yaml -o output/example.h5
swingersim run configs/examples/example_2p2.yaml --backend secular --t-max-yr -1000 -o
 ↪  output/past.h5
```

6.2.2 `swingersim view`

Writes the interactive 3D viewer of saved runs: one run, several, or every run in folders. `run -o` already writes it; `view` rebuilds it, e.g. with more animation frames.

Option	Meaning
`h5 …`	HDF5 files of `run`, or folders searched for them
`-o, --output`	HTML file of a single run (default `<run>_viewer.html`)
`--frames N`	number of animation frames (default 150)
`--force`	folders: rebuild every viewer (default: only missing ones and those older than their run)
`--cdn`	load plotly.js from the internet: a smaller file that needs a connection

Table 19. Options of `swingersim view`.

```
swingersim view output/example.h5 --frames 300
swingersim view output                # every run under output/ that has no up-to-date viewer
```

6.2.3 `swingersim stability`

Evaluates the stability of every nested triple of the hierarchy with the criteria of Mardling & Aarseth (2001) and Vynatheya et al. (2022), the neural classifiers of Vynatheya et al. (2023) for every sub-quadruple when they are installed, and runs the N-body ghost test of Vynatheya et al. (2023) (*Physics and Methods*, chapter Stability and fates).

Option	Meaning
`--orbits N`	outer orbits of the N-body test (default 100)
`--no-nbody`	skip the N-body test

Table 20. Options of `swingersim stability`.

```
swingersim stability configs/examples/example_2p2_zlk.yaml
```

6.3 Observables

6.3.1 `swingersim timings`

Finds every eclipse of the inner binaries in an N-body integration (the conjunction of the two stars of a pair, with the light-travel time of the pair's centre of mass) and writes the times and their O−C against a linear ephemeris fitted to the primary eclipses.

Option	Meaning
`--years Y`	end of the series [yr after $t = 0$] (required)
`--from-years Y0`	start of the series [yr, ≤ 0]; negative values add the eclipses before $t = 0$, integrated into the past
`--bjd-t0 BJD`	BJD of model time $t = 0$: adds the column `bjd_obs`

Option	Meaning
`-o, --output`	CSV file (required)
`--binaries …`	orbit names (default: all binaries of two stars)
`--samples N`	samples per binary period (default 40)
`--plot`	PNG of the O−C diagram

Table 21. Options of `swingersim timings`.

```
swingersim timings configs/examples/example_timings_czev4315.yaml --years 5 \
  -o output/timings_czev4315.csv --plot output/timings_czev4315.png
swingersim timings configs/squadra_test3/v907_sco_published.yaml --from-years -10 --years
↪  10 \
  --bjd-t0 2459360.0 -o output/v907_timings.csv
```

6.3.2 `swingersim lightcurve`

Computes the flux of all stars in a passband from the N-body positions: limb darkening, mutual eclipses of every pair of stars, ellipsoidal variation, Doppler beaming, reflection, the light-travel time, pulsations, rotating and Roche-shaped stars and dust discs as configured (*Physics and Methods*, chapter Observables).

Option	Meaning
`--days D`	length of the series [d] (required)
`--start-day S`	first time [d after $t = 0$] (default 0)
`--cadence-min C`	sampling [min] (default 2)
`--band B`	passband: `U`, `B`, `V`, `R`, `I`, `TESS`, `Gaia_G`, `Gaia_BP`, `Gaia_RP` (default `TESS`)
`--exposure-min E`, `--supersample N`	exposure time [min] and sub-exposures per point, for the smearing of the eclipses
`--noise-ppm N`, `--seed S`	white noise [ppm] and its random seed
`--ld FILE`	YAML file: star → four limb-darkening coefficients, instead of the tables
`--no-proximity`	no ellipsoidal variation, beaming and reflection
`--no-light-time`	ignore the light-travel time
`--engine swingersim` or `phoebe`	the light of the inner binaries from SWINGERsim's own model (default) or from PHOEBE 2
`--phoebe-atm`, `--phoebe-distortion`, `--phoebe-ntriangles`, `--no-irradiation`	PHOEBE settings: atmosphere (`auto`, `ck2004`, `blackbody`), shape (`roche`, `sphere`), mesh size (default 1500), no reflection
`-o, --output`	CSV file (required)
`--plot`	PNG of the light curve

Table 22. Options of `swingersim lightcurve`.

```
swingersim lightcurve configs/squadra_test3/v907_sco_published.yaml --days 30 --cadence-min
↪  10 \
  -o output/v907_lc.csv --plot output/v907_lc.png
```

6.3.3 `swingersim minima`

Computes a synthetic light curve and measures every minimum on it with the phenomenological eclipse profile of Mikulášek (2015), with an error, as an observer would measure real data.

It takes the light-curve options of Table 22 (`--band`, `--cadence-min`, `--noise-ppm`, `--seed`, `--no-proximity`, `--no-light-time`, the PHOEBE options) and:

Option	Meaning
`--years Y`	length of the series [yr] (required)
`-o, --output`	CSV file of the minima (required)
`--binaries …, --samples N`	orbit names; samples per binary period
`--window W`	width of the fitted window in eclipse durations (default 2.5)
`--season-days S, --seasons FILE`	fit one eclipse shape per season of this length and write the shapes to a CSV
`--blended`	measure on the total light (default: each binary on its own light plus third light)
`--bjd-t0 BJD`	BJD_TDB of $t = 0$ for the time column (default `fit.stage2.bjd_t0`, else J2000)
`--plot`	PNG of the O−C of the minima and of the conjunctions

Table 23. Options of `swingersim minima` beyond those of `lightcurve`.

```
swingersim minima configs/benchmarks/qzcar_minima.yaml --years 2 --binaries A \
  -o output/qzcar_minima.csv --plot output/qzcar_minima.png
```

6.3.4 `swingersim rv`

Radial-velocity curves of the stars from the N-body model, at the emission times, with measured velocities for comparison.

Option	Meaning
`--start, --end`	first and last time [BJD_TDB] (required)
`--bjd-t0`	BJD_TDB of $t = 0$ (default `fit.stage2.bjd_t0`)
`--step`	step of the curves [d] (default 0.01)
`--stars …`	stars (default all)
`--gamma`	systemic velocity [km/s] (default `fit.rv.gamma`)
`--measured FILE`	CSV of measured velocities (default `fit.rv.file`)
`-o, --output`	output folder (default `output/rv`)

Table 24. Options of `swingersim rv`.

```
swingersim rv configs/squadra_test3/v907_sco_fit_lc_v2.yaml --start 2459700 --end 2459710
 ↪  -o output/v907_rv
```

6.3.5 `swingersim apsidal`

The apsidal motion of every binary of two stars split into its contributions: the tidal and the rotational distortion of both stars after Sterne (1939) for synchronous, pseudo-synchronous and the configured rotation, and the relativistic advance; optionally the total advance measured in an N-body run, which adds the perturbations by the other stars; and the k_2 that a target apsidal period would need.

Option	Meaning
`--target A=194 B=98`	target apsidal periods [yr] per pair
`--nbody-years Y`	also run the N-body model for Y years and measure the total advance
`-o, --output`	CSV file of the table

Table 25. Options of `swingersim apsidal`.

```
swingersim apsidal configs/squadra_test3/v907_sco_published.yaml -o output/v907_apsidal.csv
```

The table lists per pair the period, eccentricity, the two k_2, the tidal advance, the rotational advance for the three rotation assumptions, the relativistic advance, the total in deg/yr and the apsidal period U in years.

6.3.6 `swingersim pulsation`

Prints for every star with `stars.<m>.pulsation` its type and mode, the period of the main mode, dP/dt from the evolving stellar model, its radius and $T_{\rm eff}$, and notes on validity.

```
swingersim pulsation configs/examples/example_pulsating_triple.yaml
# m1: delta_scuti (F)  P = 0.0711228 d  dP/dt = 5.35e-06 s/yr  R = 2.03 Rsun  Teff = 7931 K
# m3: lpv (F)  P = 334.992 d  dP/dt = 135 s/yr  R = 348 Rsun  Teff = 2867 K
```

6.4 Fits of measured data

6.4.1 `swingersim fit`

Fits the model to the measurements listed in the `fit` section: stage 1 is an analytic O−C model of every eclipsing pair, stage 2 the joint N-body model of the whole system fitted to all minima at once, optionally with radial velocities, light curves and inclinations (workflow in Workflows, keys in The configuration file).

Option	Meaning
`config`	configuration with a `fit` section
`-o, --output`	output folder (default `fit.output.dir`)
`--stage 1` or `2`	1: analytic O−C (default without `fit.stage2`); 2: joint N-body model (default with it)
`--processes N`	stage 2: cores for the derivatives (0 = all; default `fit.stage2.processes`)
`--sampler none` or `emcee`	replace the sampler of the stage that runs
`--anchor`	only anchor the period and phase of every eclipsing pair to the minima and write `anchored.yaml`
`--bjd-t0 BJD`	with `--anchor`: BJD_TDB of $t = 0$ (default `fit.stage2.bjd_t0`)
`--no-plots`	do not draw the figures

Table 26. Options of `swingersim fit`. `--sampler`, `--processes` and `--stage` do not apply to `--anchor`; `--processes` applies to stage 2 only.

```
swingersim fit configs/examples/example_fit.yaml                    # stage 1, seconds
swingersim fit configs/examples/example_fit.yaml --sampler emcee     # stage 1 with MCMC
swingersim fit configs/squadra_test/tic120362137_fit.yaml --processes 0 # stage 2
swingersim fit configs/squadra_test/bu_cmi_fit.yaml --anchor         # anchoring only
```

6.4.2 `swingersim applegate`

Can the magnetic activity cycle of a star explain a periodic O−C variation of its binary (Applegate 1992)? The command gives the largest O−C semi-amplitude the star can power over the modulation period in the thin-shell model of Tian et al. (2009) and, for an observed amplitude, the ratio of the energy needed to the energy available (below 1: possible).

Option	Meaning
`--mass`, `--radius`, `--teff`	the active star [$M_\odot$, $R_\odot$, K]
`--a`, `--pbin`	separation [$R_\odot$] and period [d] of the binary
`--pmod`	modulation period [yr]
`--oc`	observed O−C semi-amplitude [s] (optional)

Table 27. Options of `swingersim applegate`; all except `--oc` are required.

```
swingersim applegate --mass 0.93 --radius 0.96 --teff 5040 --a 3.3 --pbin 0.522 --pmod 30.5
↪  --oc 137.2
# largest O - C semi-amplitude the star can power over 30.5 yr: 1145 s (0.0132 d; ...)
# observed 137.2 s: dP/P = 1.79e-06, dE/E_sec = 0.0144 -> possible
```

The example is V471 Tau with the values of Völschow et al. (2016).

6.5 Ensembles and inference

6.5.1 `swingersim ensemble`

Draws the members described in the `ensemble` section, runs each forward and/or backward in parallel and writes one summary row per member and direction (workflow in Workflows, output in Outputs).

Option	Meaning
`-o`, `--output`	output folder (required)
`-j`, `--processes`	parallel processes (0 = all cores)
`--n N`	replace `ensemble.n`
`--backend`	replace `run.backend`
`--resume`	continue an interrupted ensemble (finished members are kept)
`--retry-errors`	with `--resume`: run again the members that ended with an error

Table 28. Options of `swingersim ensemble`.

```
swingersim ensemble configs/examples/example_v994_ensemble.yaml -o output/v994 --n 40 -j 0
swingersim ensemble configs/examples/example_v994_ensemble.yaml -o output/v994 --n 400 -j 0
↪  --resume
```

6.5.2 `swingersim ensemble-report`

Statistics of a finished ensemble: the fractions of the fates, the distribution of the largest eccentricities, mutual inclinations, eclipses, rates of change of the inclinations and the stability verdicts, separately forward and backward, with 68 % Wilson intervals; with observations, every statistic is weighted and the drawn parameters are summarised before and after the conditioning.

Option	Meaning
`directory`	the ensemble folder
`--observed FILE`	YAML file: summary column → `{value, sigma}` or limits; replaces `ensemble.observed`
`--prior`	ignore the observations (unconditioned report)
`--thresholds …`	report the fraction of members with $1 - e_{\max}$ below these values (default 0.1, 0.01, 0.001)
`--plot FILE`	an overview figure
`--x`, `--y`	drawn parameter and summary column of the last panel of the figure

Table 29. Options of `swingersim ensemble-report`.

```
swingersim ensemble-report output/v994 --plot output/v994/overview.png --x orbits.out.i --y
↪  B_didt0_deg_yr
```

6.5.3 `swingersim emulator-train` and `emulator-report`

`emulator-train` trains a neural emulator (a deep ensemble of multi-layer perceptrons of scikit-learn) that predicts summary columns of an ensemble from its drawn parameters, or a classifier of a yes/no column, and prints its scores on held-out members. `emulator-report` draws the emulator inputs from the distributions of `ensemble.vary`, weights them by `ensemble.observed` and prints the posterior of every input (median, 68 % and 95 % intervals) – with millions of draws at negligible cost.

Option	Meaning
`ensemble`, `-o`	the ensemble folder and the model file (`.pkl`)
`--inputs …`	input columns (drawn parameters)
`--targets …`, `--classes …`	regression columns; binary columns or `column==value` (e.g. `fate==intact`)
`--log-inputs`, `--log-targets`	use $\log_{10}$
`--angle-inputs`	angles in degrees, used as cos and sin
`--scale-targets target=input^power`	fit the target divided by a known power law
`--hidden`, `--models`, `--activation`, `--alpha`, `--learning-rate`, `--batch-size`, `--max-iter`, `--test-fraction`, `--seed`	network settings (defaults: four layers of 50 units, 5 networks, ReLU, L2 10^{-4}, rate 10^{-3}, 2000 iterations, 20 % held out)
`--plot`	PNG of the training and held-out diagnostics
`emulator-report model config`, `--n`, `--seed`	the model, the ensemble configuration, prior draws (default 10^6)

Table 30. Options of the emulator commands.

```
swingersim emulator-train output/v994 -o output/v994/emulator.pkl \
  --inputs orbits.out.i orbits.out.Omega orbits.B.Omega \
  --angle-inputs orbits.out.i orbits.out.Omega orbits.B.Omega \
  --targets B_didt0_deg_yr --plot output/v994/emulator.png
swingersim emulator-report output/v994/emulator.pkl my_v994_with_observed.yaml
```

The inputs must be columns that the ensemble drew (`ensemble.vary`); `emulator-report` needs a configuration whose `ensemble` section also has `observed` entries. An emulator is only as good as its training set: trained on the 40 members of the first `swingersim ensemble` example above,

the held-out score of this example is $R^2 = -0.25$, i.e. worse than predicting the mean; it needs thousands of members.

6.5.4 `swingersim sbi`

Simulation-based inference: a neural posterior estimator (the package `sbi`) is trained on a finished ensemble and returns the posterior of chosen parameters for measured values; a trained model is reused for new measurements with the same errors.

Option	Meaning
`directory`	the ensemble folder (training set)
`--observed FILE`	YAML: column → `{value, sigma}` and/or `{min, max}`
`--params …, --log …`	parameters to infer; those inferred in $\log_{10}$
`--model FILE, --retrain`	trained model file (reused if it matches); train anew
`--epochs, -n`	maximum training epochs (2000); posterior samples (20 000)
`--save-samples, --plot`	CSV of the samples; PNG corner plot
`--coverage, --diagnostics`	calibration test on 20 % held-out members; PNG of loss, coverage and calibration

Table 31. Options of `swingersim sbi` (needs the `sbi` group, Table 8).

```
cat > output/v994/observed.yaml <<'EOF'
B_didt0_deg_yr: {value: 0.05, sigma: 0.02}
EOF
swingersim sbi output/v994 --observed output/v994/observed.yaml \
  --params orbits.out.i orbits.out.Omega --model output/v994/sbi_model.pt \
  --plot output/v994/corner.png --coverage
```

The observed value in this example is illustrative; a posterior is only as good as the number of members the network was trained on (*Physics and Methods*, chapter Ensembles and inference).

6.6 Stability, chaos and scattering

6.6.1 `swingersim chaosmap`

Runs one N-body integration per point of a grid of one or two configuration parameters and computes the chaos indicator MEGNO (Mean Exponential Growth factor of Nearby Orbits; 2 for regular motion, larger for chaos).

Option	Meaning
`-o, --output`	output folder (required)
`--x PATH, --x-range MIN MAX, --nx N, --log-x`	first parameter (e.g. `orbits.out.P`), its range, points (30), logarithmic spacing
`--y, --y-range, --ny, --log-y`	optional second parameter
`--orbits N`	length in periods of the widest orbit (default 1000)
`--t-max T`	length in years (replaces `--orbits`)
`--max-wall S`	seconds per grid point (default 300)
`-j, --resume`	parallel processes; continue an interrupted map

Table 32. Options of `swingersim chaosmap`.

```
swingersim chaosmap configs/validation/quarles18_circumbinary.yaml -o output/chaosmap_cb \
  --x orbits.out.a --x-range 1.8 4.0 --nx 23 --y orbits.out.e --y-range 0 0.5 --ny 11
  ↪  --t-max 3000 -j 0
```

6.6.2 `swingersim scatter`

Few-body scattering experiments: a binary or a single star meets another binary or single star; the encounters are integrated until the outcome is clear (flyby, exchange, ionisation, a bound triple or quadruple, a collision). The configuration has its own section `scatter` (The configuration file).

Option	Meaning
`-o`, `--output`	output folder (required)
`-j`, `--processes`, `--n`, `--resume`	parallel processes, number of encounters (replaces `scatter.n`), continue

Table 33. Options of `swingersim scatter`.

```
swingersim scatter configs/examples/example_scatter_2p2.yaml -o output/scatter_2p2 --n 200
 ↪  -j 0
```

6.6.3 `swingersim escape-theory`

The statistical theory of the break-up of a non-hierarchical three-body system (Stone & Leigh 2019): which star escapes and the eccentricity of the binary left behind, for given masses, binary, velocity at infinity and impact parameter (or angular momentum); optionally compared with a `scatter` folder of binary–single runs.

```
swingersim escape-theory --masses 1 1 1 --a 1 --v-inf 1 --b 2
```

Options: `--masses m1 m2 m3` (binary star 1, binary star 2, single star, $M_\odot$), `--a` [au], `--e`, `--v-inf` [km/s], `--b` [au] or `--L0-ratio`, `--alpha`, `--criterion AE|SE`, `--compare FOLDER`.

6.7 Validation and own stellar models

6.7.1 `swingersim validate`

Reproduces published results and prints PASS, EXPLAINED or FAIL for each benchmark; `--all` includes the slow ones (minutes each); benchmark ids select some. The report is written to `output/validation/`. The benchmarks are the subject of *Testing, Part I*.

```
swingersim validate           # the fast benchmarks
swingersim validate --all     # all of them
```

6.7.2 `swingersim mesa-grid` and `mesa-reduce`

Compute an own grid of stellar models with MESA from a grid file (`configs/mesa/*.yaml`) and reduce its profiles to the quantities SWINGERsim needs (Installation). Options: `--root` (the SWINGERsim folder), `-j` (parallel MESA runs), `--test` (one 1 $M_\odot$ track only).

6.8 Running and controlling

6.8.1 `swingersim worker`

Runs job files dropped into a folder (Installation, Workflows).

Option	Meaning
`jobdir`	job folder (default `<root>/jobs`)
`--root`	the SWINGERsim folder (default `.`)
`--poll S`	seconds between checks (default 10)
`--once`	run the waiting jobs and stop
`--cores N`	cores the jobs may use together (default all)

Table 34. Options of `swingersim worker`.

6.8.2 `swingersim gui`

Starts the web application; its options are in Table 11.

Chapter 7

Workflows

This chapter describes the typical tasks from start to finish. Each workflow begins with a table of its inputs, steps and outputs and ends with a short worked example. The commands are those of chapter The command line; every step can also be done in the web application (chapter The web application).

7.1 Forward model of a published system

Question. A paper gives the periods, eclipses, masses and perhaps an outer orbit of a multiple system. What does this system do, and what will observers see next?

Inputs	Steps	Outputs
published masses, radii, temperatures; elements of every orbit; the epoch they refer to; measured apsidal motion or inclination changes for comparison	1. write the configuration; 2. check stability; 3. run with N-body and, for long times, secular; 4. compute eclipse times and a light curve; 5. compare with the measurements	run files, eclipse windows, O−C diagram, light curve, stability verdict; the differences from the measurements

Table 35. The forward-model workflow. The left column lists what you need, the middle the steps, the right what you get.

1. **Collect the parameters** with their sources: inner periods, eccentricities, inclinations, radii (from light-curve solutions), masses (spectroscopic if available), the outer orbit (from the LTTE, radial velocities or astrometry), apsidal periods and measured rates of change. Note the epoch of the elements: it becomes $t = 0$.
2. **Write the configuration.** Start from an example in `configs/` (a catalog of published systems is in `configs/catalog/`). Mark every value that is not measured with a comment (`# ASSUMED`, `# ESTIMATED`). Give the radii (`stars.<m>.R`) when they are known; otherwise zero-age main-sequence radii are used, which are too small for evolved stars. Unknown nodes: keep `Omega = 0` for the eclipsing pair and set the mutual inclination through `i` and `Omega` of the other orbits. For close binaries switch on `physics.gr`, `tides` and `rotation`; they set the apsidal motion and can suppress ZLK cycles.
3. **Check stability**: `swingersim stability config.yaml`. An unstable nominal system means that a value is wrong or the system is not what was assumed.
4. **Run** with the N-body backend over a few precession periods and look at the result page or the viewer. For evolution over millions of years see below.
5. **What would observers see?** `swingersim timings` gives the O−C of every binary with the light-travel time and the dynamical effects; `--bjd-t0` gives each minimum its BJD, and a negative `--from-years` adds the minima before the epoch of the elements. `swingersim`

`lightcurve` gives the light curve in a passband. If measured minima disagree with the model, fit them (next section).

Worked example. V907 Sco is a 3.78-d eclipsing binary with a third star on a 142-d orbit; its eclipses stopped and restarted during the 20th century. The configuration `configs/squadra_test3/v907_sco_published.yaml` takes the solution of Zasche et al. (2023):

```
swingersim stability configs/squadra_test3/v907_sco_published.yaml
swingersim run configs/squadra_test3/v907_sco_published.yaml -o output/v907/run.h5
swingersim timings configs/squadra_test3/v907_sco_published.yaml --from-years -100 --years
↪  100 \
  -o output/v907/timings.csv
```

The stability check gives ratios of 1.67 (Mardling–Aarseth) and 2.2 (Vynatheya et al. 2022) and a stable ghost test over 100 outer orbits. The 200-yr run (27 s) shows a mutual inclination of 25.6–26.3°, a nodal precession period of 63 yr and eclipses of the inner pair during 50.5 % of the time, in windows of about 30 years (Figures 9–11); the eclipse times show the same windows (Figure 13). The result pages of these three jobs are the ones shown in chapter The web application.

7.2 Fitting measured data

Question. Which parameters of the system reproduce the measured times of minima of all eclipsing pairs – and, if available, radial velocities, light curves and inclinations measured at different epochs?

Inputs	Steps	Outputs
a configuration with start values; a table of minima; optionally radial velocities, light-curve segments, inclinations; the BJD of $t = 0$; priors from published solutions	1. prepare the tables; 2. stage 1 (analytic O−C of each pair); 3. anchoring; 4. stage 2 (joint N-body model) with error scaling, outlier rejection, priors, red noise; 5. check residuals and warnings; 6. MCMC for reliable errors; 7. predictions and the fitted configuration	fitted parameters with errors and correlations, residuals of every measurement, predicted minima with their uncertainty, `stage2_fitted.yaml`

Table 36. The fitting workflow (“the fitting chain”). The steps are described below; every option is a key of the `fit` section (The configuration file).

7.2.1 The data

The **times of minima** are a CSV table with one row per minimum and the columns `pair` (the name of the eclipsing orbit), `type` (`p` primary, `s` secondary), `time`, `sigma` (both in days), `source` and `method`. The column names of the measured-minima files of the web application (`binary`, `kind`, `bjd_tdb`, `sigma_d`, `ref`) are accepted as well. Times are BJD_TDB; `fit.time_system: HJD_UTC` or `JD_UTC` converts them (with `fit.coordinates`). A *primary* minimum is an eclipse of the first star of the pair (`m1` of `inner`), whatever its depth; if the observer’s primary is the other star, swap the types or the stars.

Radial velocities (`fit.rv`): columns `star`, `time` (BJD_TDB), `rv` and `sigma` (km/s) and `source`. Each source can have its own zero point (`fit.rv.offsets`, free as `rv.offset.<source>`); the systemic velocity is `rv.gamma`. For semi-detached and contact binaries `centre: light` uses the flux-weighted velocity of the visible surface, including the Rossiter–McLaughlin effect in eclipse, instead of the velocity of the centre of mass.

Light curves (`fit.lc`): columns `time`, `flux`, `sigma` and optionally `segment`. Each segment gets its own scale, zero point and polynomial trend, solved linearly at every evaluation; with `scale: false` the depths follow from the stars as given, so the light curve constrains the inclination and radii.

Inclinations at epochs (`fit.inclination`): columns `mid_hjd` (or `time`), `i_deg`, `sigma_deg` and optionally `limit` and `source`, e.g. inclinations from light-curve solutions of archival data. Only $|\cos i|$ is compared, because eclipses do not tell the sense of revolution. The column `limit` makes a row one-sided: `upper`, `lower`, `no_eclipse` (the model must be below its eclipse limit then) or `eclipse` (it must be above).

7.2.2 Stage 1: the analytic O−C of each pair

Stage 1 describes the O−C of every pair with analytic terms: the light-time effect of an outer orbit (Irwin 1952), apsidal motion of an eccentric pair, and a quadratic term (`fit.stage1.terms` or per pair `fit.stage1.pairs`). With `shared_outer: true` all pairs of a 2+2 share one outer orbit (common period, epoch and eccentricity, opposite pericentres). It is fast (seconds) and gives start values and a first look: the outer period, the amplitude and mass function, the apsidal period. `sampler: emcee` adds an MCMC run (Foreman-Mackey et al. 2013).

7.2.3 Anchoring

Stage 2 converges only from a start that numbers the minima correctly: an error of the inner period accumulated over the span of the data must stay well below half a period. Published elements from a photodynamical solution often drift in an N-body model with slightly different physics. *Anchoring* adjusts only the period P and the mean anomaly M_0 of every eclipsing pair so that the model minima fall on the measured ones; `swingersim fit --anchor` writes the result as `anchored.yaml`, and `fit.stage2.anchor: true` does it automatically before the fit.

7.2.4 Stage 2: the joint N-body model

Stage 2 computes the times of all minima of all pairs from one N-body integration of the whole system – the same physics as `run` and `timings` – forward and backward from $t = 0$ (the BJD `fit.stage2.bjd_t0`), and fits the parameters listed in `fit.stage2.free` by weighted least squares. Light-time effect, apsidal motion and dynamical perturbations are in the model at once; no term has to be chosen. Free parameters are elements of orbits (`orbits.<o>.P`, `M0`, `e`, `omega`, `Omega`, `i`, or `ecosw` and `esinw` for nearly circular orbits, and `Pdot`), masses (`masses.<m>`), properties of stars (`stars.<m>.R`, `Teff`, `apsidal_k2`, `rg2`) and the radial-velocity zero points (`rv.gamma`, `rv.offset.<source>`). The derivatives are computed by finite differences, two N-body integrations per free parameter and iteration, in parallel on `processes` cores. The method is described in *Physics and Methods*, chapter Fitting observed data.

What the minima constrain: the periods, phases, eccentricities and arguments of pericentre of all orbits and, through the light-time amplitude and the dynamical perturbations, the masses of the components relative to each other. The inclinations of the eclipsing pairs and the absolute masses need light curves and radial velocities; keep them fixed at the values of such a solution, fit them together with those data, or give them priors. Without astrometry only the difference of the nodes of two orbits is measurable: keep the node of one orbit fixed.

Errors and jitter. Minima from many sources rarely have realistic errors. `error_scale: per_source` gives every source (column `source`) its own correction so that its reduced χ^2 is 1, never below its quoted errors; `global` uses one factor for all. The correction is a factor on the errors (`error_model: scale`) or an error floor added in quadrature (`error_model: jitter`), suited to precise timings with optimistic errors. Sources with fewer than `min_source_minima` minima share one correction. The fit is then repeated from its solution until the corrections settle, at most `refits` times.

Outliers. `clip_sigma: 5` removes, after each fit, the minima that deviate by more than five robust standard deviations of their source (at most `max_clip_fraction` of them) and fits again; they stay in the residual table with the status `outlier`. Alternatively a robust loss (`loss: soft_l1`, `huber` or `cauchy`, from `f_scale` σ on) reduces their weight without removing them.

Priors. `priors: {orbits.out.e: [0.47, 0.05]}` adds a Gaussian term $((x-\mu)/\sigma)^2$ to χ^2 for a free parameter, e.g. from a published light-curve or radial-velocity solution. Priors keep weakly constrained parameters (an outer orbit seen through one eclipsing pair, a light-time orbit covered once) from wandering to solutions that fit the minima equally well but contradict other data. The report gives the deviation of every prior in σ; several σ mean a real tension between the data and the prior.

Red noise. Residuals of a light curve are often correlated (instrumental trends, spots, oscillations), so the formal errors are too small. `red_noise: true` on a light-curve entry multiplies the errors of each segment by the factor β of the time-averaging method (Pont et al. 2006; Winn et al. 2008): how much more the residuals averaged over 20–120 min scatter than white noise would. It is applied between the fits in a row, so it needs `error_scale` or `clip_sigma`.

MCMC. The errors from the covariance matrix assume a straight valley of χ^2. When the report lists strongly correlated parameters ($|r| > 0.95$), `sampler: emcee` runs an affine-invariant MCMC (Goodman & Weare 2010; Foreman-Mackey et al. 2013) around the solution with `walkers` × `mcmc_steps` model evaluations; the first half is discarded. With `mcmc_tau_target: 50` the chain is extended until it is at least 50 integrated autocorrelation times long, as the emcee authors recommend, or reaches `mcmc_max_steps`. The fit page shows the ratio of steps to autocorrelation time.

Predictions. `fit.output.predict: {years: 5}` writes the next minima of every pair with their uncertainty: parameter sets are drawn from the MCMC samples or the covariance matrix and each is integrated, so the errors include the correlations of the parameters.

Checking the result. Read, in this order: the warnings; χ^2 from the start to the fitted model and the reduced χ^2; the rms of every data type; the mean residual of primary and secondary minima (a difference of several σ means a wrong $e\cos\omega$); systematic patterns in the residuals (a parameter that should be free is fixed, or the fit sits in a local minimum); parameters on their bounds; deviations from priors.

Worked example. The fit of V907 Sco in *Testing, Part III* combines 58 TESS minima, 172 radial velocities from two sources, 2562 points of two TESS light-curve segments and 14 inclinations measured between 1900 and 2022, with 16 free parameters and 3 priors:

```
swingersim fit configs/squadra_test3/v907_sco_fit_lc_v2.yaml
```

Its configuration anchors the inner pair (`anchor: true`), corrects the errors per source with a jitter, removes outliers beyond 5σ and applies red-noise factors to the light curves. The fit lowers χ^2 from 11 920 to 1052 in 4384 model evaluations; the rms of the minima is 2.6 min, of the radial velocities 3.85 km/s and of the light curves 0.666 ppt (Figures 19–23).

7.3 Ensembles: what we do not know

Question. Some parameters of the system are unknown (an unseen companion, the orientation of an outer orbit, the sense of revolution). Which values are possible, what do the measurements allow, and what follows for the future and the past?

Inputs	Steps	Outputs
the nominal configuration; distributions of the unknown parameters (`ensemble.vary`); measured values or limits (`ensemble.observed`); stability requirements	1. write the `ensemble` section; 2. run a few hundred members locally, more on the worker; 3. condition on the measurements; 4. read n_{eff}, the conditioned distributions and the outcomes; 5. open the best members as configurations	`summary.csv` (one row per member and direction), the report, the weighted distributions and fractions, best-fitting members

Table 37. The ensemble workflow.

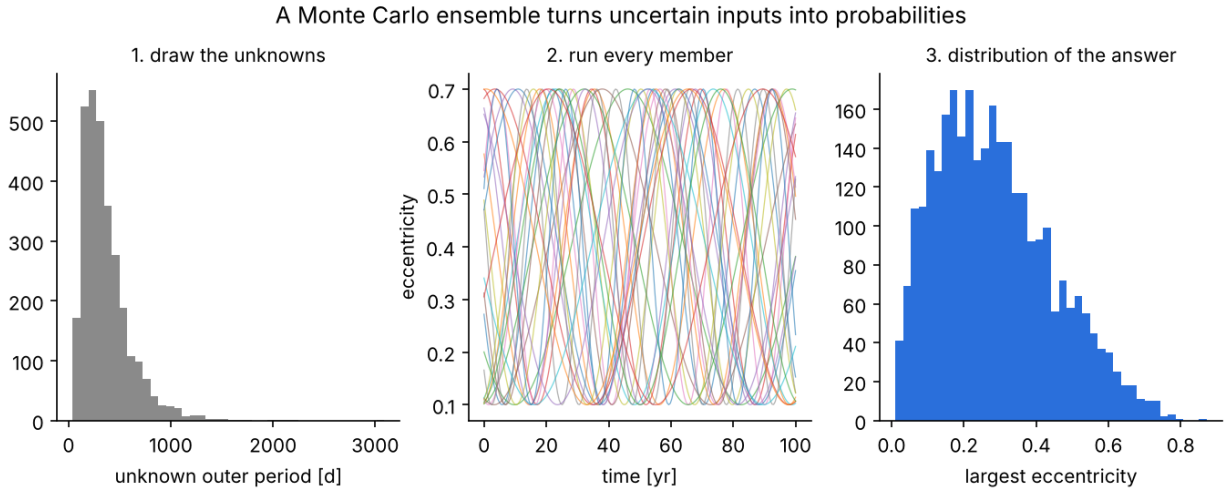


Figure 25. The idea of an ensemble, schematic. Left: the unknown parameter (here an outer period in days) is drawn many times from its distribution (histogram of the draws). Middle: every member is integrated; the curves are the eccentricity of the inner binary against time in years for a few members. Right: the distribution of the answer over all members (here the largest eccentricity reached). The weighting by measurements described below turns the left histogram into a posterior and the right one into a conditioned prediction.

1. **What we do not know** (Figure 25). Draw only what is unknown; keep measured values at their values or draw them within their errors (`normal`). Typical choices: an unknown orientation is `isotropic` (uniform in $\cos i$); an unknown node or phase is `angle` (uniform 0–360°); a scale parameter such as an outer period or a companion mass is `loguniform`; an eclipsing pair whose sense of revolution is unknown gets `mirror: true` (i or $180° - i$). `require: [stable_ma01]` redraws members that a stability criterion classifies as unstable. Write down every choice: the result depends on it.

2. **Run** forward and backward (`directions: [forward, backward]`): the rates of change "now" are best measured from both sides of $t = 0$, and the past is half of the answer.

3. **Observation weights.** Every entry of `ensemble.observed` is a summary column of the ensemble with either a measurement `{value, sigma}`, limits `{min, max}`, or a required text value `{equals: intact}`. A member gets the weight $w = \exp(-\chi^2/2)$ with $\chi^2 = \sum((x - \text{value})/\sigma)^2$, and $w = 0$ when it violates a limit; the weighted members approximate the posterior distribution of the drawn parameters (importance weighting with the prior as the proposal). Present rates of change are by default *central* – the mean of the forward and backward fits – as an observer measures them; prefix a column with `forward:`, `backward:` or `central:` to choose.

4. **How much is left.** The effective number of members $n_{\text{eff}} = (\sum w)^2 / \sum w^2$ (Kish 1965) says how many equally weighted members the weighted ensemble is worth. Below about 20 the conditioned distributions are noise: run more members, or narrow the drawn ranges to the region the measurements allow.

5. **Evidence.** The mean weight of the members, $Z = \langle w \rangle$, is the probability of the measurements under the hypothesis of the ensemble (the evidence). The ratio of Z of two ensembles compares two hypotheses (e.g. 2+2 against 2+(2+1)) – but only when the shared parameters are drawn from the same priors; otherwise the ratio measures the priors. `ensemble-report` prints Z with its Monte Carlo error $\text{std}(w)/\sqrt{n}$ next to n_{eff}; when fewer than two members match, it asks you to quote the upper bound $Z < 1/n$. The same numbers in a few lines of Python:

```
import numpy as np
from swingersim.config import load_config
from swingersim.ensemble import load_rows, observation_weights
```

```
cfg = load_config("configs/squadra_test3/qx_cas_ensemble.yaml")
rows = load_rows("output/squadra_test3/qx_cas/ensemble")
w = np.array(list(observation_weights(rows, cfg.ensemble.observed).values()))
print("evidence Z =", w.mean(), " n_eff =", w.sum() ** 2 / (w ** 2).sum())
```

6. **Use the members.** On the result page, the members with the largest weight can be opened as configurations and run with a light curve or eclipse times. For a fast posterior of new measurements, train an emulator or a neural posterior on the ensemble (`emulator-train`, `sbi`).

Worked example. The eclipsing binary QX Cas (6.0 d) changes its inclination by −0.23 ± 0.08 °/yr; the cause is an unseen third body. `configs/squadra_test3/qx_cas_ensemble.yaml` draws its mass (0.3–6 $M_\odot$, log-uniform), its period (30–11 000 d, log-uniform), eccentricity (0–0.6) and the orientation of its orbit, runs 2000 members 30 years forward and backward, and weights them by the present inclination (82 ± 1.5°), its central rate and the survival of the system both ways:

```
swingersim ensemble configs/squadra_test3/qx_cas_ensemble.yaml -o output/qx_cas_ensemble -j
↪ 0
swingersim ensemble-report output/qx_cas_ensemble
```

The conditioned ensemble has $n_{\mathrm{eff}} = 186.7$; it narrows the outer period from the drawn 118–4429 d (16–84 %) to 193–1028 d, leaves the mass and the angles almost unconstrained, and predicts that in 19.5 % of the systems that eclipse now the eclipses end within 30 years (Figures 15–18).

7.4 Evolution over millions of years

Question. How do the stars and orbits change over the life of the system: when does a star fill its Roche lobe, and what follows?

Inputs	Steps	Outputs
masses, orbits, the age and metallicity of the system; the physics of the stars and their interaction	1. choose the stellar model and age; 2. check the stars with a short run; 3. switch on evolution, tides, winds, mass transfer; 4. run with the secular backend forward or backward; 5. read the events and compare light curves at different ages	runs with evolving stars (`_stars.png`), events of the mass-transfer history, mergers, the system at later ages

Table 38. The evolution workflow.

1. **Stellar model.** `stellar_model: {kind: mist, age_myr: …, feh: …}` uses the MIST tracks from the pre-main sequence to the end of the asymptotic giant branch (AGB); `kind: sse` the faster SSE formulae up to helium ignition. All stars are coeval; `stars.<m>.age_offset_myr` gives a star an apparent age (a stripped donor looks older, a rejuvenated gainer younger). A short run (small `t_max_yr`) lists T_{eff}, radius, phase and age of every star in the summary.
2. **Physics.** `physics.evolution: true` lets the stars follow their tracks during the run; `tides: true` with `tidal_lag: hurley2002` takes the tidal dissipation from the envelope of every star; `mass_loss` adds winds; `mass_transfer` follows Roche-lobe overflow (stable transfer, contact binaries, common envelopes, transfer from a tertiary); `mergers` continues a run after two stars merge; `magnetic_braking` spins down stars with convective envelopes. The models are described in *Physics and Methods*, chapters Stellar models and evolution, Tides and rotation, and Mass transfer, common envelope and mergers.

3. **Run** with `run.backend: secular` (the N-body backend cannot follow millions of years of a close binary); a negative `t_max_yr` integrates back towards the birth of the stars. Without `mass_transfer` the run stops when a star fills its Roche lobe; it also stops at the end of the tracks, and the summary says why (`EVENT`).
4. **Look at the system at a later age**: a configuration with the later age and the orbits at that time gives its light curve and eclipse times.

Keep in mind the systematic uncertainty of the stellar tracks; when a conclusion depends on the age, draw `stellar_model.age_myr` in an ensemble. The worked example is in Worked examples.

7.5 Stability, chaos maps and scattering

Question. Is the system stable? Where in the space of its unknown parameters is it stable or chaotic? Can it have formed by a capture?

Inputs	Steps	Outputs
a configuration; one or two parameters and their ranges; for scattering, the two colliding systems	1. `stability` (seconds); 2. a long `run`; 3. `chaosmap` over the uncertain parameters; 4. `scatter` and `escape-theory` for formation	stability ratios and verdicts, MEGNO maps with the stability limit, outcome fractions and cross-sections

Table 39. The stability workflow.

1. **Criteria.** `swingersim stability` applies the criterion of Mardling & Aarseth (2001) and its update by Vynatheya et al. (2022) to every nested triple (an orbit and the body that perturbs it); a ratio above 1 means stable. The N-body ghost test integrates the system for a number of outer orbits and compares it with a slightly displaced copy (Vynatheya et al. 2023). The criteria are approximate near 1; a long run decides.
2. **Chaos maps.** `swingersim chaosmap` integrates a grid of configurations and computes MEGNO (2 for regular motion, growing for chaos); escapes and collisions are marked. The map shows where the system sits relative to unstable regions and resonances.
3. **Scattering.** `swingersim scatter` sends a binary towards another binary or a single star from a range of impact parameters and records the outcomes; `escape-theory` gives the statistical prediction of Stone & Leigh (2019) for comparison.

Worked example. The chaos map of a circumbinary orbit is in Worked examples, a binary–binary scattering experiment in Worked examples.

7.6 Long computations on the worker

Ensembles of 10^4 members, N-body ensembles, stage-2 fits with MCMC and MESA grids take hours to days. They run as **jobs** on the worker (Installation). A job is a YAML file in `jobs/inbox/`:

```yaml
command: ensemble                      # any sub-command except gui and worker
config: configs/squadra_test3/qx_cas_ensemble.yaml
output: output/qx_cas_ensemble
args: {n: 2000, processes: 0, resume: true}
note: QX Cas, unseen third body, 2000 members forward and backward
```

Key	Meaning
`command`	the sub-command

Key	Meaning
`config` (and other positional arguments by their name: `ensemble`, `directory`, `h5`, `ids`, `model`)	the positional arguments of the command
`output`	the output argument (`-o`, or the folder of `ensemble-report` and `view`)
`args`	the options by their long name, with `-` written as `_` (`retry_errors: true`, `t_max_yr: 3000`)
`cores`	cores the job occupies (a number or `all`); default from `processes`
`note`	free text, shown in the web application

Table 40. Keys of a job file. Every path must lie inside the SWINGERsim folder; the worker runs only SWINGERsim commands with the arguments their parser allows.

Rules that matter in practice:

- One-core jobs (`run`, `lightcurve`, `timings`, `minima`, a stage-1 `fit`) run side by side up to the number of cores; a job that needs all cores runs alone. Every job receives the number of cores it was booked for (environment variable `SWINGERSIM_CORES`), so `processes: 0` means the job's cores, not the whole computer.
- Use `resume: true` for ensembles, scattering and chaos maps: an interrupted job continues where it stopped. A job interrupted by stopping the worker or by a restart of the computer goes back to the inbox automatically. `retry_errors: true` runs again only the members that ended with an error.
- A file with another suffix (e.g. `.yaml.hold`) stays in the inbox without being run.
- When the free memory falls below 8 % of the total (at least 1.5 GB), the worker stops the job that uses the most memory and marks it failed with an explanation; its results so far stay.
- Restart the worker after every update of SWINGERsim.

7.7 Reproducibility

Every result can be traced back to its input and recomputed:

- the HDF5 file of a run stores the complete configuration (with every default filled in), the versions of SWINGERsim, Python, NumPy, SciPy, h5py, REBOUND, REBOUNDx and emcee, and the time of its creation; the CSV and fit tables carry the same provenance in their header lines;
- an ensemble folder stores the configuration and a fingerprint of it; `--resume` refuses to mix rows of a changed configuration, and the drawn members are fixed by `ensemble.seed`;
- a fit writes the fitted configuration with its epoch (`fit_epoch_bjd_tdb`), the history of the optimisation and, with MCMC, the samples (`seed` fixes the walkers);
- the worker keeps every job file, its log and its status in `jobs/done/` or `jobs/failed/`, and the web application keeps a frozen copy of every submitted configuration in `jobs/submitted/`.

To share a result, share the configuration, the data tables it refers to and the version of SWINGERsim; with the seeds in the configuration anyone can recompute it. The *Methods paragraph* of a result page writes the description of the computation for a paper.

Chapter 8

Worked examples

The examples of this chapter go from one simple system to ensembles, fits, scattering and evolving stars. The configurations are in `configs/examples/` (the chaos map in `configs/validation/`); the parameters of examples 1–3 are illustrative, not those of a real system. The numbers quoted are those printed by the commands; computing times are for one core of a desktop computer.

8.1 Example 1: a precessing 2+2 quadruple

The configuration `configs/examples/example_2p2.yaml` is the system of the Quick start: binary A (2.2 $M_\odot$, 6.74 d) is nearly edge-on to us ($i = 85°$), binary B (1.7 $M_\odot$, 2.68 d) is inclined 60° to the sky, and the two orbit each other every 500 d on an eccentric orbit ($e = 0.3$). From the angles, the mutual inclinations are about 18° between `out` and A and 29° between `out` and B.

```
swingersim run configs/examples/example_2p2.yaml -o output/example_2p2.h5
```

Line of the summary	Value	Meaning
`out a, out e`	1.940–1.942 au; 0.2995–0.3007	the 500-d period with 3.9 $M_\odot$ gives $a_{\rm out} = 1.94$ au (Kepler's third law); the 0.1 % spread is the osculating wiggle, not a real change
`A a, A e`	0.0908 au; 0.044–0.054	19.5 $R_\odot$, constant to 0.07 %; a small oscillation of e forced by the outer orbit; no ZLK because 18° is below 39.2°
`i_mut out-A, out-B`	16.3–18.1°; 28.5–32.9°	each binary precesses around the outer orbit keeping its tilt
`i_mut A-B`	25.6–46.7°	A and B precess at different rates, so the angle between their planes changes
max |dE/E|	3.2×10^{-13}	the integration is accurate
`precession period`	283 yr (A), 598 yr (B) estimated	the 200-yr run is shorter than one cycle; a 1000-yr run measures 255 yr and 590 yr
`stability ratio`	3.67 (Mardling–Aarseth), 4.52 (Vynatheya+22) for A; 6.80 and 8.46 for B	comfortably stable
`binaries A and B`	$P_A/P_B = 2.515$, nearest 5:2 (+0.6 %)	the periods are not in resonance
`eclipses`	A 2 %, B 43 %, both 0 %	in these 200 years A eclipses only briefly at the start, B for almost half of the time

Line of the summary	Value	Meaning
`binary A`	apsidal period 25 660 yr from the binary itself, 270 yr measured	without tides only relativity turns the periastron of an isolated binary; in the quadruple the outer orbit makes it circulate a hundred times faster
wall time	80 s	it scales with the shortest period and the length of the run

Table 41. The summary of example 1, line by line. Left: the line of the summary; middle: the printed value; right: its meaning.

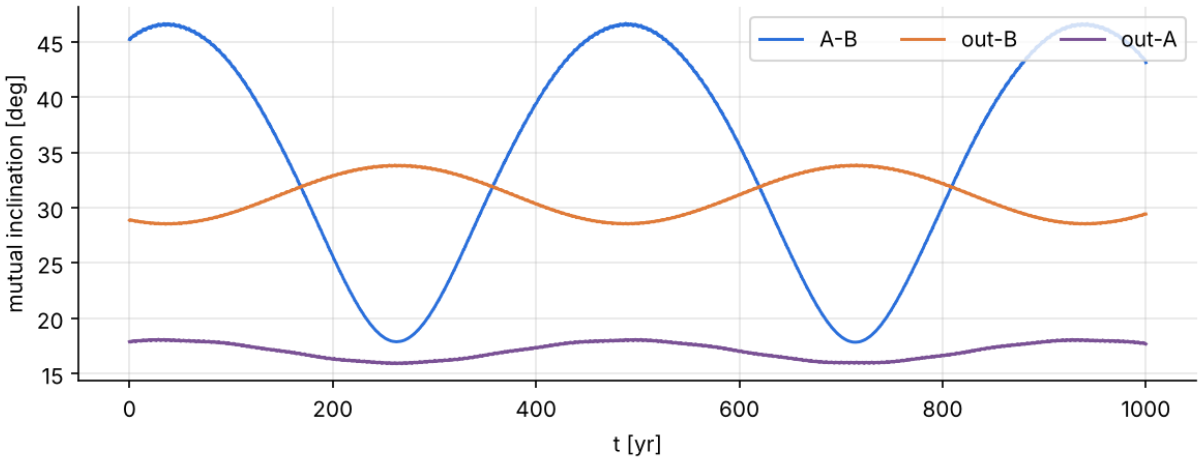


Figure 26. Mutual inclinations of the three pairs of orbits of example 1 over 1000 years (`run.t_max_yr: 1000`). Horizontal axis: time in years; vertical axis: mutual inclination in degrees. Blue: A–B; orange: out–B; violet: out–A. The angle between A and B oscillates between 17.8° and 46.7° with a period of about 450 years; the angle of B to the outer orbit oscillates between about 29° and 34° and that of A between about 16° and 18°, in the same rhythm.

The 450-year cycle of the angle between A and B (Figure 26) is the beat of the two precession periods, $1/(1/255 - 1/590) \approx 450$ yr: the two planes are alternately on the same side and on opposite sides of the outer orbit, so the angle between them swings between the difference and the sum of their tilts to the outer orbit.

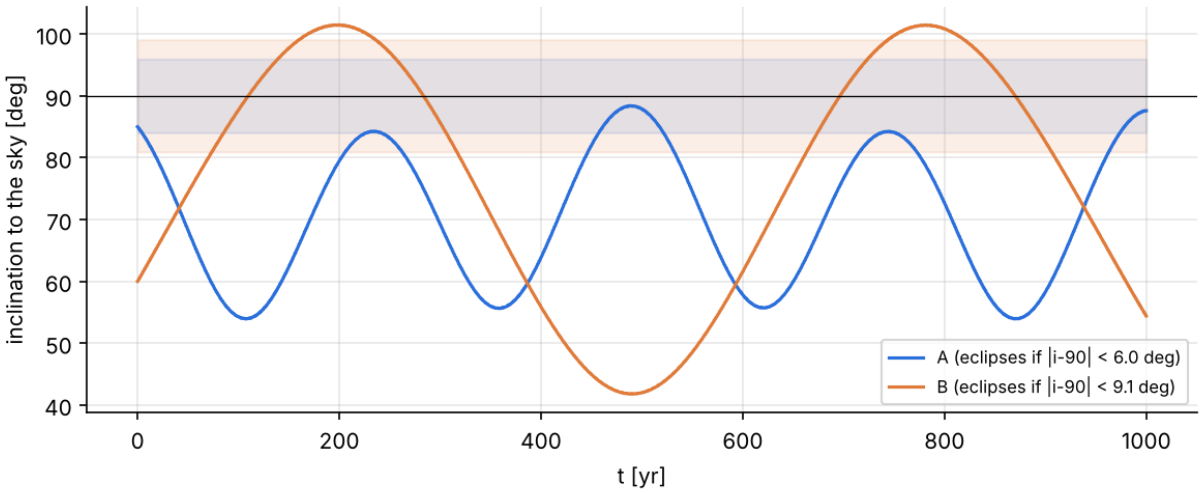


Figure 27. Inclination of A (blue) and B (orange) to the plane of the sky over 1000 years. Horizontal axis: time in years; vertical axis: inclination in degrees. The shaded bands are the ranges in which each binary eclipses ($|i - 90°| < 6.0°$ for A, $< 9.1°$ for B, from the default radii). A swings between 54° and 88° and eclipses only near its maxima; B swings between 42° and 102°.

Figure 27 is what an observer sees of the precession: a binary eclipses only while its inclination is inside its band, so A eclipses in windows a few decades long every 255 years and is inside such a window today ($t = 0$), and B eclipses about a third of the time. Inside each window the eclipse depth changes continuously. The same run with `--backend secular` takes 2.5 s instead of 80 s and gives the mutual inclinations to a few hundredths of a degree.

8.2 Example 2: ZLK cycles

`configs/examples/example_2p2_zlk.yaml` has all nodes aligned, so the mutual inclinations are differences of the inclinations: B is tilted by 65° to the outer orbit (200 d), inside the ZLK window of 39.2–140.8°; A by only 5°.

```
swingersim run configs/examples/example_2p2_zlk.yaml -o output/zlk.h5
```

```
  B in out: i_mut0 = 65.0 deg, g1_0 = 90 deg
    quadrupole theory : e = 0.010..0.807, i_mut = 39.7..65.0 deg, e-cycle 18.9 yr (inside
↪   ZLK window)
    simulation        : e = 0.007..0.811, i_mut = 40.6..65.1 deg, 6 ZLK cycles of ~14 yr
    stability ratio Mardling-Aarseth = 1.64 (stable), Vynatheya+22 = 1.73 (stable), eps_oct
↪   = 0.002, eps_SA = 0.032  (mild hierarchy: secular estimates approximate)
```

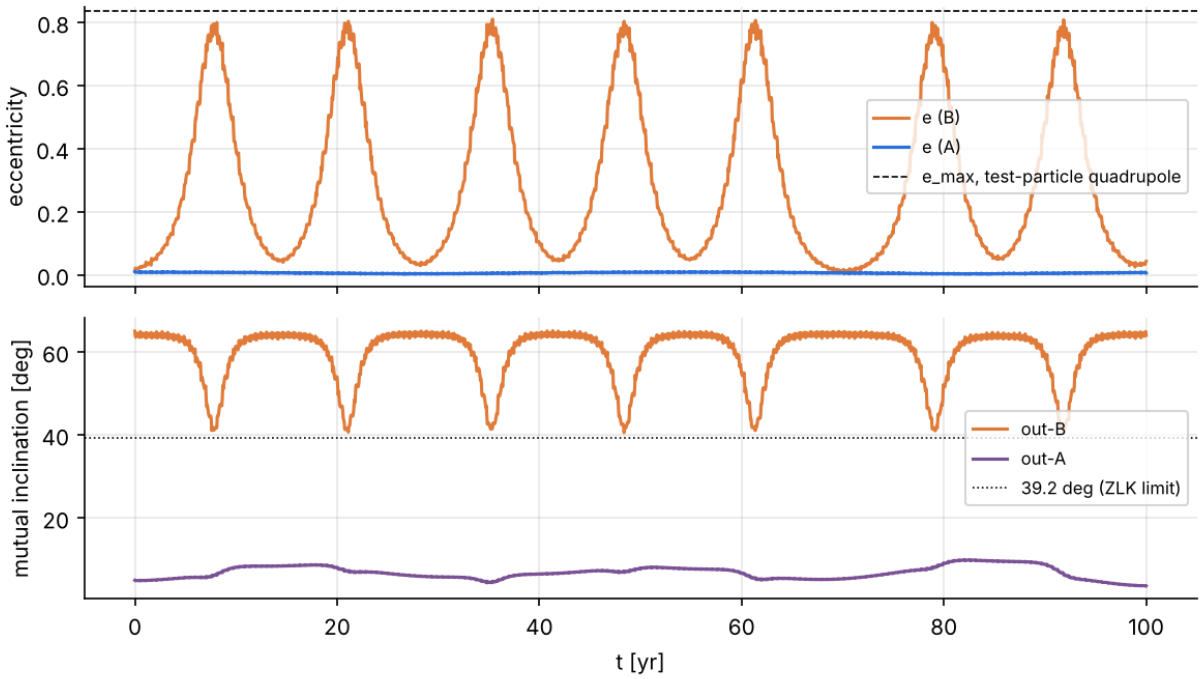


Figure 28. ZLK cycles of binary B in example 2. Top: eccentricity of B (orange) and of A (blue) against time in years; the dashed line is the largest eccentricity of the test-particle quadrupole approximation, $\sqrt{1-\frac{5}{3}\cos^2 65^\circ} = 0.84$. Bottom: mutual inclination of B (orange) and of A (violet) to the outer orbit in degrees; the dotted line marks 39.2°, the lower edge of the ZLK window. In every cycle the eccentricity of B rises to 0.81 while its inclination drops to about 41°, then both return.

The eccentricity range agrees with the orbit-averaged quadrupole theory, but the cycle is about 25 % shorter than predicted (14 yr against 18.9 yr; Figure 28). The summary warns about this: $\varepsilon_{\mathrm{SA}} = 0.032$ means a mild hierarchy (the outer period is only 20 times the period of B), for which the averaged equations lose accuracy (Luo et al. 2016); with a two-times wider outer orbit the averaged period agrees to 1.5 %. At the eccentricity maximum the stars of B approach to $a(1-e) = 4.4\ \mathrm{R}_\odot$, three times the sum of their radii.

8.3 Example 3: stars with structure

`configs/examples/example_2p2_tides.yaml` is a compact 2+2 of two eclipsing binaries with periods of 1.8 and 2.9 days, with `physics: {gr: true, tides: true, rotation: true}`: the stars are extended bodies distorted by tides and rotation, rotating synchronously with their binary.

```
swingersim run configs/examples/example_2p2_tides.yaml -o output/example_tides.h5
```

Line	Value	Meaning
`tidal+rotational apsidal period` (A)	64 yr vs ZLK time 949 yr	the distortion turns the periastron of A fifteen times faster than the outer orbit could build up eccentricity, so ZLK cycles of A are suppressed
`stars`	$k_2 = 0.0144$, P_{rot} 1.8 → 1.8 d	structure from the default polytropes; without dissipation the spin rates do not change
`Roche lobe filled up to`	0.29–0.51	detached binaries
`binary A`	62.3 yr expected, 59.4 yr measured	dominated by its own stars; the perturbation by B and the outer orbit adds a few per cent

Line	Value	Meaning
`binary B`	1116 yr expected, 505 yr measured	B is wider, its own distortion is weak; the other binary doubles the rate
max |dE/E|	2.2×10^{-11}	conserved because the tidal lag is zero

Table 42. Key lines of the summary of example 3.

In a quadruple, the observed apsidal period of an eclipsing binary need not agree with the value expected from the stellar structure alone (Table 42). Giving a tidal time lag (`stars: {m3: {lag_s: 10}}`) switches on tidal dissipation; the summary then lists the circularisation and synchronisation time scales. For apsidal-motion studies of stars of 2–3 $M_\odot$ use `stellar_model.kind: mist` or `sse` (the Claret 2023 values of k_2) instead of the default polytropes, whose k_2 is about three times too large for such stars.

8.4 Example 4: into the past, and stability

The same system as example 1 integrated 200 years into the past needs only a negative end time, `run: {t_max_yr: -200}` or `--t-max-yr -200`:

```
integrated to t = -200 yr (2000 outputs, osculating elements)
  i_mut A-B          17.82..  45.32 deg
  eclipses (observer along z): A 27 %, B 0 %, both at once 0 % of the run
```

In the 200 years before today A eclipsed about a quarter of the time and B not at all; in the next 200 years it is the other way round. The stability check of example 2:

```
swingersim stability configs/examples/example_2p2_zlk.yaml
```

```
nested triples (ratio > 1 = stable):
  A in out (+B)   a_in/a_out = 0.0706, e_in = 0.010, e_out = 0.300, i_mut = 5.0 deg, q_out =
 ↪  0.773: Mardling-Aarseth 2.38, Vynatheya+22 3.04
  B in out (+A)   a_in/a_out = 0.1029, e_in = 0.010, e_out = 0.300, i_mut = 65.0 deg, q_out
 ↪  = 1.294: Mardling-Aarseth 1.64, Vynatheya+22 1.73
N-body ghost test (100 outer orbits, Newtonian): stable after 100.0 outer orbits, max
 ↪  relative divergence 1.70e-03
```

Binary B, tilted by 65° and in ZLK cycles, is the more critical one; both criteria and the direct test agree that the system is stable.

8.5 Example 5: an ensemble – V994 Her

For V994 Her, photometry fixes the periods, eccentricities and inclinations of both binaries (84° and 85°), but not how the orbits are turned around the line of sight. `configs/examples/example_v994_ensemble.yaml` draws 200 systems: masses within 2 %, inclinations within 1°, binary B at 85° or 95° (sense of revolution unknown), its node relative to A anywhere, the outer orbit oriented at random; every member is integrated 10 000 years forward and backward with the secular backend.

```
swingersim ensemble configs/examples/example_v994_ensemble.yaml -o output/v994/ensemble -j
 ↪  4
swingersim ensemble-report output/v994/ensemble --plot output/v994/overview.png --x
 ↪  orbits.out.i --y B_didt0_deg_yr
```

```
forward: 200 members (fractions with 68 % Wilson intervals)
  fate intact           100.0 % (99.5-100.0) [200/200]
  i_mut A_B     now 90.3 deg (16-84 %: 27.7-150.9), range over the run: median 40.0...141.9
 ↪  deg, members that cross 90 deg: 69.5 %
  A      di/dt now +0.00462 deg/yr (16-84 %: -0.0765..+0.0723); of the members eclipsing
 ↪  now, eclipses end within the run in 92 % (median after 440.4 yr)
```

```
  eclipses (time averaged over members): A 48.9 %, B 51.6 %, both 27.9 %
  stable by Vynatheya+22       100.0 % (99.5-100.0) [200/200]
```

The 400 runs take about 10 minutes on one core with Numba. The system is safe in every member, and its eclipses are temporary: in 92 % of the members the eclipses of A stop within 10 000 years, typically after about 400 years, and both binaries eclipse at once only 28 % of the time – the doubly eclipsing state observed today is a phase. The present rate of change of the inclination depends on the unknown orientation of the outer orbit, so a measured rate constrains it; that is what the conditioning of Workflows does. The past looks statistically like the future, as it must without a measured direction of change.

8.6 Example 6: a fit of times of minima, stage 1

`configs/examples/example_fit.yaml` is a 2+2 whose minima are synthetic: the table `example_fit_minima.csv` (50 minima of A, 354 of B over 17.7 years; made by `tools/make_example_fit_minima.py`) was made with an outer orbit of 2200 d ($e_3 = 0.35$, $\omega_3 = 1.2$ rad), an apsidal motion of A of 0.0015 rad per cycle and $a \sin i$ of 1.1 au (A) and 0.95 au (B), at randomly chosen cycles, with errors of $2{\cdot}10^{-4}$ d. The fit uses the light-time effect for both pairs, apsidal motion for A and one shared outer orbit:

```
swingersim fit configs/examples/example_fit.yaml
```

```
stage 1: chi^2 = 385.8 for 391 degrees of freedom (reduced 0.99)
  out.P3        2199.8 ± 1.2       d
  out.e3        0.3508 ± 0.0058
  out.w3         1.191 ± 0.015     rad
  A.e          0.24986 ± 0.00040
  A.dw       0.0015010 ± 0.0000028 rad/cycle
  A.asini       1.0965 ± 0.0073    AU
  B.asini       0.9461 ± 0.0032    AU
```

All values are recovered within 1.3 standard deviations, and the reduced χ^2 is 0.99, as it must be for errors that are correct by construction. The cycles are random on purpose: minima spaced by a fixed number N of cycles cannot tell an apsidal rate $\dot\omega$ from $\dot\omega + 2\pi k/N$ (an alias). An earlier version of this table had the minima of A every 40 cycles, and the fit returned 0.1586 rad per cycle ($0.0015 + 2\pi/40$) with a reduced χ^2 of 14. Real data with a regular cadence (one minimum per season, say) can fall into the same trap; a reduced χ^2 far above 1 for trusted errors is the warning sign.

8.7 Example 7: two binaries meet

Can a 2+2 quadruple be assembled from two binaries that meet in a young cluster? `configs/examples/example_scatter_2p2.yaml` sends a binary with $a = 0.5$ au at 1 km/s towards one with $a = 1$ au (all stars 1 $M_\odot$ and 1 $R_\odot$, eccentricities drawn from the thermal distribution, pericentres up to 3 au):

```
swingersim scatter configs/examples/example_scatter_2p2.yaml -o output/scatter_2p2 --n 200
 ↪  -j 0
```

```
  outcome                          n  fraction    1-sigma interval  sigma [AU^2]
  collision                       97     0.485  [0.450, 0.520]  3.245e+04
  flyby                           56     0.280  [0.249, 0.313]  1.874e+04
  exchange + ionization           19     0.095  [0.076, 0.118]  6357
  triple + single (stable)        12     0.060  [0.045, 0.079]  4015
  single ionization               10     0.050  [0.037, 0.068]  3346
  exchange                         2     0.010  [0.005, 0.020]  669.2
  triple + single (unstable)       2     0.010  [0.005, 0.020]  669.2
  bound 3+1 (unstable)             1     0.005  [0.002, 0.013]  334.6
  bound 2+2 (unstable)             1     0.005  [0.002, 0.013]  334.6
```

The columns are the outcome, the number of encounters, the fraction with its 68 % interval and the cross-section. At 1 km/s the two binaries are strongly bound once they meet: almost every close encounter becomes a long resonant interaction, and in half of them two stars touch. A stable triple with an escaping star is the most common dynamical product (6 %); a bound 2+2 or 3+1 appears only as a temporary, unstable state. Captures into stable quadruples need a third party or dissipation (tides, gas). The 200 encounters took 22 minutes on two cores.

8.8 Example 8: where is a system stable? A chaos map

`configs/validation/quarles18_circumbinary.yaml` puts a test particle around an equal-mass binary ($a = 1$ au); the map scans its semi-major axis and eccentricity:

```
swingersim chaosmap configs/validation/quarles18_circumbinary.yaml -o output/chaosmap_cb \
  --x orbits.out.a --x-range 1.8 4.0 --nx 23 --y orbits.out.e --y-range 0 0.5 --ny 11
  ↪  --t-max 3000 -j 0
```

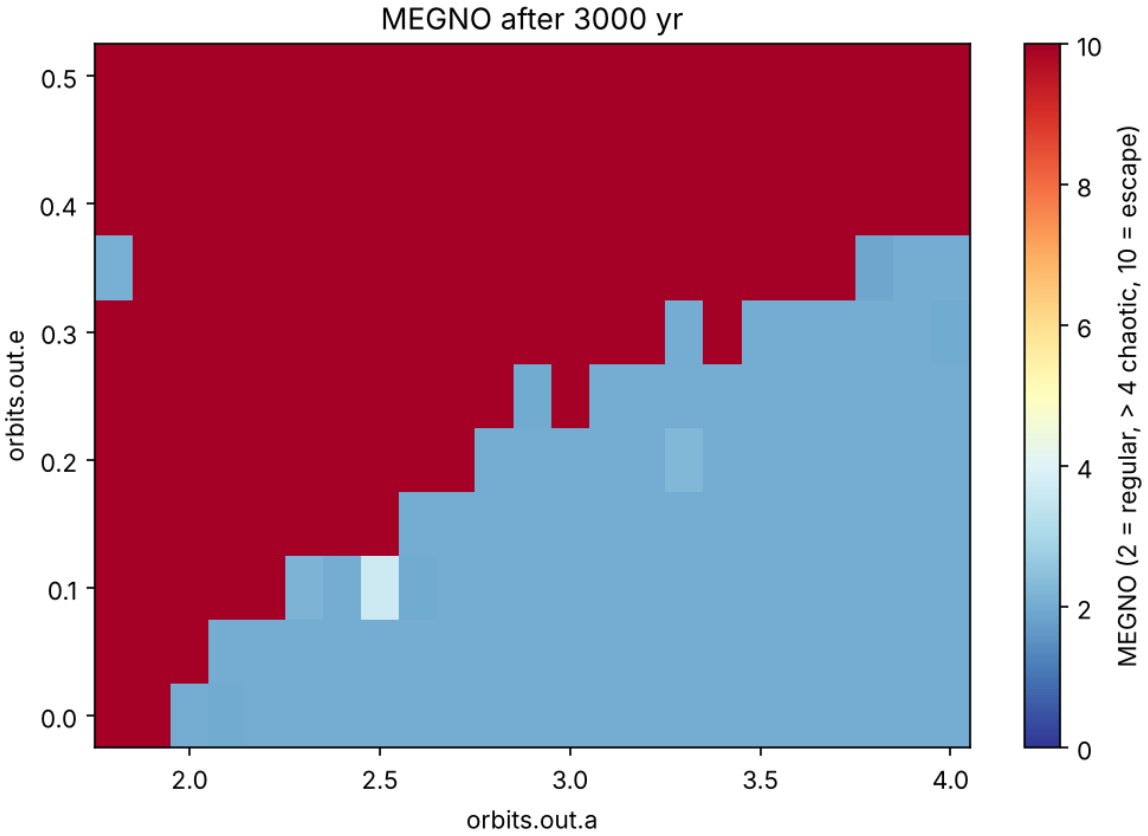


Figure 29. MEGNO map of a circumbinary orbit after 3000 years. Horizontal axis: semi-major axis of the outer orbit in au; vertical axis: its eccentricity. Colour: MEGNO (scale on the right), 2 for regular motion (blue), above 4 chaotic, 10 marks an escape or a collision (dark red). The boundary between the regular region (lower right) and the escapes (upper left) follows a nearly constant pericentre distance $a(1 - e)$.

The 253 runs take 7 minutes on two cores. At $e = 0$ the orbit is regular from 2.0 au outwards for this single initial phase; the published limit, which considers all phases, is reproduced in *Testing, Part I*. The same command maps any parameter pair of a quadruple, e.g. `orbits.out.P` against `orbits.B.P` near a 3:2 ratio of the periods of the binaries.

8.9 Example 9: what would observers measure? Eclipse timings

`configs/examples/example_timings_czev4315.yaml` is the doubly eclipsing quadruple CzeV4315 with one possible mutual orbit (120 d, close to the 119-d signal in the timings of B; the orientation is illustrative):

```
swingersim timings configs/examples/example_timings_czev4315.yaml --years 5 \
  -o output/timings_czev4315.csv --plot output/timings_czev4315.png
```

```
eclipse timings over 5 yr
  A: 45 primary and 73 secondary eclipses, P = 6.74571156 d, O-C of primaries -840.62 ..
↪  +787.62 min (semi-amplitude 814.12 min)
    eclipses missing between 0.330 and 3.158 yr
    no primary eclipses after t = 3.635 yr
  B: 1987 primary and 1988 secondary eclipses, P = 0.91885817 d, O-C of primaries -4.30 ..
↪  +3.59 min (semi-amplitude 3.94 min)
```

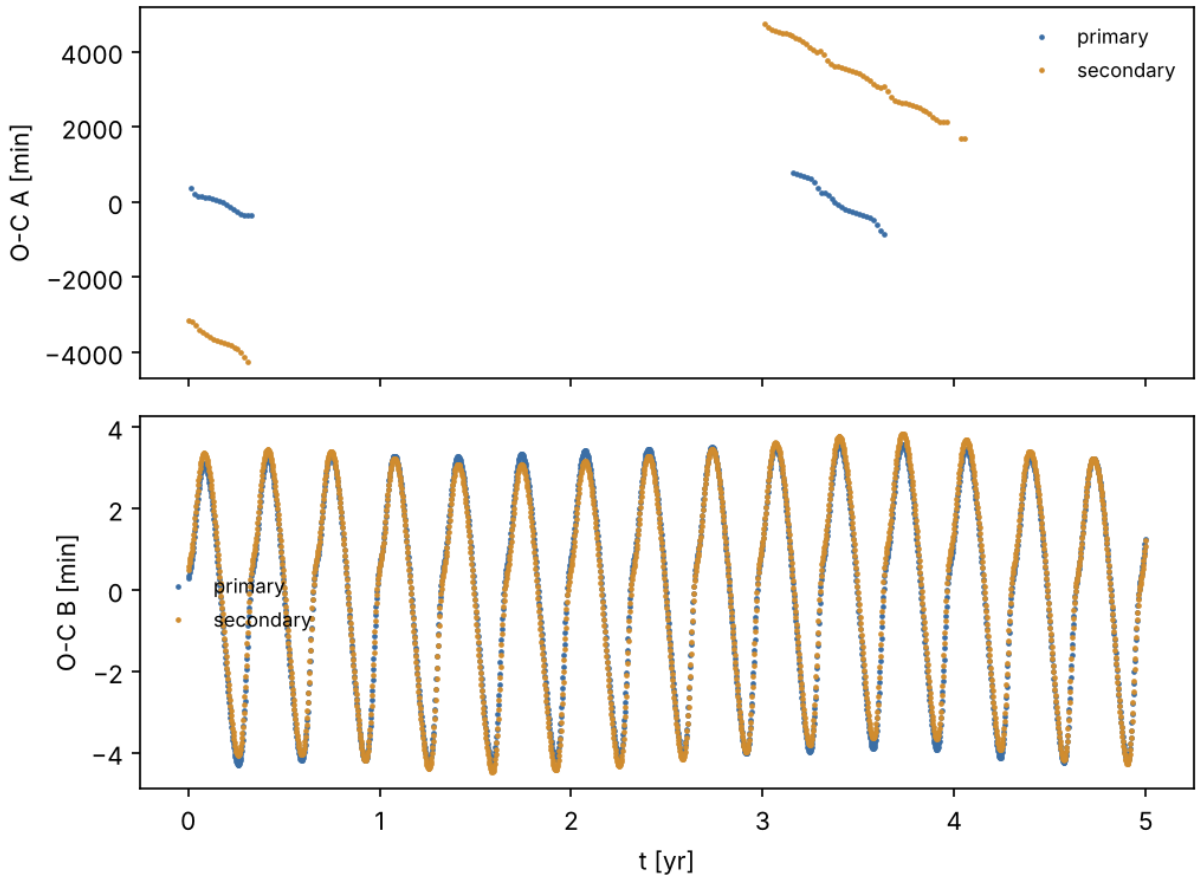


Figure 30. Synthetic O−C of CzeV4315 for a 120-d mutual orbit over 5 years. Top: binary A, bottom: binary B; horizontal axis: time in years from $t = 0$; vertical axis: O−C in minutes; blue dots are primary, orange dots secondary eclipses. B shows a wave of ±4 min with the 120-d period. A eclipses only in two intervals; its primaries and secondaries are separated by thousands of minutes and drift.

B shows the expected 120-day oscillation (light time plus the dynamical effect of A; Figure 30). A, eccentric and only 18 times faster than the mutual orbit, reacts strongly: its apsides turn within years, its secondary eclipses move by about ±3 days relative to the primaries, and for almost three years it does not eclipse at all. CzeV4315 A does eclipse in the data, so this particular orbit is excluded – the kind of test that an ensemble applies to thousands of orbits.

8.10 Example 10: six stars – V994 Her

V994 Her is a triply eclipsing sextuple: the doubly eclipsing quadruple A+B (mutual orbit 1062 d) and a third eclipsing binary C on a wide orbit (Zasche et al. 2023b). The preset `2+2+2` describes it; `configs/examples/example_v994_sextuple.yaml` takes the masses, radii and orbits of the published solution and marks what is assumed.

```
swingersim stability configs/examples/example_v994_sextuple.yaml --no-nbody
swingersim run configs/examples/example_v994_sextuple.yaml -o output/v994_sextuple.h5
```

The stability check applies the criteria to the three nested triples (AB in `out` with C: Mardling–Aarseth 21.00; A in AB with B: 4.43; B in AB with A: 5.65) and, with the classifiers installed, to every sub-quadruple. The run (40 s for 30 years) measures apsidal periods of A, B and C that correspond to 1.42, 3.51 and 0.71 °/yr against the observed 1.86, 3.66 and 1.68 °/yr: B, whose apsidal constant was derived from these data, agrees to 4 %; A and C turn too slowly with the assumed theoretical k_2, so their structure constants or radii are what the data would constrain.

8.11 Example 11: an evolving quadruple and its light curve

`configs/examples/example_evolution.yaml` is an illustrative 2+2 of age 1.1 Gyr: binary A with a 2 $M_\odot$ primary (4 d), binary B of two solar-type stars (5 d). With `stellar_model.kind: sse` and `physics.evolution: true` the radii, luminosities and structure constants follow the age of the stars, and `tidal_lag: hurley2002` takes the tidal dissipation of every star from its envelope.

```
swingersim run configs/examples/example_evolution.yaml -o output/evolution.h5
swingersim lightcurve configs/examples/example_evolution.yaml --days 30 -o
 ↪  output/evolution_lc.csv
```

```
EVENT: roche_overflow at t = 7.445e+07 yr {'age_myr': 1174.45, 'stars': ['m1']}
  max |dE/E| = 2.22e-01  (tidal dissipation: energy decreases)
  max |dL/L| = 1.51e-05
  m1  ... P_rot = 4 -> 3.509 d, Roche lobe filled up to 1.01
      evolving: age 1100 -> 1174.3 Myr, R 3.72 -> 5.96 Rsun, Teff 6761 -> 5057 K, main
 ↪  sequence -> giant branch
```

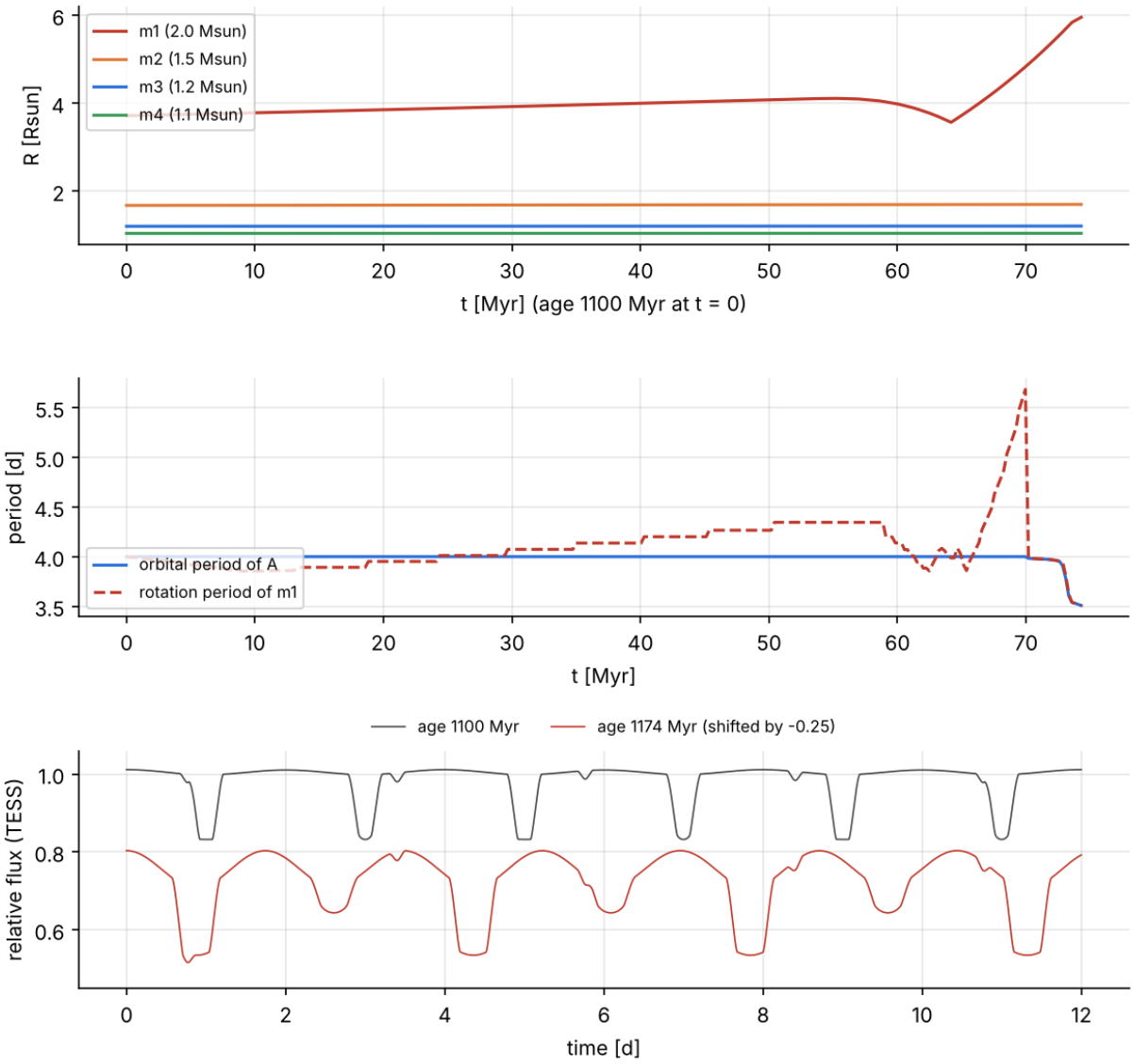


Figure 31. Example 11. Top: radii of the four stars in $R_\odot$ against the time of the run in Myr (age 1100 Myr at $t = 0$); the 2 $M_\odot$ primary m1 (red) leaves the main sequence after 64 Myr and expands to 6 $R_\odot$. Middle: orbital period of A (blue) and rotation period of m1 (dashed red) in days; the steps are the operator-splitting segments, the drop at the end is the tidal synchronisation that takes angular momentum from the orbit. Bottom: TESS light curves of 12 days at the start (age 1100 Myr, grey) and at the end (1174 Myr, red, shifted by −0.25), relative flux against time in days.

The primary of A crosses the Hertzsprung gap and fills its Roche lobe at 1174 Myr; there the run stops (`roche_overflow`), because this example does not switch on `physics.mass_transfer` (Figure 31). While the star expands, its spin slows down; once its envelope becomes convective, the tides synchronise it within a few hundred thousand years and take the angular momentum from the orbit: the period of A shrinks from 4.0 to 3.5 d. The energy is therefore not conserved (the tides dissipate it), while the total angular momentum is kept to 1.5×10^{-5}. The run takes 3 minutes for 74 Myr. The large ellipsoidal variation of the late light curve is at the edge of the small-distortion approximation of spherical stars; `stars.<m>.shape: roche` gives the star its Roche surface.

8.12 Example 12: more kinds of systems

`configs/examples/` holds a ready configuration for each kind of system; run any of them with `swingersim run <file> -o <name>.h5`.

Kind of system	Configuration	What happens
young stars in their gas	`example_gas_2p2.yaml`	the gas shrinks, circularises and aligns the outer orbit, then disperses (secular backend)
a triple with a fourth star	`example_3p1.yaml`	precession of the nested orbits
a giant beside a binary	`example_giant_triple.yaml`	an AGB tertiary loses mass in its wind; the outer orbit widens
a giant feeding a binary	`example_tertiary_transfer.yaml`	the tertiary transfers mass onto the inner binary
mass transfer (Algol)	`example_algol_triple.yaml`	the donor overflows onto its companion; the mass ratio reverses
common envelope	`example_ce_triple.yaml`	unstable transfer engulfs the binary; the orbit shrinks
contact binary	`example_contact_evolution.yaml`	a close binary comes into contact, shrinks and merges
pulsating stars	`example_pulsating_triple.yaml`, `example_cepheid_triple.yaml`	δ Scuti, long-period and Cepheid pulsations in the light curve
massive stars, Wolf–Rayet	`example_wr_triple.yaml`	a massive tertiary becomes a Wolf–Rayet star; its wind widens the outer orbit
winds that destabilise	`example_mass_loss.yaml`	mass loss pushes a triple over the stability limit

Table 43. Further example configurations. Left: the kind of system; middle: the file in `configs/examples/`; right: what the run shows. The physics each needs is described in *Physics and Methods*, chapter Stars in multiple systems.

Chapter 9

The configuration file

A configuration is a YAML text file that describes one system and what to do with it. This chapter is its complete reference: every section, every key with its type, default, allowed values and meaning, as checked by the program. A key that is not listed here is refused, and so is a value outside its range; the error message names the key.

9.1 Units and conventions

Quantity	Unit in the file
masses	$M_\odot$
periods of orbits and stars	days
semi-major axes, disc radii	au
stellar radii	$R_\odot$
angles (inclination, node, argument of pericentre, mean anomaly, obliquity)	degrees
times of runs, time scales	years; ages in Myr
times of measurements	BJD_TDB (unless `fit.time_system` says otherwise)
velocities	km/s

Table 44. Units of the configuration file. Internally SWINGERsim works in $M_\odot$, au and years.

Reference frame. Inclinations and nodes are measured from the plane of the sky; the z axis is along the line of sight, pointing away from the observer, so $i = 90°$ is edge-on (eclipses possible) and $i = 0°$ face-on. Values of i above 90° describe the same orbit revolving in the opposite sense; this matters for mutual inclinations, not for eclipses. Each orbit is the relative orbit of its second component around its first, both taken as their centres of mass (Jacobi coordinates). The elements given are osculating elements at $t = 0$, the epoch of the configuration.

Names. Stars are `m1`, `m2`, … without gaps. Orbits have the names of the hierarchy: `A`, `B`, `out` for 2+2; `inner`, `mid`, `out` for 3+1; your names for `custom`.

9.2 Top level

Key	Type, default	Meaning
`topology`	text, required	`2+2`, `3+1`, `2+2+1`, `3+1+1`, `2+(2+1)`, `2+2+2`, `3+3`, or `custom`
`hierarchy`	map orbit → [child 1, child 2]; only with `custom`	the tree: each orbit has two children, a star or another orbit; exactly one orbit is nobody's child
`masses`	map star → $M_\odot$, required	one positive mass per star
`orbits`	map orbit → orbit, required	one entry per orbit of the hierarchy (below)
`stars`	map star → star, default empty	properties beyond the mass (below)
`discs`	map name → disc, default empty	dust discs and rings that occult stars in light curves (below)
`stellar_model`	section	where radii, temperatures and structure come from (below)
`physics`	section	the physics beyond Newtonian point masses (below)
`run`	section, required (except in a configuration with only a `fit`)	how the motion is computed (below)
`ensemble`	section, optional	a Monte Carlo ensemble (`swingersim ensemble`); ignored by `run`
`fit`	section, optional	a fit of measured data (`swingersim fit`); ignored by `run`
`fit_epoch_bjd_tdb`	number, optional	written by a stage-2 fit into `stage2_fitted.yaml`: the BJD_TDB of $t = 0$ of the fitted elements; ignored by runs

Table 45. Keys at the top level of a configuration.

The presets of `topology` are shortcuts for these trees (Figure 1):

Preset	Tree
`2+2`	`out: [A, B], A: [m1, m2], B: [m3, m4]`
`3+1`	`out: [mid, m4], mid: [inner, m3], inner: [m1, m2]`
`2+2+1`	`out5: [out, m5], out: [A, B], A: [m1, m2], B: [m3, m4]`
`3+1+1`	`out5: [out, m5], out: [mid, m4], mid: [inner, m3], inner: [m1, m2]`
`2+(2+1)`	`out: [A, Bt], A: [m1, m2], Bt: [B, m5], B: [m3, m4]`
`2+2+2`	`out: [AB, C], AB: [A, B], A: [m1, m2], B: [m3, m4], C: [m5, m6]`
`3+3`	`out: [T1, T2], T1: [A, m3], A: [m1, m2], T2: [B, m6], B: [m4, m5]`

Table 46. The preset hierarchies. Each orbit is written as `name: [first child, second child]`. A triple is written as `custom`, e.g. `hierarchy: {out: [inner, m3], inner: [m1, m2]}`. Trees of up to ten stars are tested.

9.3 `orbits.<name>`

Key	Type, default	Meaning
`P`	number > 0	period [d]; give exactly one of `P` and `a`
`a`	number > 0	semi-major axis [au]
`e`	0 ≤ e < 1, default 0	eccentricity
`i`	0–180°, default 0	inclination to the plane of the sky
`Omega`	degrees, default 0	longitude of the ascending node
`omega`	degrees, default 0	argument of pericentre
`M0`	degrees, default 0	mean anomaly at $t = 0$
`Pdot`	d/yr, default 0	imposed secular period change of an orbit of two stars, $P(t) = P + \dot{P}\,t$ (mass transfer or angular-momentum loss of semi-detached and contact binaries); N-body backend; free in a fit as `orbits.<name>.Pdot`
`Pdot_mode`	`angular_momentum` (default) or `conservative`	what changes the period: angular-momentum loss at constant masses, or conservative transfer between the two stars ($\dot{M}_1 = (\dot{P}/P)\,M_1M_2/(3(M_1 - M_2))$)
`contact_fillout`	0–1, optional	light curves: the two stars of this binary are a contact binary in a common envelope with fill-out factor f (Lucy 1968); their radii follow from f and they share one temperature

Table 47. Keys of an orbit.

9.4 `stars.<name>`

Everything left out comes from the stellar model.

Key	Type, default	Meaning
`R`	$R_\odot$ > 0	radius
`Teff`	K > 0	effective temperature (light curves; with `R` it fixes the luminosity)
`apsidal_k2`	0 < k2 < 0.75	apsidal-motion constant (Love number = 2 k_2)
`rg2`	0 < rg2 ≤ 0.4	moment of inertia $I/(MR^2)$
`lag_s`	s ≥ 0	constant tidal time lag (Mignard 1979); 0 = no dissipation
`P_rot`	days, `sync` (default) or `pseudosync`	rotation period; `sync`: the mean motion of its binary; `pseudosync`: pseudo-synchronous rotation of an eccentric binary (Hut 1981)
`obliquity`	0–180°, default 0	angle between the spin and the orbit normal
`spin_azimuth`	degrees, default 0	azimuth of the spin around the orbit normal, from the node
`age_offset_myr`	Myr, default 0	`sse`, `mist`: apparent age of this star minus `stellar_model.age_myr` (products of binary interaction)
`pulsation`	section, optional	pulsations in light curves (below)

Key	Type, default	Meaning
`shape`	`sphere` (default), `rotating`, `roche`	light curves: a sphere; an oblate, gravity-darkened rigid rotator; the Roche equipotential in its circular synchronised binary
`radius_inflation`	−0.5 to 1, default 0	fractional radius change against the stellar model at fixed luminosity (e.g. 0.05 for magnetically active M dwarfs)

Table 48. Keys of a star.

9.4.1 `stars.<name>.pulsation`

Key	Type, default	Meaning
`type`	`manual` (default), `lpv`, `delta_scuti`, `gamma_dor`, `beta_cep`, `spb`, `rr_lyrae`, `cepheid`	kind of pulsator; the period of `lpv`, `delta_scuti`, `beta_cep`, `cepheid` and `rr_lyrae` follows from published relations, the others need `period_d` or `Q`
`mode`	`F` (default), `10`, `20`, `30`	radial mode
`period_d`	days > 0	period; overrides the relation
`Q`	days > 0	pulsation constant, $P = Q/\sqrt{\rho/\rho_\odot}$
`amplitude_mag`	mag ≥ 0, default 0.01	semi-amplitude in the band
`phase`	cycles, default 0	phase at emission time $t = 0$
`period_spacing_s`, `n_modes`	s; 1–100, default 1	gravity modes (`spb`, `gamma_dor`): a series of `n_modes` modes $P_0 + k\,\Delta P$
`extra_modes`	list	further modes, each with `period_d` or `frequency_ratio`, `amplitude_mag` (required) and `phase`

Table 49. Keys of a pulsation.

9.5 `discs.<name>`

A dust disc or ring around a star or a binary that occults stars behind it: a slab of outer radius `r_out_au`, optional inner hole and full thickness, dimming starlight by $\exp(-\tau L/\text{thickness})$ along each line of sight.

Key	Type, default	Meaning
`host`	star or orbit, required	the disc is centred on the star or on the centre of mass of the stars of that orbit
`r_out_au`, `r_in_au`	au; `r_in_au` default 0	outer radius (required) and inner hole
`thickness_au`	au, required	full thickness
`tau`	> 0, default none (opaque)	optical depth through the full thickness, perpendicular to the disc
`i`, `Omega`	degrees, defaults 90 and 0	orientation of the disc plane, as for orbits
`relative_to`, `tilt`, `tilt_node`	orbit; degrees	tilt the disc from the plane of an orbit (then `i`, `Omega` are ignored)

Key	Type, default	Meaning
`precession_period_yr`	years	with `relative_to`: precession of the disc normal about the orbit normal (negative = retrograde)
`dynamics`	`prescribed` (default) or `rigid`	the precession as given, or computed for a rigid disc torqued by the binary or by the companion of its host (needs `relative_to`)
`sigma_index`, `damping_yr`	0–2.5, default 1; years	`rigid`: surface-density exponent; alignment time scale by viscous dissipation
`edge_au`	au, optional	soft outer edge (needs a finite `tau`)
`emission`	default false	add the thermal light of a flat reprocessing disc
`albedo`, `g`	0–1, default 0; −1 to 1, default 0.5	scattered starlight: single-scattering albedo and asymmetry parameter

Table 50. Keys of a disc. Discs affect light curves and minima only, not the dynamics.

9.6 `stellar_model`

Key	Type, default	Meaning
`kind`	`zams` (default), `sse`, `mist`, `table`, `python`	`zams`: zero-age main sequence (Tout et al. 1996) and polytropes; `sse`: stars of age `age_myr` after Hurley et al. (2000), up to helium ignition, 0.1–8 $M_\odot$; `mist`: MIST tracks (Choi et al. 2016) from the pre-main sequence to the end of the AGB; `table`: own CSV table; `python`: own function
`age_myr`	Myr ≥ 0	`sse`, `mist` (required there): age of the stars at $t = 0$; `sse` counts from the ZAMS, `mist` from the start of the pre-main sequence
`feh`	default 0	`mist`: metallicity [Fe/H] of an installed grid (−1.50 to +0.50 in steps of 0.25)
`vvcrit`	0 or 0.4 (default)	`mist`: initial rotation of the models
`z`	0.004–0.03, default 0.02	`sse`: metallicity Z
`structure`	`auto` (default), `polytrope`, `claret2023`, `mesa:<grid>`	source of k_2 and $I/(MR^2)$: the Claret (2023) models at the star's mass and log g, polytropes of index 1.5 or 3, or an own MESA grid; `auto` = `claret2023` for `sse`, polytropes for `zams`
`file`	path	`table`: the CSV file (relative to the configuration)
`entry`	`file.py:function` or `module:function`	`python`: the function

Table 51. Keys of the stellar model. `mist` always uses the Claret structure (Claret 2004 above 8 $M_\odot$).

9.7 `physics`

Key	Type, default	Meaning
`gr`	default true	1PN relativity (N-body), periastron advance (secular)
`gw`	default false	gravitational-wave emission of orbits of two stars (Peters 1964)
`tides`	default false	tidal distortion of the stars (apsidal motion; dissipation with a lag)
`rotation`	default false	rotational flattening and spin–orbit coupling (needs `tides`)
`evolution`	default false	the stars follow their tracks during the run (needs `stellar_model.kind` `sse` or `mist`; no fixed `R` or `Teff`)
`tidal_lag`	`fixed` (default) or `hurley2002`	the lag from `stars.<m>.lag_s`, or from the envelope of every star (Hurley et al. 2002; needs `tides`)
`tidal_efficiency`	> 0, default 1	`hurley2002`: factor on the dissipation rate
`tidal_convective_reduction`	`quadratic` (default) or `linear`	`hurley2002`: reduction of the turbulent viscosity for tidal periods shorter than the convective turnover time
`k2_rotation`	default false	correct k_2 for the rotation of the stars (Claret 1999)
`inertial_waves`	default false	add the dissipation of inertial waves in convective envelopes (needs `hurley2002`)
`gas`	section	parametric gas phase (secular backend): `t_end_yr`, `t_decay_yr`, `plane` (`invariable` or `{i, Omega}`), `orbits: {<orbit>: {tau_a_yr, tau_e_yr, tau_i_yr}}`
`mass_loss`	section	stellar winds in the dynamics (below)
`mass_transfer`	section	Roche-lobe overflow in two-star binaries (below)
`mergers`	section	continue after two stars merge (below)
`galactic_tide`	section	tide of the Galactic disc: `rho0_msun_pc3` (0.185), `pole_i_deg` (0), `pole_Omega_deg` (0)
`encounters`	section	passing stars: `n_star_pc3` (0.1), `sigma_kms` (40), `m_perturber` (0.5), `b_max_au` (default 30 × the outermost a), `seed` (1)
`activity`	map star → cycle	magnetic activity cycle (Applegate 1992) in the eclipse times: `p_mod_yr`, `amplitude_s` (required), `phase_deg` (0)
`magnetic_braking`	section	wind braking of spins (Matt et al. 2015; needs `evolution` and `rotation`): `chi` (10), `p` (2), `t0_erg` (9.5×10^{30}), `stars` (default all), `max_step_yr` (10^6)

Table 52. Keys of `physics`. Default values of sub-keys in brackets.

9.7.1 `physics.mass_loss`

Key	Type, default	Meaning
`source`	`track` (default) or `rates`	masses from the tracks (needs `evolution`), or constant rates
`rates`	map star → $M_\odot$/yr	`rates`: dM/dt (negative = loss)
`reimers_eta`	default 0.5	SSE stars: coefficient of the Reimers wind

Key	Type, default	Meaning
`masses`	`initial` (default) or `current`	`track`: the configured masses are initial masses, or present masses at `age_myr`
`branch`	`lowest` (default) or `unique`	`current`: which initial mass to take when several tracks reach the present mass
`max_psi`	default 0.05	secular backend: refuse runs whose mass-loss index $\Psi = \lvert\dot{M}\rvert/(nM)$ exceeds this

Table 53. Keys of `physics.mass_loss`.

9.7.2 `physics.mass_transfer`

Needs `physics.evolution: true` with `stellar_model.kind` `sse` or `mist`.

Key	Type, default	Meaning
`rejuvenation`	`hurley` (default) or `fractional_age`	rejuvenation of accreting main-sequence stars
`stability`	`bse` (default) or `zeta`	critical mass ratios of Hurley et al. (2002), or the adiabatic exponents of giant donors
`q_crit_ms`	default 3.0	critical mass ratio of main-sequence and core-helium-burning donors
`accretion`	`bse` (default) or `fractions`	accretion limited by the thermal time scale of the accretor, or fixed fractions
`alpha`, `beta`	0–1, defaults 0	`fractions`: mass lost from the donor's and the accretor's vicinity ($\alpha + \beta \leq 1$)
`dynamic_onset`	`angular_momentum`, `periastron` or none (default)	continue when the dynamics drives a star over its Roche lobe: circularise the binary and start the transfer
`tertiary`	section (default present) or null	transfer from a star onto the binary it orbits (secular backend): `accretion` (`auto`), `split` (`auto`), `inner_orbit` (`energy` or `angular_momentum`), `alpha_lambda` (5), `h` (0)
`contact`	section or none (default)	follow contact binaries: `braking` (true), `q_min` (0.09)
`common_envelope`	section or none (default)	follow a common envelope with the energy formalism: `alpha` (1), `lam` (0.5), `duration_yr` (1)

Table 54. Keys of `physics.mass_transfer`. The prescriptions are in *Physics and Methods*, chapter Mass transfer, common envelope and mergers.

9.7.3 `physics.mergers`

Key	Type, default	Meaning
`mass_loss`	`glebbeek2013` (default) or a fraction	mass ejected by the merger
`rejuvenation`	`glebbeek` (default), `hurley` or a number	fractional main-sequence age of the product relative to the mass-weighted ages of the two stars
`spin_fraction`	0–1, default 0.3	rotation of the product as a fraction of break-up

Table 55. Keys of `physics.mergers`; `mergers: {}` uses the defaults.

9.8 `run`

Key	Type, default	Meaning
`t_max_yr`	years, required, ≠ 0	end of the run; negative = into the past
`backend`	`nbody` (default) or `secular`	the backend (How SWINGERsim works)
`integrator`	`ias15`	the N-body integrator
`n_output`	≥ 2, default 1000	number of stored outputs
`collisions`	default true	stop when two stars touch (N-body)
`stop_on_hierarchy_change`	default false	stop at the first confirmed change of the hierarchy
`hierarchy_persistence`	≥ 1, default 3	outputs a new hierarchy must persist before it is confirmed
`stop_on_roche_overflow`	default false	stop when a star fills its Roche lobe
`stop_when_e_reached`	map orbit → e	N-body: stop once every listed orbit has reached its eccentricity
`max_wall_s`	seconds	stop after this computing time (event `wall_time`)
`max_energy_error`	default 10^{-5}	abort if $\lvert\Delta E/E\rvert$ exceeds this
`extremes_every_orbits`	≥ 0, default 10	internal sampling of $e_{\max}$ and the closest periastron, in periods of the shortest orbit; 0 = only at outputs
`rate_window_yr`	years	length of the linear fits for the present rates of change of the inclinations, on each side of $t = 0$ (default: two periods of the perturbing orbit, at least 6 outputs)
`megno`	default false	compute MEGNO (Newtonian; not with mass loss or transfer)
`secular_octupole`	default true	secular backend: include octupole terms
`secular_rtol`	default 10^{-10}	secular backend: relative tolerance
`evolution_max_dr`	0–0.2, default 0.01	`evolution`: largest relative change of a radius within one segment

Table 56. Keys of `run`.

9.9 `ensemble`

Key	Type, default	Meaning
`n`	≥ 1, required	number of members
`seed`	default 1	random seed of the draws
`vary`	map parameter → value or distribution	the drawn parameters: `masses.<m>`, `orbits.<o>.<key>`, `stars.<m>.<key>` (numeric keys only)

Key	Type, default	Meaning
`directions`	list of `forward`, `backward`; default `[forward]`	integrate each member to $+\lvert t_{\max}\rvert$, $-\lvert t_{\max}\rvert$ or both
`require`	list of `stable_ma01`, `stable_v22`, `hierarchical`	redraw members that a criterion classifies as unstable (`hierarchical`: every inner apocentre inside the pericentre of the orbit around it)
`max_tries`	default 1000	draws allowed per member
`ma01_unstable_keep`, `ma01_window`	0–1, default 1; [low, high]	keep only a fraction of the draws classified unstable, or only draws whose smallest MA01 ratio lies in a window (training sets for classifiers)
`mlp`	default false	add the verdict of the classifiers of Vynatheya et al. (2023) to every member
`ghost_test`, `ghost_orbits`, `ghost_max_wall_s`	default false; 100; none	add the N-body ghost test to every member
`save_runs`	default false	also store the HDF5 file of every member
`observed`	map column → observation	measurements that weight the members (below)

Table 57. Keys of `ensemble`.

A value of `vary` is a number (fixed), `isotropic` (an inclination with $\cos i$ uniform), `angle` (uniform 0–360°) or a distribution with exactly one of the kinds of Table 58 and the optional keys `min`, `max` (truncation: the value is drawn again) and `mirror` (an inclination is replaced by $180^\circ - i$ with probability ½, for eclipsing binaries of unknown sense of revolution).

Kind	Form	Meaning
`uniform`	`[low, high]`	uniform
`loguniform`	`[low, high]`, positive	uniform in the logarithm
`normal`	`[mean, sigma]`	Gaussian
`isotropic`	`[i1, i2]`	inclination with $\cos i$ uniform between two limits
`choice`	list	one of the listed values

Table 58. Distributions of `ensemble.vary`.

An entry of `ensemble.observed` names a column of the ensemble summary (Outputs) and is one of:

- a measurement `{value, sigma}` – Gaussian weight;
- limits `{min, max}` (either or both) – weight 0 outside; `missing_ok: true` lets a missing value pass;
- a text value `{equals: intact}` or a list of values – for text columns such as `fate`.

The column name may start with `forward:`, `backward:` or `central:` (the mean of both) to take the value from that run; the default is `central` for the present rates of change (`*_didt0_deg_yr`, `*_dtiltdt0_deg_yr`) and `forward` for everything else.

9.10 `fit`

Key	Type, default	Meaning
`minima`	path, required	CSV table of minima: `pair`, `type` (`p`/`s`), `time`, `sigma`, `source`, `method` (also `binary`, `kind`, `bjd_tdb`, `sigma_d`, `ref`)
`time_system`	`BJD_TDB` (default), `HJD_UTC`, `JD_UTC`	time system of the table; the others are converted (needs `coordinates`; times before 1960 are taken as Universal Time, TT = UT + ΔT of Espenak & Meeus 2006)
`coordinates`	`{ra_deg, dec_deg}`	ICRS coordinates of the system
`ephemerides`	map pair → `{T0, P}`	linear ephemeris for the epochs and the O−C (default: fitted to the primary minima)
`exclude`	`{before, after, sources, pairs}`	drop minima by time, source or pair
`error_scale`	`none` (default), `global`, `per_source`	rescale the errors so that the reduced χ^2 is 1
`rv`	section	radial velocities (Table 60)
`lc`	list	light-curve segments (Table 61)
`inclination`	list	inclinations at epochs: `file` (columns `mid_hjd` or `time`, `i_deg`, `sigma_deg`, optional `limit` and `source`), `pair`, `label`, `min_sigma_deg` (0)
`stage1`	section	the analytic O−C model (Table 62)
`stage2`	section	the joint N-body model (Table 63)
`output`	section	`dir` (`output/fit`), `tables` (`[csv]`, also `latex`), `plots` (`[oc, oc_components]`, also `corner`), `predict` (`years` 5, `pairs`, `start`, `samples` 100)

Table 59. Keys of `fit`. Paths are relative to the configuration.

Key of `fit.rv`	Type, default	Meaning
`file`	path, required	CSV: `star`, `time` (BJD_TDB), `rv`, `sigma` [km/s], `source`
`gamma`	km/s, default 0	systemic velocity (start value; free as `rv.gamma`)
`offsets`	map source → km/s	zero-point offsets (start values; free as `rv.offset.<source>`)
`stars`	list	stars used (default all)
`max_sigma`	km/s	leave out velocities with a larger error
`centre`	`mass` (default) or `light`	velocity of the centre of mass, or flux-weighted velocity of the visible surface (distortion, rotation, Rossiter–McLaughlin effect)
`band`	default `V`	passband of the flux weights for `centre: light`

Table 60. Keys of `fit.rv`.

Key of `fit.lc[]`	Type, default	Meaning
`file`	path, required	CSV: `time` (BJD_TDB), `flux` (normalised), `sigma`, optional `segment`
`label`	text	name in the report
`band`	default `TESS`	passband
`exposure_min`, `n_sub`	min ≥ 0, default 0; 1–15, default 1	exposure or binning time and model points per exposure

Key of `fit.lc[]`	Type, default	Meaning
`baseline_degree`	0–2, default 1	polynomial trend per segment
`scale`	default true	scale the model per segment (dilution); false: depths from the stars as given
`gap_d`	default 0.5	without a `segment` column: split segments at gaps longer than this [d]
`segments`	list	use only these segments
`shapes`	`auto` (default), `as_configured`, `spheres`	star shapes: Roche surfaces where a star fills its lobe beyond `roche_threshold`, as in the configuration, or spheres
`roche_threshold`	0–1, default 0.5	volume radius / Roche-lobe radius above which `auto` uses a Roche surface
`mesh_resolution`, `coarse_resolution`	default (64, 128) and (24, 48)	mesh directions of the accurate and the coarse model
`fidelity`	`mapped` (default) or `direct`	fit on the coarse mesh corrected by the accurate one, or the accurate mesh in every evaluation
`red_noise`	default false	inflate the errors per segment by the red-noise factor β (needs `error_scale` or `clip_sigma`)

Table 61. Keys of an entry of `fit.lc`.

Key of `fit.stage1`	Type, default	Meaning
`terms`	list of `lite`, `apsidal`, `quadratic`; default `[lite]`	terms of every pair
`pairs`	map pair → terms	terms per pair
`shared_outer`	default false	one outer orbit for all pairs
`P3_range_d`	[min, max]	search range of the outer period
`clip_sigma`, `max_clip_fraction`	none; 0.1	outlier rejection
`sampler`	`none` (default) or `emcee`	MCMC after the least squares
`walkers`, `steps`, `seed`	48, 4000, 1	MCMC settings

Table 62. Keys of `fit.stage1`.

Key of `fit.stage2`	Type, default	Meaning
`bjd_t0`	BJD_TDB, required	the epoch of $t = 0$ of the configuration
`free`	list, required	free parameters: `orbits.<o>.<key>` with key `P`, `M0`, `e`, `omega`, `Omega`, `i`, `ecosw`, `esinw` or `Pdot`; `masses.<m>`; `stars.<m>.<key>` with key `R`, `Teff`, `apsidal_k2` or `rg2`, `rv.gamma`, `rv.offset.<source>`
`pairs`	list	pairs whose minima are fitted (default all)
`max_sigma_d`	days	leave out minima with a larger error
`anchor`	default false	first anchor P and M_0 of every eclipsing pair to the minima

Key of `fit.stage2`	Type, default	Meaning
`priors`	map parameter → `[mean, sigma]` or `{normal: [mean, sigma]}`	Gaussian priors on free parameters (on `e`, `omega` also when `ecosw`, `esinw` are free, and the reverse)
`bounds`	map parameter → [low, high]	default: P within 0.5–2 × the start, e 0–0.95, i 0–180°, masses 0.05–20 ×
`fd_steps`	map parameter → step	finite-difference steps (default: P 10^{-6} relative, angles 0.01°, e 10^{-4}, masses 10^{-4} relative)
`loss`, `f_scale`	`linear` (default), `soft_l1`, `huber`, `cauchy`; 3	robust loss from `f_scale` σ
`error_scale`	`none`, `global`, `per_source`; default `fit.error_scale`	error correction after a first fit
`error_model`	`scale` (default) or `jitter`	factor on the errors or an error floor in quadrature
`min_source_minima`	default 5	sources with fewer minima share one correction
`clip_sigma`, `max_clip_fraction`	none; 0.1	outlier rejection
`refits`	1–10, default 4	fits in a row while corrections or outliers change
`max_iterations`	default 30	iterations of the optimiser
`processes`	default 0 (all)	cores for the derivatives
`sampler`	`none` (default) or `emcee`	MCMC after the least squares
`walkers`, `mcmc_steps`	0 (= 2 × free + 2), 300	MCMC size; the first half of the chain is discarded
`mcmc_tau_target`, `mcmc_max_steps`	0, 5000	extend the chain until it is this many autocorrelation times long
`seed`	default 1	seed of the MCMC

Table 63. Keys of `fit.stage2`. `bounds`, `fd_steps` and `priors` must name free parameters; MCMC keys have no effect with `sampler: none` (a warning says so).

9.11 The `scatter` configuration

A scattering experiment (`swingersim scatter`) uses a configuration of its own with the keys `name`, `description` and `scatter`:

Key of `scatter`	Type, default	Meaning
`target`, `projectile`	body, required	a single star or a binary: `masses` (one or two, $M_\odot$), `a_au` (binaries), `e` (0), `e_thermal` (draw e from $f(e) = 2e$), `radii_rsun` (enables collisions), `polytrope_n` (for tidal capture)
`v_inf_kms` or `v_inf_vc`	one required	velocity at infinity [km/s] or in units of the critical velocity
`b_au`, `b_max_au` or `r_peri_max_au`	one required	fixed impact parameter, largest one, or largest pericentre (with gravitational focusing)

Key of `scatter`	Type, default	Meaning
`b_distribution`	`area` (default) or `linear`	uniform in b^2 or in b
`r_start_factor`	default 50	start distance in units of the largest semi-major axis
`t_max_yr`, `max_wall_s`, `check_periods`	none, 600, 5	limits and the interval of hierarchy checks
`gr`, `tidal_capture`	default false	1PN relativity; tidal energy loss at close passages
`n`, `seed`	100, 1	number of encounters and random seed

Table 64. Keys of a scattering configuration.

9.12 A complete example

The following configuration is a triple with an eclipsing inner binary and an unseen third star. It runs as it is with `run`, `timings`, `lightcurve` and `stability`, and its `ensemble` and `fit` sections are used by `ensemble` and `fit` (the table `minima.csv` must exist next to it).

```yaml
topology: custom
hierarchy:
  out: [inner, m3]          # the inner binary and a third star
  inner: [m1, m2]
masses: {m1: 2.0, m2: 1.8, m3: 1.0}
orbits:
  inner: {P: 3.5, e: 0.0, i: 88.0, Omega: 0.0}
  out:   {P: 150.0, e: 0.3, i: 70.0, Omega: 20.0, omega: 45.0, M0: 10.0}
stars:
  m1: {R: 2.1, Teff: 9000}
  m2: {R: 1.8, Teff: 8200}
stellar_model: {kind: mist, age_myr: 100, feh: 0.0}
physics: {gr: true, tides: true, rotation: true}
run: {backend: nbody, t_max_yr: 100, n_output: 2001}
ensemble:
  n: 500
  seed: 7
  directions: [forward, backward]
  require: [stable_ma01]
  vary:
    masses.m3: {loguniform: [0.3, 3.0]}
    orbits.out.P: {loguniform: [50.0, 2000.0]}
    orbits.out.e: {uniform: [0.0, 0.6]}
    orbits.out.i: isotropic
    orbits.out.Omega: angle
  observed:
    inner_i0: {value: 88.0, sigma: 1.0}
    inner_didt0_deg_yr: {value: -0.05, sigma: 0.02}
    "backward:fate": {equals: intact}
fit:
  minima: minima.csv
  time_system: BJD_TDB
  stage2:
    bjd_t0: 2459000.0
    anchor: true
    free: [orbits.inner.P, orbits.inner.M0, orbits.out.P, orbits.out.M0,
           orbits.out.e, orbits.out.omega, masses.m3]
    error_scale: per_source
```

```
    error_model: jitter
    clip_sigma: 5.0
  output:
    dir: output/example_triple_fit
    predict: {years: 5}
```

Chapter 10

Outputs

This chapter describes every file SWINGERsim writes, the meaning of its columns, how to interpret the values and how to read the files in Python. All tables are CSV files whose first lines may start with #: they record the software versions, the units, the time system and, for some, the configuration; the first line that does not start with # is the header.

10.1 The summary of a run

`swingersim run` prints a summary, which is also saved as `<name>_summary.txt`. Example 3 of Worked examples (a 2+2 with tides and rotation) gives, shortened:

```
integrated to t = 300 yr (1500 outputs, osculating elements)
  out    a = 2.442..2.4431 AU  e = 0.3998..0.4002
  A      a = 0.041764..0.041767 AU  e = 0.0494..0.0502
  i_mut A-B           8.44..  25.08 deg
  max |dE/E| = 2.22e-11
  max |dL/L| = 1.83e-07
hierarchy: configured [[1,2],[3,4]]
  no change detected
orbit pairs:
  A in out: i_mut0 = 11.6 deg, g1_0 = 98 deg
    quadrupole theory : e = 0.048..0.050, i_mut = 11.6..11.6 deg, e-cycle 323 yr (outside
 ↪  ZLK window)
    simulation        : e = 0.049..0.050, i_mut = 11.6..13.6 deg
    precession period : 1.29e+03 yr estimated, not completed in the run, closest periastron
 ↪  = 2.78 x (R1+R2)
    stability ratio Mardling-Aarseth = 7.97 (stable), Vynatheya+22 = 10.17 (stable),
 ↪  eps_oct = 0.00054, eps_SA = 0.0017
    GR apsidal period ~ 2.31e+03 yr
   tidal+rotational apsidal period 64 yr, ZLK time 949 yr -> faster than ZLK, ZLK suppressed
binaries A and B: P_A/P_B = 0.6207 (nearest 3:5, +3.45 %), precession ratio A/B = 1.97
 ↪  estimated
  mutual inclination of A and B: 8.44..25.08 deg, analytic limit 104.21 deg (Vokrouhlicky
 ↪  2016, circular A and B)
  eclipses (observer along z): A 100 %, B 91 %, both at once 91 % of the run
secular coupling beta (Hamers & Lai 2017): A 1.45, B 0.365
stars:
  m1  R = 1.7 Rsun, k2 = 0.0144, rg2 = 0.0754, lag = 0 s, P_rot = 1.8 -> 1.8 d, obliquity
 ↪  0.0 -> 0.1 deg, Roche lobe filled up to 0.51
  binary A (m1+m2): apsidal period 62.29 yr from the binary itself, tides+rotation 64.01 yr,
 ↪  GR 2311 yr; measured 59.43 yr (with the other stars)
  wall time 152.9 s
```

Line	Content
`integrated to`	time reached, number of outputs and the kind of elements (osculating for N-body, mean for secular); shorter than `t_max_yr` if an event stopped the run
`EVENT`	only when the run stopped early: `collision` (with the stars), `roche_overflow`, `hierarchy_change`, `energy_error`, `wall_time`, `e_reached`, `stellar_evolution_end`, `stellar_birth`, `gw_unresolved`, or `mass_transfer: <kind>` (below)
`MERGER`, `MASS TRANSFER STARTED BY THE DYNAMICS`	with `physics.mergers` or `mass_transfer.dynamic_onset`: what happened and how the run continued
one line per orbit	minimum..maximum of the semi-major axis [au] and the eccentricity
`i_mut X-Y`	minimum..maximum of the mutual inclination of two orbits [deg]
max |dE/E|, max |dL/L|	largest relative error of the energy and of the total angular momentum; a measure of numerical accuracy (with dissipation the energy changes physically and a note says so)
`hierarchy`	the configured hierarchy and every confirmed change: time, kind, old → new structure
`X in Y`	each inner orbit in its parent orbit: initial mutual inclination and argument of periastron measured from the line of nodes
`quadrupole theory`	prediction of the orbit-averaged quadrupole equations: ranges of e and i_{mut} and the period of the eccentricity cycle; whether the orbit is inside the ZLK window (39.2–140.8°)
`simulation`	the same ranges measured in the run (the maximum of e includes peaks between outputs) and the number of completed ZLK cycles
`precession period`	estimated and measured nodal precession period; closest periastron in units of the sum of the stellar radii
`stability ratio`, `eps_oct`, `eps_SA`	stability ratios of Mardling & Aarseth (2001) and Vynatheya et al. (2022) for this nested triple (above 1 stable); octupole strength; single-averaging parameter – above about 0.01 the averaged estimates lose accuracy (Luo et al. 2016) and a note says so
`GR apsidal period`, `tidal+rotational apsidal period`	relativistic and (with `physics.tides`) tidal and rotational apsidal periods, compared with the ZLK time scale; a note says when ZLK cycles are suppressed
`binaries A and B`	2+2 cores: period ratio and the nearest small-integer commensurability, ratio of the precession periods; the range of the mutual inclination of A and B and its analytic limit (Vokrouhlický 2016)
`eclipses`	fraction of the run during which each binary, and both at once, eclipse for an observer along z
`secular coupling beta`	parameter β of Hamers & Lai (2017) for each inner orbit of a 2+2
`stars`	radius, k_2, $I/(MR^2)$, lag, rotation period and obliquity at start and end, largest Roche-lobe filling; with evolving stars the age, radius, T_{eff} and phase at start and end
`binary X`	each binary of two stars: apsidal period expected from the binary itself and measured in the run (which includes the perturbation by the other stars)
`tidal time scales`	with a tidal lag: circularisation, orbital decay and synchronisation times, pseudo-synchronous rotation period
`MEGNO`	with `run.megno`: the chaos indicator and the Lyapunov time
`wall time`	computing time

Table 65. Lines of the summary of a run. Left: the start of the line; right: its content.

The kinds of `mass_transfer` events are, for a binary, `common_envelope`, `contact`, `contact_merger_darwin`, `contact_merger_l2`, `common_envelope_merger`, `post_common_envelope` and `donor_stripped`; for transfer from a tertiary, `triple_common_envelope`, `inner_roche_overflow`, `dynamical_instability` and `donor_stripped`.

10.2 Files of a run

`swingersim run config.yaml -o out/sys.h5` writes, next to each other:

File	Content
`sys.h5`	all data of the run (below)
`sys.html`	the HTML report: command line, configuration, summary, interactive plots, the figures
`sys_summary.txt`	the summary and the list of the files of the run
`sys_timeseries.csv`	one row per output time (at most 20 000 rows; every k-th output for longer runs, all of them in the HDF5 file)
`sys_viewer.html`	the interactive 3D viewer (needs plotly)
`sys_orbits.png`	a/a_0, e and the inclination to the sky of every orbit over time
`sys_inclinations.png`	the mutual inclinations over time
`sys_conservation.png`	relative errors of the energy and of the angular momentum
`sys_stars.png`	mass, radius and $T_{\rm eff}$ of the stars over time (only with evolving stars)

Table 66. The standard products of a run. Every run started by the worker or the web application writes the same files; `--no-viewer`, `--no-plots` and `--no-html` leave out the viewer, the figures and the report.

The columns of `sys_timeseries.csv` (orbit names as in the configuration; angles in degrees, unlike the radians of the HDF5 file):

Column	Content
`t_yr`	time [yr] from $t = 0$ (negative for backward runs)
`a_<orbit>_au`, `e_<orbit>`	semi-major axis and eccentricity of every orbit
`i_<orbit>_deg`, `i_inv_<orbit>_deg`	inclination to the sky and to the invariable plane (the plane perpendicular to the total angular momentum)
`Omega_<orbit>_deg`, `Omega_inv_<orbit>_deg`, `omega_<orbit>_deg`	node on the sky and on the invariable plane, argument of periastron
`imut_<o1>-<o2>_deg`	mutual inclination of every pair of orbits
`dE_over_E`, `dL_over_L`	relative errors of the energy and of the angular momentum
`M_<star>_msun`, `R_<star>_rsun`, `Teff_<star>_k`	mass, radius and $T_{\rm eff}$ (only with evolving stars)

Table 67. Columns of the time-series CSV.

10.2.1 The HDF5 file

HDF5 is a standard binary format for numerical data, readable from Python (`h5py`) and tools such as HDFView, MATLAB or Julia.

Path	Content	Unit
`time_yr`	output times	yr
`energy_error`, `angmom_error`	relative errors of the energy and of the total angular momentum (orbits + spins)	–
`orbits/<orbit>/a`, `e`	semi-major axis, eccentricity	au, –
`orbits/<orbit>/inc`, `Omega`, `omega`, `M`	inclination to the sky, node, argument of periastron, mean anomaly	**rad**
`orbits/<orbit>/inc_inv`, `Omega_inv`	inclination and node with respect to the invariable plane	**rad**
`mutual_inclination/<o1>-<o2>`	mutual inclination	**rad**
`stars/<m>` (attributes)	values used: `radius_rsun`, `apsidal_k2`, `rg2`, `lag_s`, `mass_msun`, `mass_msun_config`, Roche-lobe filling (`roche_fill_max`, its time), and from the stellar model `luminosity_lsun`, `teff_k`, `age_myr`, `phase`, envelope mass and radius, convective turnover time	$R_\odot$, –, –, s, $M_\odot$ …
`stars/<m>/P_rot_d`, `obliquity_deg`	rotation period and spin–orbit angle at each output (with `physics.tides`)	d, deg
`stars/<m>/radius_rsun`, `luminosity_lsun`, `teff_k`, `phase`, `age_myr`, `mass_msun`	the evolving star at each output (with `physics.evolution` / `mass_loss`); phase 0 pre-main sequence, 1 main sequence, 2 Hertzsprung gap, 3 giant branch, 4 core helium burning, 5 early AGB, 6 thermally pulsing AGB	$R_\odot$, $L_\odot$, K, –, Myr, $M_\odot$
`megno`	MEGNO at each output (with `run.megno`)	–
`extremes/<orbit>` (attributes)	`e_max`, `t_e_max_yr`, `q_min_au`, `t_q_min_yr`: extremes found between outputs	–, yr, au
`hierarchy/detected`	the detected hierarchy at each output, e.g. `[[1,2],[3,4]]`	text
`hierarchy` (attributes)	`configured`, `final`, `transitions_json` (confirmed changes: time, kind, from, to), `detected_changes_json`	–
`diagnostics/<inner orbit>` (attributes)	the diagnostics of each orbit pair (Table 69)	–

Table 68. Content of the HDF5 file of a run. Angles are stored in radians.

File attributes: `config_yaml` (the complete configuration with all defaults, so that the run can be repeated), `swingersim_version` and the versions of Python, NumPy, SciPy, h5py, REBOUND, REBOUNDx and emcee, `created_utc`, `wall_s`, `units_json` (the conventions: Julian years from $t = 0$, au, $M_\odot$, radians, the sky frame), `backend`, `event`, `event_time_yr` and the stars and age of the event, `dissipative`, `gas_phase`, `mass_loss` (true when the energy is not conserved physically), `binaries_json` (per binary of two stars: the expected and measured apsidal periods, the tidal time scales, the eclipse fraction), `system_json` (period ratios and commensurabilities of 2+2 cores, the analytic limit of the mutual inclination, eclipse fractions, the coupling parameters, the mass-loss indices, the mass-transfer and common-envelope history of every orbit, the numbers of passing-star encounters) and `lyapunov_time_yr` (with `run.megno`).

Hierarchy notation. Brackets group bound pairs: `[[1,2],[3,4]]` is a 2+2, `[[[1,2],3],4]` a 3+1; unbound parts are separated by `|`, e.g. `[[1,2],3] | 4` is a triple with an escaping star. The kinds of change are `exchange` (same type, different partners), `reconfiguration` (the type

changes, e.g. 2+2 → 3+1), `ejection` (one star becomes unbound), `breakup` (several unbound parts) and `capture` (unbound parts become bound).

Field	Meaning
`i_mut0_deg`, `g1_0_deg`	initial mutual inclination to the parent orbit; initial argument of periastron from the node on the local invariable plane
`in_zlk_window`, `e_max_quadrupole`	whether $i_{\mathrm{mut},0}$ lies between 39.2° and 140.8°; the test-particle estimate $\sqrt{1-\frac{5}{3}\cos^2 i_0}$
`t_zlk_yr`, `t_prec_yr`, `t_gr_yr`, `gr_suppresses_zlk`	ZLK time scale, nodal precession period, relativistic apsidal period, and whether relativity is faster than ZLK
`epsilon_oct`, `epsilon_sa`	octupole strength (Naoz 2016) and single-averaging parameter (Luo et al. 2016)
`stability_ratio`, `stability_ratio_v22`	Mardling–Aarseth and Vynatheya et al. (2022) ratios
`e_min`, `e_max`, `i_mut_min_deg`, `i_mut_max_deg`	ranges in the run
`e_max_true`, `t_e_max_yr`, `q_min_au`, `q_min_over_radii`	largest eccentricity including internal samples, its time; smallest periastron, also in units of $R_1 + R_2$
`quad_e_min`, `quad_e_max`, `quad_i_min_deg`, `quad_i_max_deg`, `quad_period_yr`	prediction of the averaged quadrupole equations (Naoz et al. 2013)
`n_zlk_cycles`, `zlk_period_yr`, `prec_period_measured_yr`	completed eccentricity cycles and their period; precession period measured about the invariable plane

Table 69. Diagnostics of an orbit pair (attributes of `diagnostics/<inner orbit>`).

10.3 Ensembles

`swingersim ensemble config.yaml -o folder` writes into the folder:

File	Content
`ensemble.yaml`	the full configuration and its fingerprint (a resumed ensemble must use the same one)
`rows.jsonl`	one line per finished member and direction, written as members finish; `--resume` continues from it
`summary.csv`	the same rows as a table, sorted by member and direction
`summary.parquet`	the same table for pandas (with the `ensemble` group installed)
`runs/member_<n>_<direction>.h5`	the full run of every member (only with `save_runs: true`)
`report.html`	the report

Table 70. Files of an ensemble.

Column	Meaning
`member`, `direction`	number of the member; `forward` or `backward`
`masses.m3`, `orbits.out.P`, …	the drawn values of the `vary` parameters
`fate`	`intact` (no event, hierarchy unchanged), a hierarchy change (`hierarchy_change`, `exchange`, `ejection`, `breakup`, `reconfiguration`, `capture`), an event (`collision`, `merger`, `roche_overflow`, `energy_error`, `wall_time`, `e_reached`, `gw_unresolved`) or `error`
`event`, `event_time_yr`, `t_end_yr`	the event, its time, the end of the run
`n_transitions`, `first_transition_yr`, `final_hierarchy`	changes of the hierarchy
`<orbit>_e0`, `_e_min`, `_e_max`, `_log1me_min`	initial, smallest, largest eccentricity; $\log_{10}(1-e_{\max})$
`<orbit>_t_e_max_yr`, `_q_min_au`, `_a_min`, `_a_max`	time of the largest eccentricity, closest periastron, range of a
`<orbit>_t_1me_lt_1e-1_yr` … `_1e-4_yr`	first time at which $1-e$ fell below 10^{-1} … 10^{-4} (empty if never)
`<orbit>_U_measured_yr`	apsidal period measured over the run (empty when the periastron advanced too little)
`imut_<o1>_<o2>_0`, `_min`, `_max`	initial, smallest, largest mutual inclination [deg]
`<binary>_eclipse_fraction`, `eclipse_fraction_both`, `eclipse_fraction_all`	fraction of the run during which the binary (both binaries of the first 2+2; all binaries) eclipse
`<binary>_i0`, `_didt0_deg_yr`, `_dtiltdt0_deg_yr`	inclination to the sky now; its rate of change; the rate of change of $\lvert i-90^\circ\rvert$ (negative: turning towards edge-on, deeper eclipses); fitted one-sided from $t=0$ over `run.rate_window_yr`
`<binary>_eclipsing0`, `_eclipse_change_yr`	1 if the binary eclipses now; the first time the eclipse state changes
`<star>_roche_fill_max`, `roche_first`, `roche_first_kind`	largest Roche-lobe filling; with `roche_overflow`, the star that filled its lobe first and whether its companion is a star or a subsystem
`mass_transfer_*`	the scalar mass-transfer and common-envelope entries of the run
`ma01_min`, `v22_min`	smallest stability ratio of all nested triples at $t=0$
`mlp_stable`, `ghost_stable`, `ghost_delta_max`	with `mlp` / `ghost_test`: 1 stable, 0 unstable
`max_energy_error`, `wall_s`	quality and cost of the run

Table 71. Columns of the ensemble summary. These names are the ones used in `ensemble.observed`.

`swingersim ensemble-report` prints, separately forward and backward, the fraction of every fate, the distribution of the largest eccentricities and the fractions with $1-e$ below 0.1, 0.01 and 0.001, the mutual inclinations, the present rates of change and when eclipses end or began, the eclipse fractions and the stability verdicts; fractions come with their 68 % Wilson interval (Wilson 1927). With observations every number is weighted, the header gives the effective number of members, and the drawn parameters are listed as prior → conditioned median and 16–84 % range.

10.4 Eclipse times, minima, light curves and radial velocities

`swingersim timings` writes one row per eclipse:

Column	Meaning
`binary`, `kind`	orbit name; `primary` (the first star of the binary eclipsed) or `secondary`
`epoch`	cycle number of the linear ephemeris (half-integer for secondary eclipses)
`t_yr`, `t_obs_yr`	time of mid-eclipse [yr from $t = 0$], without and with the light-travel time
`o_c_d`	observed minus calculated [d] against the linear ephemeris of the primary eclipses
`impact`	projected separation at mid-eclipse / $(R_1 + R_2)$: 0 central, near 1 grazing
`T0_yr`, `P_d`	the fitted linear ephemeris
`bjd_obs`	with `--bjd-t0`: the BJD of the minimum

Table 72. Columns of the eclipse-times CSV. The summary on screen gives the number of eclipses, the period, the range of the O−C and the intervals without eclipses; with `physics.activity` it also reports the energy check of every active star.

`swingersim minima` writes one row per measured minimum:

Column	Meaning
`binary`, `kind`, `epoch`	as for the eclipse times
`time`, `t_min_d`, `sigma_d`	time of minimum (BJD_TDB, and days after $t = 0$) from the Mikulášek (2015) profile and its standard error
`t_geom_d`	the geometric conjunction with light time [d]
`o_c_d`, `o_c_geom_d`	O−C of the minima and of the conjunctions against their own linear ephemerides [d]
`depth_mag`, `width_d`, `rms_mag`	depth and width of this eclipse, residuals of its fit

Table 73. Columns of the minima CSV. `--seasons` writes the shape per season (`t_mid_d`, `n`, `depth_mag`, `width_d`, `gamma`, `corr`, `rms_mag`).

`swingersim lightcurve` writes one row per time: `t_day` (days after $t = 0$, observer's time), `flux` (total flux normalised to the sum of the stars outside eclipse), `flux_m1`, … (the contribution of every star; pulsating stars include their pulsation, stars behind a disc its absorption) and `flux_disc_<name>` (the light of discs with emission or scattering). With `--engine phoebe` the columns are `flux_A`, `flux_B`, … (each inner binary), the single stars and `flux_mutual` (light lost in eclipses between different binaries). The header lines give the band, the time convention and the BJD of $t = 0$ if known.

`swingersim rv` writes `rv_model.csv` (columns `bjd_tdb`, `rv_m1`, `rv_m2`, …, km/s) and a figure `rv_model.png` with the measured velocities if given.

`swingersim apsidal` writes per pair the period, eccentricity, k_2, radii and rotation periods of both stars, the tidal advance, the rotational advance for synchronous, pseudo-synchronous and configured rotation, the relativistic advance, the total (deg/yr) and the apsidal period U (yr).

10.5 Fits

`swingersim fit` writes into `fit.output.dir`. Stage 1:

File	Content
`stage1_parameters.csv`, `.tex`	fitted parameters with 1σ errors (`<pair>.T0`, `.P`, `.Q`, `.e`, `.w0`, `.dw` [rad/cycle], `.asini` [au], `.P3`, `.T3`, `.e3`, `.w3`; `out.*` for a shared outer orbit) and derived values: `.K_s` (semi-amplitude of the light-time effect), `.f_m3` (mass function), `.P3_yr`, `.U_yr` (apsidal period), `.Pa_d` (anomalistic period), `mass_ratio_A_over_B`
`stage1_oc.csv`	every minimum: pair, type, epoch, time, sigma, source, O−C against the linear ephemeris, the model, the residual [d]
`stage1_summary.json`	χ^2, degrees of freedom, parameters, derived values, error scaling, outliers, ephemerides, software versions
`stage1_predictions.csv`	with `output.predict`: future minima and their spread
`stage1_oc.png`, `stage1_corner.png`	O−C with model and residuals and the terms separately; corner plot of the MCMC samples

Table 74. Files of a stage-1 fit.

Stage 2:

File	Content
`stage2_params.csv`	every free parameter: `parameter`, `start`, `value`, `error` (1σ; with MCMC half the 16–84 % range), `unit`, `prior_mean`, `prior_sigma`, `error_linear` (from the covariance matrix), `p16`, `p50`, `p84` (MCMC percentiles)
`stage2_covariance.csv`	the covariance matrix of the linear model
`stage2_residuals.csv`	every minimum: `pair`, `type`, `bjd_tdb`, `sigma_d` (the error used), `model_start_bjd`, `model_bjd`, `o_minus_model_min`, `source`, `sigma_quoted_d` (the error of the table), `status` (`fitted` or `outlier`)
`stage2_rv.csv`	every velocity: `star`, `bjd_tdb`, `rv`, `sigma`, `sigma_quoted`, `model_start`, `model`, `o_minus_model`, `source`, `status`
`stage2_lc.csv`	every light-curve point: `source`, `segment`, `bjd_tdb`, `flux`, `sigma`, `sigma_quoted`, `model_start`, `model`, `o_minus_model`
`stage2_inclination.csv`	every inclination epoch: `label`, `pair`, `time`, `i_obs`, `sigma`, `limit`, `i_start`, `i_model`, `i_eclipse_limit`, `z` (normalised residual), `source`
`stage2_fitted.yaml`	the configuration with the fitted values and `fit_epoch_bjd_tdb`, without the `fit` section; its header gives χ^2, the rms per pair and the epoch; it runs as it is
`stage2_report.txt`	χ^2 from start to fit, rms per pair and data type, parameters with errors, strongly correlated pairs ($\lvert r\rvert > 0.95$), error factors or jitters per source, priors and their deviations, the fits in a row and the outliers; the same text is printed
`stage2_history.json`	χ^2 and the parameters at every model evaluation
`stage2_samples.csv`	with MCMC: the samples after burn-in
`stage2_predictions.csv`, `.png`	predicted minima: `pair`, `type`, `bjd_tdb`, `sigma_min`, `bjd_16`, `bjd_84`, `duration_h`, `impact`, the observing window and, with coordinates, UTC times
`stage2_oc.png`, `stage2_rv.png`, `stage2_lc.png`, `stage2_inclination.png`	figures of the O−C, the velocities, the light curves and the inclinations with start and fitted model and residuals
`anchored.yaml`	the anchored start configuration (with `anchor` or `--anchor`)
`report.html`	the report

Table 75. Files of a stage-2 fit.

10.6 Scattering experiments and chaos maps

`swingersim scatter` writes `scatter.yaml` (the configuration and the critical velocity), `rows.jsonl` (one line per encounter: impact parameter, outcome, status, final hierarchy, end time, energy error, the bound orbits at the end, stability of a bound subsystem, colliding stars, total energy and angular momentum, the drawn initial conditions) and `summary.txt` (number, fraction with its 68 % Wilson interval and cross-section of every outcome).

`swingersim chaosmap` writes `chaosmap.json` (the grid and the configuration), `grid.jsonl` (one line per grid point: indices, values, `megno`, `lyapunov_yr`, `t_end_yr`, `status` – `ok`, `escape`, `collision` or `wall_time`), `summary.txt` (numbers of regular, chaotic and escaping points and the stability limit along x) and `map.png` (MEGNO over the grid, clipped at 10; escapes and collisions drawn as 10). `lyapunov_yr` (and `lyapunov_time_yr` of a run) is the Lyapunov time $1/\lambda$: REBOUND's `sim.lyapunov()` is the slope of the mean MEGNO, $\langle Y \rangle \approx \lambda t/2$, so SWINGERsim reports $1/(2\,\text{slope})$. The initial tangent vector of MEGNO has a fixed seed, so a map is the same on every run.

10.7 Machine learning

`emulator-train` writes the model (`.pkl`) and `model.json` next to it (columns, scores, settings) and, with `--plot`, its diagnostics; `emulator-report` prints the posterior of every input with the effective sample size. `sbi` prints the posterior (median, 68 % and 95 % intervals) and optionally writes the network (`--model`), the samples (`--save-samples`), a corner plot and the calibration diagnostics.

10.8 How to interpret the values

- **Osculating elements** of the N-body backend wiggle on orbital time scales (typically below 1 % in a and 0.01 in e); look at trends and at extremes over many outer orbits. The secular backend gives mean elements without these wiggles.
- **Semi-major axes** stay constant on average in point-mass dynamics (with mass loss aM stays constant). A drift together with a growing energy error points to a numerical problem; a sudden jump to a close encounter.
- **Inclination to the sky** decides whether a binary eclipses; **mutual inclination** decides the dynamics (precession, ZLK).
- **Ω and ω** can wrap around 360°; unwrap them before plotting.
- **NaN** means that an orbit was not bound at that output, e.g. after a break-up.
- **Apsidal periods** measured in a run include the perturbation by the other stars and can differ strongly from the value of the isolated binary.
- **Roche-lobe filling** near 1 means that a star nearly fills its lobe at periastron.
- **Energy error.** Without relativity 10^{-14}–10^{-12} is normal; with relativity the 1PN energy is checked and 10^{-13}–10^{-11} is typical; values approaching `max_energy_error` (10^{-5}) mean that the result should not be trusted. With dissipation, gas or mass loss the energy changes physically.
- **Primary eclipse** in SWINGERsim is the eclipse of the first star of the binary, whatever the depth; an observer usually calls the deeper eclipse primary.
- **Model time** starts at $t = 0$, the epoch of the elements; measured BJDs are compared after giving the BJD of $t = 0$ (`--bjd-t0`, `fit.stage2.bjd_t0`).

10.9 Reading the files in Python

```
import h5py
import numpy as np
```

```
import pandas as pd

with h5py.File("output/example.h5") as f:
    t = f["time_yr"][:]
    e_A = f["orbits/A/e"][:]
    i_AB = np.degrees(f["mutual_inclination/A-B"][:])      # radians in the file
    print(f.attrs["event"] or "no event")
    print(f.attrs["config_yaml"])

ts = pd.read_csv("output/example_timeseries.csv", comment="#")
ens = pd.read_csv("output/v994/summary.csv")
fwd = ens[ens.direction == "forward"]
print(fwd.fate.value_counts(normalize=True))
res = pd.read_csv("output/example_fit/stage1_oc.csv", comment="#")
```

Chapter 11

Troubleshooting and FAQ

The problems met most often, ordered by where they appear. Each table gives the symptom, its usual cause and the fix. General questions are answered at the end.

11.1 The configuration is refused

SWINGERsim checks the whole configuration before it starts and stops with a message that names the key. In the web application the same message appears next to the field.

Message (shortened)	Cause	Fix
`give exactly one of P (days) or a (AU)`	an orbit has both or neither	keep one
`hierarchy: unknown child 'x'`	a name in `hierarchy` is neither a star (`m1`, …) nor an orbit	check the spelling; stars are numbered from 1 without gaps
`hierarchy: needs exactly one outermost orbit`	two orbits are not joined	add the orbit that joins them
`topology … needs orbits [...], masses must be exactly [...]`	the orbits or masses do not match the hierarchy	use *Add stars and orbits* in the web application, or compare with Table 46
`stellar_model.kind = mist needs 'age_myr'`	evolving models need an age	give `stellar_model.age_myr`
`physics.rotation needs physics.tides`	rotation and tides share one model of the stellar distortion	add `tides: true`
`physics.evolution: stars [...] have a fixed radius R or Teff`	with evolution, radii and temperatures follow from mass and age	remove `R` and `Teff`, or switch evolution off
`physics.mass_transfer needs physics.evolution: true and stellar_model.kind sse or mist`	transfer histories come from the tracks	switch on evolution with `sse` or `mist`
`physics.gas needs run.backend: secular`	the gas phase is implemented in the secular backend	set `run.backend: secular`
`run.megno assumes constant masses`	MEGNO with mass loss or transfer	switch one of them off
`give value and sigma, or min and/or max (or equals)`	an `observed` entry is incomplete	an observation is a value with an error, a limit, or a text value

Message (shortened)	Cause	Fix
`fit.stage2.priors.<name>: not a free parameter`	a prior on a fixed parameter has no effect	free it or remove the prior
`fit.lc red_noise: true … needs fit.stage2.error_scale … or clip_sigma`	red-noise factors are applied between fits in a row	add `error_scale` or `clip_sigma`
`Extra inputs are not permitted`	a typo in a key; keys are case-sensitive	compare with The configuration file

Table 76. Common messages when a configuration is refused.

11.2 The run is slow or stops early

Symptom	Cause	Fix
an N-body run takes hours	every revolution of the shortest orbit needs tens of steps	use the secular backend if the hierarchy stays intact; shorten `t_max_yr`; install Numba (`fast`) for the secular backend
`EVENT: collision` early	two stars touch at periastron	expected for high eccentricities; check the radii and whether tides and relativity suppress ZLK cycles
`EVENT: roche_overflow`	a star fills its Roche lobe	switch on `physics.mass_transfer` to follow it, or take the time of the event as the answer
`EVENT: stellar_evolution_end`	a star reached the end of its track	the run cannot continue past the models
`EVENT: wall_time`	`run.max_wall_s` reached	raise it or use the secular backend
`EVENT: energy_error`	$\lvert\Delta E/E\rvert$ exceeded `run.max_energy_error`, usually at a very close approach	look for near-collisions; check with the stability tools
the secular backend refuses a run with mass loss	the mass-loss index Ψ exceeds `max_psi`: the orbit-averaged adiabatic response is not valid	use the N-body backend

Table 77. Problems during a run.

11.3 The results look wrong

Symptom	Cause	Fix
no eclipses although the binary is observed to eclipse	the default zero-age main-sequence radii are too small for the inclination	give the measured `stars.<m>.R` or an evolved stellar model; the Quick look shows the eclipse limit
apsidal motion much too fast	the polytropic k_2 of `zams` is too large for stars of 2–3 $M_\odot$	use `mist` or `sse`, or give measured `apsidal_k2`
ZLK cycles reach $e \approx 1$ and the stars collide	point masses: no tidal or relativistic precession	switch on `gr`, `tides` and `rotation`

Symptom	Cause	Fix
osculating elements wiggle	they are instantaneous ellipses, perturbed on every orbit	normal; the secular backend reports the averages
a rate of change differs from a published one	different time window or definition	set `run.rate_window_yr` to about half of the observing baseline
primary and secondary eclipses swapped against a paper	SWINGERsim's primary is the eclipse of the first star of the pair	swap the stars or the types
the O−C of a model has the wrong phase against measured minima	the model has no calendar	give the BJD of $t = 0$ (`--bjd-t0`)
different ensemble result each time	the seed changed	fix `ensemble.seed`

Table 78. Results that look wrong.

11.4 Ensembles

Symptom	Cause	Fix
an interrupted ensemble starts from zero	`--resume` missing	run the same command with `--resume`
members ended with `error`	a configuration that fails for some draws	read the message in `rows.jsonl`; after a fix use `--resume --retry-errors`
the conditioned result rests on a few members	precise observations, broad prior	run more members, narrow the prior, or use an emulator or SBI
fractions with wide intervals	too few members	more members narrow the 68 % interval as $1/\sqrt{n}$
`require` rejects most draws	the prior produces mostly unstable systems	narrow the drawn ranges
`observed columns not in the summary`	a typo in a column name, or a column the run does not produce	compare with Table 71

Table 79. Problems with ensembles.

11.5 Fits of measured data

Symptom	Cause	Fix
stage 2 jumps to the wrong cycle: residuals of whole periods, χ^2 above the start	the start is too far from the solution, so the minima are numbered differently	anchor first (`anchor: true` or `--anchor`); free only `P` and `M0` in a first fit and the rest in a second one started from its `stage2_fitted.yaml`; narrow `bounds`

Symptom	Cause	Fix
warning “strongly correlated parameters”	the data do not separate them (e.g. M_0, ω and P of an outer orbit covered less than twice)	the large, correlated errors are correct; check them with `sampler: emcee`, or fix one parameter
stage 2 is slow	each evaluation integrates the span of the data; each iteration needs two more per free parameter	raise `processes`; leave out imprecise minima (`max_sigma_d`); fit fewer parameters
a few minima have residuals of many σ	wrong type or pair in the table, or outliers	check those rows in `stage2_residuals.csv`; `clip_sigma: 5`, a robust `loss`, or `exclude`
reduced χ^2 of tens to thousands	quoted errors far below the real scatter	`error_scale: per_source` with `error_model: jitter`
warning “the errors / outliers still changed after N fits”	the corrections did not settle within `refits`	raise `refits`, or start the next fit from `stage2_fitted.yaml`
primary and secondary residuals differ by several σ	a wrong eccentricity vector of the pair (a local minimum)	restart with $e\cos\omega$ of the opposite sign, or free `ecosw` and `esinw`
a parameter on its bound	the solution is probably not the true minimum, its error is meaningless	widen `bounds` or add a prior; check whether the data constrain it
the fitted outer orbit disagrees with a published one	the minima constrain it weakly	add priors from the published solution; the report gives each prior’s deviation in σ
a stage-1 apsidal rate far from the expected value, high reduced χ^2	minima sampled at regular cycle intervals allow aliases of the rate	add minima at other cycles, or start or constrain the rate near the expected value (Worked examples)
`sampler: emcee needs the package emcee`	the `fit` group is not installed	`pip install -e ".[fit]"`

Table 80. Problems with fits.

11.6 The worker and the web application

Symptom	Cause	Fix
jobs stay in the inbox	the worker is not running, or the jobs are on hold (`.yaml.hold`)	start the worker (Jobs page or `swingersim worker`); queue the jobs
a job runs with old code	the worker was started before an update	stop it (Jobs page or an empty file `jobs/stop`) and start it again
a job failed with an out-of-memory message	the worker stopped the job that used the most memory	fewer parallel `processes`, fewer outputs, a shorter run
a file is ignored in the inbox	its name does not end in `.yaml`	rename it
the plots of the web application do not appear	plotly is not installed and there is no internet connection	`pip install -e ".[viz]"`

Symptom	Cause	Fix
the page cannot be reached from another computer	the server listens on 127.0.0.1	`--host 0.0.0.0` (only on a trusted network)

Table 81. Problems with the worker and the web application.

11.7 Frequently asked questions

How many stars can a system have? Any hierarchy of nested pairs; systems of up to ten stars are tested. The cost grows with the number of short inner orbits, not with the number of stars.

Can SWINGERsim fit my light curve? It fits light-curve segments together with the times of minima in stage 2 of `swingersim fit`, with spherical, rotating or Roche-shaped stars and a calibration per segment; it is a photodynamical fit of the whole system, not a replacement for a detailed model of one binary's surface (spots, detailed atmospheres).

Can it run backwards in time? Yes, with a negative `run.t_max_yr` or `ensemble.directions: [backward]`. Dissipative processes (tides, winds, mass transfer) are not time-reversible, and chaotic systems make sense backwards only as ensembles (*Physics and Methods*, chapter Limitations).

Which backend should I use? N-body for anything an observer measures and for up to thousands of years; secular for millions of years and large ensembles (Two backends).

Which stellar model should I use? `zams` for quick dynamical studies of unevolved stars; `mist` for real systems and anything that ages; `sse` for fast evolution with mass transfer up to helium ignition. Measured radii and temperatures in `stars` always take precedence.

Why do the eclipse times "exact" and "as an observer" differ? The first are the conjunctions of the dynamics; the second are minima measured on a synthetic light curve. Asymmetric eclipses, the light of the other stars, pulsations, noise and sampling move a measured minimum away from the conjunction. Compare measured minima with the second.

How accurate is SWINGERsim? Every physical ingredient is compared with published results in *Testing, Part I*; applications to real systems are in Parts II–IV.

How do I cite the results? Cite the papers of the methods used (the *Methods paragraph* of a result page lists them) and give the configuration and the version of SWINGERsim, which every output file records.

Chapter 12

Glossary

Terms used in SWINGERsim and in this guide, explained for a reader from another field of astrophysics or a student starting with multiple stars. Within each group the terms are sorted alphabetically; names in code font are configuration keys or output columns.

12.1 Multiple stars and their hierarchy

Term	Meaning
2+2 quadruple	Two binaries (A and B) orbiting their common centre of mass on a wide outer orbit (`out`).
2+(2+1), 2+2+1, 3+1+1	Quintuples: in 2+(2+1) the fifth star orbits one of the binaries; in 2+2+1 and 3+1+1 it orbits the whole quadruple.
2+2+2, 3+3	Sextuples: three binaries in a hierarchy (e.g. V994 Her), or two triples orbiting each other.
3+1 quadruple	A close binary (`inner`) orbited by a third star (orbit `mid`), the triple orbited by a fourth star (`out`).
centre of mass	The mass-weighted mean position of a group of stars; in a hierarchy each subsystem moves as if its mass were concentrated there.
democratic system	Three or more stars without a clear hierarchy, all interacting strongly; short-lived.
doubly eclipsing	A system in which two binaries eclipse.
hierarchical system	A multiple star whose orbits have very different sizes, so that each is close to a Kepler ellipse slowly perturbed by the others.
hierarchy tree	The description of a hierarchy as nested pairs: each orbit joins two children, a star or another orbit.
nested triple	An orbit together with the body that perturbs it (the other component of its parent orbit); stability criteria of triples are applied to every nested triple.
preset	A named hierarchy (`2+2`, `3+1`, …) that stands for a full tree (Table 46).
sub-quadruple	Four stars inside a larger hierarchy that form a 2+2 or 3+1 on their own.
triple (2+1)	A binary with a third star on a wider orbit.

12.2 Orbits and geometry

Term	Meaning
argument of periastron (ω, `omega`)	Angle in the orbital plane from the ascending node to the periastron.
eccentricity (e)	Shape of the orbit: 0 a circle, close to 1 a very elongated ellipse.
edge-on, face-on	An orbit seen from its plane ($i = 90°$, eclipses possible) or from its pole ($i = 0°$).
inclination (i)	Angle between the orbital plane and the plane of the sky.
invariable plane	The plane perpendicular to the total angular momentum of the system; fixed in time.
Jacobi coordinates	Coordinates in which each orbit is the relative motion of two subsystems represented by their centres of mass.
longitude of the ascending node (Ω, `Omega`)	Angle in the plane of the sky to the point where the orbit crosses it moving away from the observer; only differences of Ω between orbits are measurable without astrometry.
mean anomaly (M, `M0`)	Phase of the motion along the orbit, growing uniformly in time; `M0` is its value at $t = 0$.
mean (averaged) elements	Elements averaged over the orbital motion (secular backend).
mutual inclination (i_{mut})	Angle between two orbital planes; the key quantity of the dynamics.
osculating elements	The Kepler ellipse computed from the instantaneous positions and velocities; they oscillate slightly on orbital time scales.
periastron, apoastron	Closest and farthest point of an orbit; periastron distance $q = a(1 - e)$.
prograde, retrograde	Two orbits revolving in the same sense ($i_{\mathrm{mut}} < 90°$) or in opposite senses.
semi-major axis (a)	Half of the longest diameter of the orbital ellipse.

12.3 Dynamics

Term	Meaning
1PN, 2.5PN	Orders of the post-Newtonian expansion of general relativity: 1PN gives the periastron advance, 2.5PN the loss of energy by gravitational waves.
apsidal motion	Rotation of the line of apsides of an eccentric orbit, caused by stellar distortion, relativity and other bodies; seen as diverging primary and secondary minima.
backend	The part of SWINGERsim that integrates the motion: `nbody` or `secular`.
chaos, MEGNO	Sensitive dependence on initial conditions; MEGNO (Mean Exponential Growth factor of Nearby Orbits; Cincotta & Simó 2000) tends to 2 for regular motion and grows for chaos.
Galactic tide	The tidal field of the Milky Way's disc acting on very wide orbits.
hierarchy change	A change of which stars are bound to which (exchange, ejection, break-up, reconfiguration, capture).
mean-motion resonance	A near-integer ratio of two orbital periods.
N-body integration	Direct numerical solution of the equations of motion for all stars, without averaging.

Term	Meaning
nodal precession	Slow rotation of an orbital plane around the total angular momentum; it changes the inclination to the sky and therefore the eclipses.
octupole, quadrupole	Terms of the expansion of the interaction in the ratio of inner to outer orbit size; quadrupole is the leading one.
secular dynamics	The slow evolution that remains after averaging over the orbital motions.
ZLK cycles (von Zeipel–Lidov–Kozai)	Periodic exchange between eccentricity and inclination of an inner orbit perturbed by a distant inclined body, for mutual inclinations between about 39.2° and 140.8°.

12.4 Stars, tides and interaction

Term	Meaning
AGB	Asymptotic giant branch: a late stage of low- and intermediate-mass stars with a huge, cool envelope and a strong wind.
apsidal-motion constant (k_2)	Measure of the central concentration of a star; sets how strongly its tidal and rotational bulges turn the periastron.
Applegate mechanism	A magnetic activity cycle that changes the shape of a star and thereby the period of its binary (Applegate 1992).
common envelope	Unstable mass transfer in which gas engulfs both stars; the orbit shrinks until the envelope is ejected or the stars merge.
contact binary	A binary whose two stars both overfill their Roche lobes and share one envelope (W UMa stars); fill-out factor f from 0 (inner critical surface) to 1 (outer).
effective temperature ($T_{\rm eff}$)	Temperature of a black body with the same surface flux; $L = 4\pi R^2 \sigma T_{\rm eff}^4$.
gravity darkening	A rotating or distorted star is brighter where its surface gravity is higher.
limb darkening	Decrease of the surface brightness of a star from the centre of its disc to the edge.
mass transfer	Flow of gas from a star that fills its Roche lobe (donor) to its companion (accretor).
MIST, SSE, BSE	Stellar models: MIST tracks (Choi et al. 2016); single- and binary-star evolution formulae (Hurley et al. 2000, 2002).
moment of inertia ($I/(MR^2)$, `rg2`)	Squared gyration radius of a star.
pre-main sequence	The youngest stage of a star, still contracting towards the main sequence.
pulsating star	A star whose radius and brightness change periodically (δ Scuti, γ Doradus, β Cephei, SPB, RR Lyrae, Cepheids, long-period variables).
rejuvenation	An accreting or merged main-sequence star mixes fresh hydrogen into its core and behaves as a younger star.
Roche lobe	The region around a star in a binary within which matter is bound to it; its radius after Eggleton (1983).
tidal time lag (`lag_s`)	Delay of the tidal bulge behind the companion; sets the tidal dissipation (Mignard 1979; Hut 1981).
tertiary mass transfer	Transfer from a third star that fills its Roche lobe onto the inner binary it orbits.

Term	Meaning
Wolf–Rayet star	A hot, massive star that has lost most of its hydrogen envelope and drives a dense wind.
ZAMS	Zero-age main sequence: the start of hydrogen burning; the default radii of SWINGERsim.

12.5 Stability and fates

Term	Meaning
fate	The end state of a run: intact, a hierarchy change, collision, merger, Roche overflow, or stopped by a limit.
ghost test	N-body stability test of Vynatheya et al. (2023): a system is compared with a slightly displaced copy; fast divergence signals instability.
Mardling–Aarseth criterion (MA01)	Analytic stability limit of a hierarchical triple (Mardling & Aarseth 2001); ratio above 1 = stable.
stability ratio	The ratio of the outer periastron distance to its critical value of a criterion; above 1 stable.
Vynatheya+22 criterion (V22)	Updated triple stability criterion (Vynatheya et al. 2022).

12.6 Observations

Term	Meaning
BJD_TDB, HJD, JD	Julian dates: barycentric in Barycentric Dynamical Time (the time system of SWINGERsim's measured data); heliocentric and geocentric in UTC (converted with `fit.time_system`).
centre of light	The flux-weighted velocity of the visible surface of a star, which a spectrograph measures (`fit.rv.centre: light`).
eclipse limit	The smallest inclination at which a pair still eclipses: $\lvert\cos i\rvert < (R_1 + R_2)/a$ for a circular orbit.
ellipsoidal variation	Brightness change of a binary with tidally distorted stars, two maxima per orbit.
ephemeris	Formula predicting times of minima, $T = T_0 + nP$.
light curve	Brightness against time in a passband.
LTTE (LiTE)	Light-travel-time effect: eclipses come earlier or later as the binary moves on its outer orbit (Irwin 1952).
mass function	Combination of masses and inclination measured from the LTTE or radial velocities.
minimum (primary, secondary)	Time of an eclipse. In SWINGERsim the primary is the eclipse of the first star of the pair; observers usually call the deeper one primary.
O−C, ETV	Observed minus calculated time of minimum (eclipse timing variation) against a linear ephemeris.
passband	The wavelength sensitivity of a filter and detector: UBVRI, TESS, Gaia G, BP, RP.
photodynamical model	A light curve computed directly from the positions of the stars in an N-body integration.

Term	Meaning
Rossiter–McLaughlin effect	The distortion of the measured radial velocity during an eclipse, when part of the rotating star is hidden.
third light	Light of stars that do not take part in an eclipse; it makes the eclipse shallower.

12.7 Fitting and statistics

Term	Meaning
anchoring	Adjusting the period and phase of every eclipsing pair so that the model minima fall on the measured ones, before a stage-2 fit.
autocorrelation time (τ)	Number of MCMC steps after which a walker has “forgotten” its position; a chain should be many τ long.
χ^2, reduced χ^2	Sum of squared normalised residuals; divided by the degrees of freedom it is about 1 when the model fits within the errors.
conditioning, weights	Weighting ensemble members by how well they reproduce measurements, $w = \exp(-\chi^2/2)$, 0 outside limits.
central rate	A present rate of change taken as the mean of the forward and backward fits around $t = 0$.
effective number of members (n_{eff})	$(\sum w)^2 / \sum w^2$ (Kish 1965): how many equally weighted members a weighted ensemble is worth.
error scaling, jitter	Correction of quoted errors per source: a factor, or an error floor added in quadrature.
evidence (Z)	The mean weight of an ensemble: the probability of the measurements under its hypothesis; ratios compare hypotheses with equal priors.
isotropic	Random orientation with $\cos i$ uniform.
MCMC, emcee	Markov-chain Monte Carlo sampling of the posterior; emcee is the affine-invariant ensemble sampler (Goodman & Weare 2010; Foreman-Mackey et al. 2013).
outlier rejection	Removing measurements that deviate by more than a number of robust standard deviations (`clip_sigma`).
prior, posterior	Distribution of parameters before and after taking the measurements into account.
red noise (β)	Correlated residuals; β is the factor by which binned residuals scatter more than white noise would (Pont et al. 2006; Winn et al. 2008).
seed	Number that fixes the random draws.
stage 1, stage 2	The analytic O−C model of each pair, and the joint N-body model of the whole system, of `swingersim fit`.
Wilson interval	Confidence interval of a fraction that behaves well also near 0 and 1 (Wilson 1927).

12.8 Machine learning

Term	Meaning
coverage, calibration	Whether stated intervals contain the truth as often as they claim; checked on held-out members.
deep ensemble	Several networks trained from different random starts; their spread estimates the uncertainty.
emulator	A neural network trained on an ensemble that predicts its summary columns from the drawn parameters.
neural network, MLP	A flexible function with many adjustable weights; a multi-layer perceptron is the simplest kind.
R^2	Score of a regression on held-out members: 1 perfect, 0 no better than the mean.
SBI, NPE	Simulation-based inference with neural posterior estimation: a network learns the posterior of parameters from an ensemble.

12.9 Software

Term	Meaning
configuration	The YAML file that describes one system and what to compute.
draft	A configuration written in the web application, kept in `jobs/drafts/`.
HDF5, CSV, Parquet	File formats of the run data and of the tables.
HTML report	The self-contained web page that every command writes with its results.
IAS15	The adaptive high-order integrator of REBOUND (Rein & Spiegel 2015).
job, worker	A job file describes one computation; the worker runs job files from a folder.
MESA	The stellar-evolution code used for own grids (Paxton et al. 2011).
Numba	A compiler of Python functions; it speeds up the secular equations.
operator splitting	Alternating two processes on different time scales: the stars evolve between segments, the orbits are integrated within them.
PHOEBE	A code for light curves of eclipsing binaries (Prša et al. 2016), an optional engine of SWINGERsim.
REBOUND, REBOUNDx	The N-body code and its library of additional forces (Rein & Liu 2012; Tamayo et al. 2020).
YAML	The human-readable text format of configurations and job files.

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